

Aerodynamic performance analysis of slotted airfoils for application to wind turbine blades

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1 Introduction

The study of aerofoil aerodynamics is crucial for improving the efficiency and performance of wind turbines, which play a vital role in the global transition to renewable energy. By enhancing the aerodynamic performance of aerofoils, we can improve energy extraction from wind, reduce operational costs, and optimize turbine design. To enhance the performance of the aerofoils, delaying flow separation have become a key research priority. The baseline NREL S809 aerofoil, designed for horizontal axis wind turbines (HAWT), has been studied for its aerodynamic behaviour under various operating conditions. However, at higher angle of attacks, the flow tends to separate from the aerofoil leading to stall condition. This reduced the lift of the aerofoil, affecting the energy extraction efficiency. To address the issue, passive flow control strategies such as slotted aerofoils have been used. This energizes the boundary layer, delaying the flow separation.

The study “Aerodynamic performance analysis of slotted aerofoils for application to wind turbine blades” done by Riyadh Belamadia, Abdelouaheb Djemilia, Adrian Ilincab and Ramzi Mdouki shows how the flow behaves in the presence of a slot. The results can be used to study how slot gap variation, slot position and slot angle affect the flow separation for various angle of attacks.

This current study aims to replicate the baseline two-dimensional computational fluid dynamics analysis of the S809 aerofoil shape analysis, excluding the analysis of slotted aerofoils to simplify the scope. The objective is to obtain the lift coefficient, drag coefficient and the lift/drag ratios for a various angle of attacks ranging from 0 degrees to 20 degrees. The simulation is done using an S809 aerofoil with 600mm chord length without any slots. The computational domain is a C type domain with a structured mesh. The study is carried out for Reynolds number of 1×10^6 .

The governing equations used for this study are based on Continuity equation and Navier-Stokes equations, which describes the motion of incompressible fluids. These equations are as follows.

$$\frac{\partial \rho}{\partial t} + \text{div}(\rho \mathbf{u}) = 0$$

$$\frac{\partial(\rho u)}{\partial t} + \text{div}(\rho u \mathbf{u}) = -\frac{\partial P}{\partial x} + \text{div}(\mu \text{grad } u) + S_{M_x}$$

$$\frac{\partial(\rho v)}{\partial t} + \text{div}(\rho v \mathbf{u}) = -\frac{\partial P}{\partial y} + \text{div}(\mu \text{grad } v) + S_{M_y}$$

$$\frac{\partial(\rho w)}{\partial t} + \text{div}(\rho w \mathbf{u}) = -\frac{\partial P}{\partial z} + \text{div}(\mu \text{grad } w) + S_{M_z}$$

Since we are doing incompressible, steady state simulation, these equations are converted into Reynolds Time Averaged equations as follows.

$$\frac{\partial \bar{\rho}}{\partial t} + \text{div}(\bar{\rho} \tilde{\mathbf{U}}) = 0$$

$$\frac{\partial(\bar{\rho} \tilde{U})}{\partial t} + \text{div}(\bar{\rho} \tilde{U} \tilde{\mathbf{U}}) = -\frac{\partial \tilde{P}}{\partial x} + \text{div}(\mu \text{grad } \tilde{U}) + \left[-\frac{\partial(\bar{\rho} u'^2)}{\partial x} - \frac{\partial(\bar{\rho} u' v')}{\partial y} - \frac{\partial(\bar{\rho} u' w')}{\partial z} \right] + S_{M_x}$$

$$\frac{\partial(\bar{\rho} \tilde{V})}{\partial t} + \text{div}(\bar{\rho} \tilde{V} \tilde{\mathbf{U}}) = -\frac{\partial \tilde{P}}{\partial y} + \text{div}(\mu \text{grad } \tilde{V}) + \left[-\frac{\partial(\bar{\rho} u' v')}{\partial x} - \frac{\partial(\bar{\rho} v'^2)}{\partial y} - \frac{\partial(\bar{\rho} v' w')}{\partial z} \right] + S_{M_y}$$

$$\frac{\partial(\bar{\rho} \tilde{W})}{\partial t} + \text{div}(\bar{\rho} \tilde{W} \tilde{\mathbf{U}}) = -\frac{\partial \tilde{P}}{\partial z} + \text{div}(\mu \text{grad } \tilde{W}) + \left[-\frac{\partial(\bar{\rho} u' w')}{\partial x} - \frac{\partial(\bar{\rho} v' w')}{\partial y} - \frac{\partial(\bar{\rho} w'^2)}{\partial z} \right] + S_{M_z}$$

Apart from these RANS (Reynolds Averaged Navier-Stokes) equations, the standard k-ε model uses following two equations.

$$\frac{\partial(\rho k)}{\partial t} + \text{div}(\rho k \mathbf{U}) = \text{div} \left[\frac{\mu_t}{\sigma_k} \text{grad } k \right] + 2\mu_t S_{ij} \cdot S_{ij} - \rho \varepsilon$$

$$\frac{\partial(\rho \varepsilon)}{\partial t} + \text{div}(\rho \varepsilon \mathbf{U}) = \text{div} \left[\frac{\mu_t}{\sigma_\varepsilon} \text{grad } \varepsilon \right] + C_{1\varepsilon} \frac{\varepsilon}{k} 2\mu_t S_{ij} \cdot S_{ij} - C_{2\varepsilon} \rho \frac{\varepsilon^2}{k}$$

To determine the Reynolds stresses, the Boussinesq relationship is used:

$$\overline{\rho u'_i u'_j} = \mu_t \left(\frac{\partial U_i}{\partial x_j} + \frac{\partial U_j}{\partial x_i} \right) - \frac{2}{3} \rho k \delta_{ij} = 2\mu_t S_{ij} - \frac{2}{3} \rho k \delta_{ij}$$

The geometry, mesh generation, mesh independence, simulation setup and results validation will be presented in the subsequent sections of this report.

Link to the ANSYS files: <https://dms.uom.lk/s/Hm6R3Be2XtyJcyM>

2 Assumptions and Simplifications

To simplify the scope of this study, several assumptions and simplifications were made.

2.1 Nature of the problem

The analysis of the aerofoil is two dimensional (2D), steady state and incompressible. One of the main differences between a 2D aerofoil (Infinite span) and a 3D aerofoil (Finite span) is the formation of wing tip vortices. This effect is reduced when the aspect ratio of the wing is increased. Our current study is about wind turbine blade in which the span of the blade is relatively large compared to the chord length. So, the aspect ratio of the blade is significantly high. Because of that, the 3D effect of vortices can be neglected, and the flow can be approximated as 2D. This assumption significantly reduces the computational cost while still capturing the flow behaviour around the aerofoil.

The incompressible flow assumption is justified as the simulation is done at relatively low flow velocity of 24.34ms^{-1} (Mach number below 0.3). At this subsonic velocity, the density variations are negligible. The incompressible form of Navier-Stokes equations represents this flow behaviour accurately without requiring additional complexity from using compressible effects.

The steady state is considered as the objective is to capture the time averaged aerodynamic characteristics of the aerofoil, rather than capturing the transient flow behaviour. For the range of angle of attacks, the steady state Reynolds Averaged Navier Stokes (RANS) method provides accurate predictions of mean flow properties.

2.2 Fluid properties

The fluid used in the simulation is air. The properties of the air are as follows.

Density (Constant) = 1.225 kg/m^3

Temperature = 288.16K

Dynamic viscosity (Constant) = $1.7894 \times 10^{-5}\text{ kg/ms}$

Ratio of specific heats = 1.4

2.3 Turbulence model

In the original research, the initial simulations for grid independence were done using four steady state RANS turbulence models: K- ϵ , K- ω Standard, K- ω SST and Spalart Allmaras. However, considering both accuracy and the computational power, K- ϵ model with standard wall functions was chosen as the turbulence model. This model is widely used in external aerodynamic applications due to its robustness, numerical stability and low computational cost. This model effectively captures the mean flow characteristics without requiring a finely refined near wall mesh.

2.4 Geometry

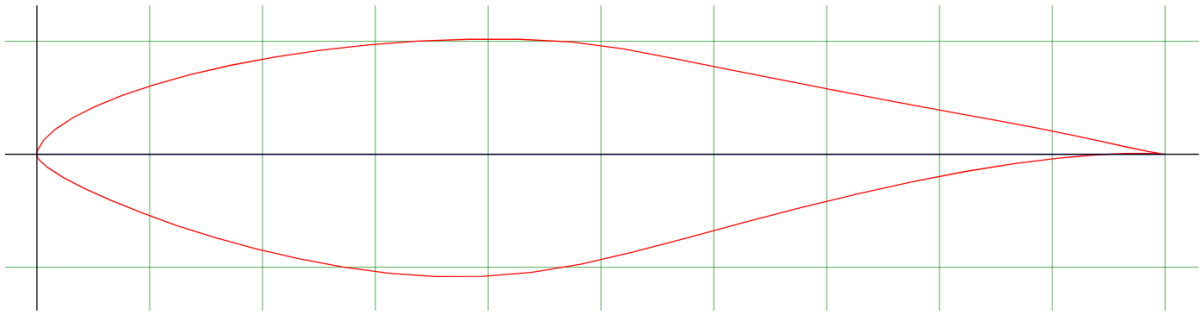


Figure 1: NREL S809 Aerofoil

The geometry used is NREL S809 aerofoil with 600mm chord length as seen in Figure 1: NREL S809 Aerofoil. The reference area for calculating the lift and drag coefficient was taken as 0.6m^2 assuming a unit span. For all simulations, the Reynolds number of 1×10^6 was used and the corresponding fixed flow velocity through the inlet is 24.34ms^{-1} .

2.5 Other assumptions

The flow from the inlet was assumed to be uniform across the boundary and the effect of gravity was neglected.

3 Flow Domain and Computational Mesh

3.1 Computational domain

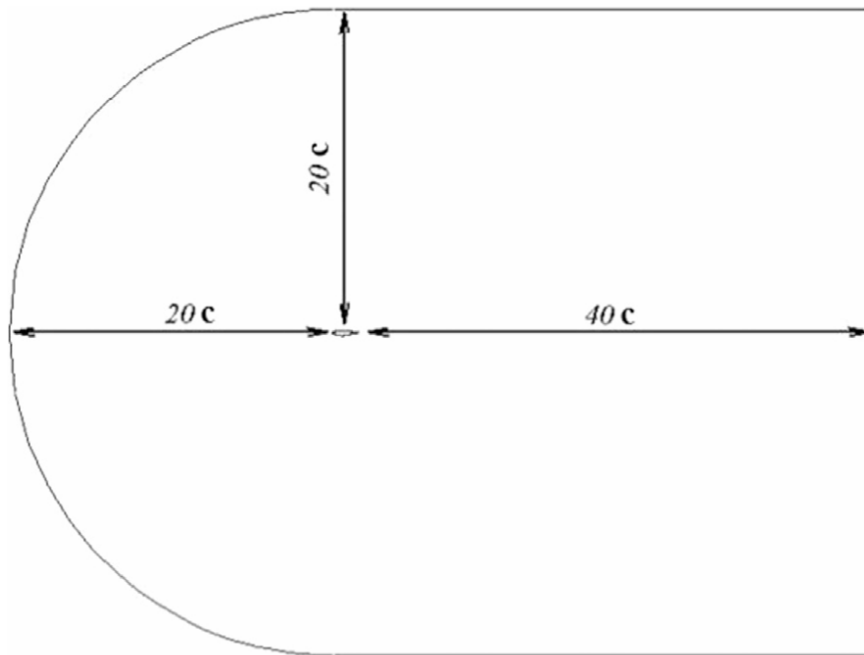


Figure 2: Computational domain dimensions

3.2 Boundaries and boundary conditions

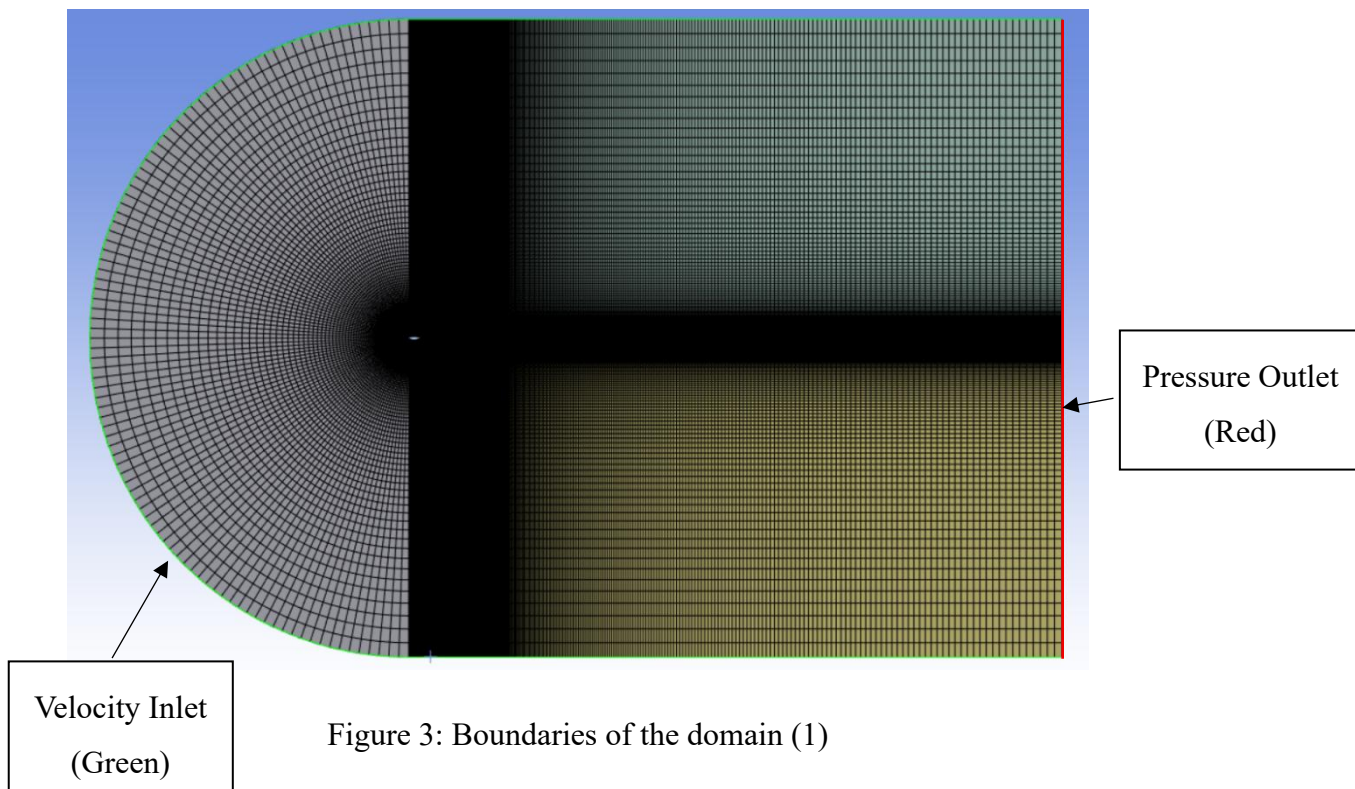


Figure 3: Boundaries of the domain (1)

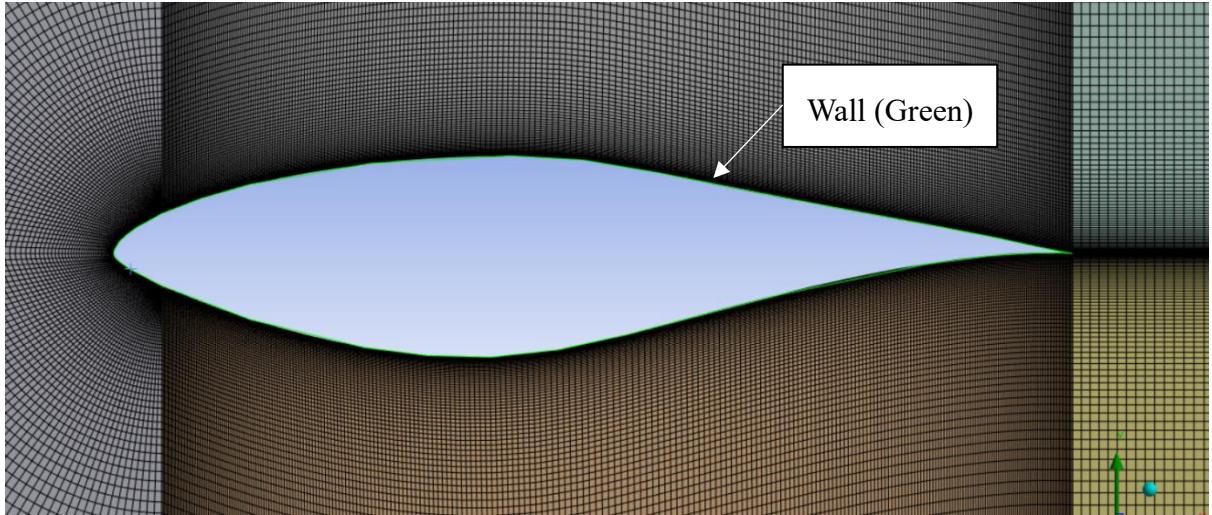


Figure 4: Boundaries of the domain (2)

Inlet: Inlet is selected as shown in green line in Figure 3. For the inlet, velocity inlet was chosen as the boundary condition. This is used to define a uniform, steady airflow. For the Reynolds number of 1×10^6 , the velocity magnitude can be calculated using the Reynolds number definition.

$$Re = \frac{\rho \times v \times c}{\mu}$$

$$v = \frac{Re \times \mu}{\rho \times c} = \frac{1 \times 10^6 \times 1.7894 \times 10^{-5} \text{ kg/ms}}{1.225 \text{ kg/m}^3 \times 0.6 \text{ m}} = 24.34 \text{ ms}^{-1}$$

This velocity was defined in the ANSYS Fluent setup as X and Y components as seen in Table 1. The angle of attack of the aerofoil is changed by changing the X and Y components as below Figure 5: Velocity captions.

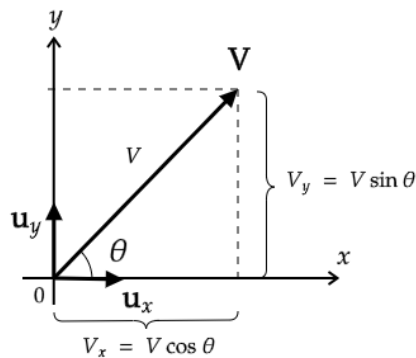


Figure 5: Velocity captions

Table 1: Velocity components for various AOA values

Velocity	AOA in Deg	AOA in Rad	X Component	Y component
24.34	20	0.3491	22.8721	8.3248
24.34	18	0.3142	23.1487	7.5215
24.34	16	0.2793	23.3971	6.7090
24.34	14	0.2443	23.6170	5.8884
24.34	12	0.2094	23.8081	5.0606
24.34	10	0.1745	23.9702	4.2266
24.34	8	0.1396	24.1031	3.3875
24.34	6	0.1047	24.2067	2.5442
24.34	4	0.0698	24.2807	1.6979
24.34	2	0.0349	24.3252	0.8495
24.34	0	0.0000	24.3400	0.0000

Example inlet boundary conditions for AOA of 20 degrees are defined as in Figure 6.

Velocity Inlet

Zone Name
inlet

Momentum Thermal Radiation Species DPM Multiphase Potential Structure UDS

Velocity Specification Method Components

Reference Frame Absolute

Supersonic/Initial Gauge Pressure [Pa] 0

X-Velocity [m/s] 22.8721

Y-Velocity [m/s] 8.3248

Turbulence

Specification Method Intensity and Viscosity Ratio

Turbulent Intensity [%] 5

Turbulent Viscosity Ratio 10

Apply Close Help

Figure 6: Boundary conditions of inlet for AOA of 20 degrees

Outlet: The outlet is shown in red line in Figure 3. For the outlet, pressure outlet boundary condition was used. It allows air to exit the domain smoothly and prevents reflection of disturbances back into the computational domain. The gauge pressure was set to 0 Pa as shown in Figure 7.

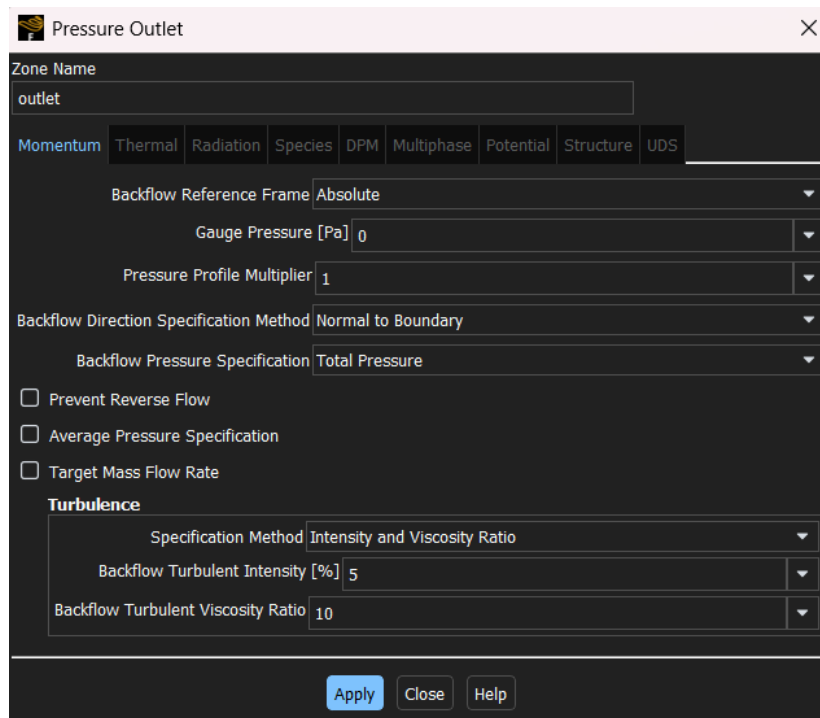


Figure 7: Boundary conditions of outlet

Aerofoil: The aerofoil surface is selected as shown in green outline in Figure 4. For the aerofoil, no slip wall condition was selected as seen in Figure 8. This condition allows zero velocity at the aerofoil surface, capturing viscous shear stresses and boundary layer effects accurately.

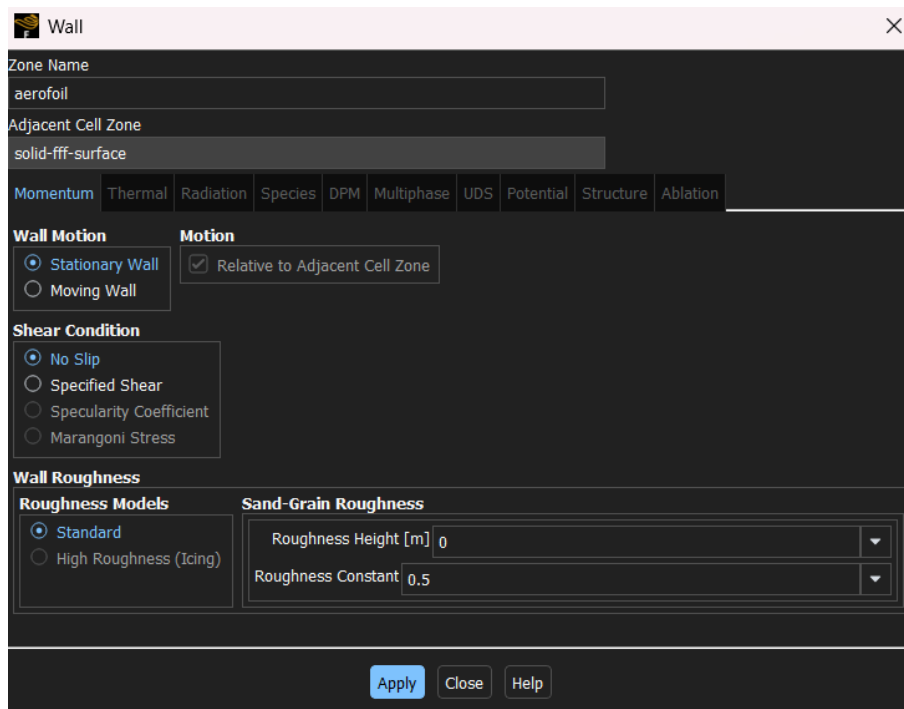


Figure 8: Boundary conditions of aerofoil

3.3 Mesh

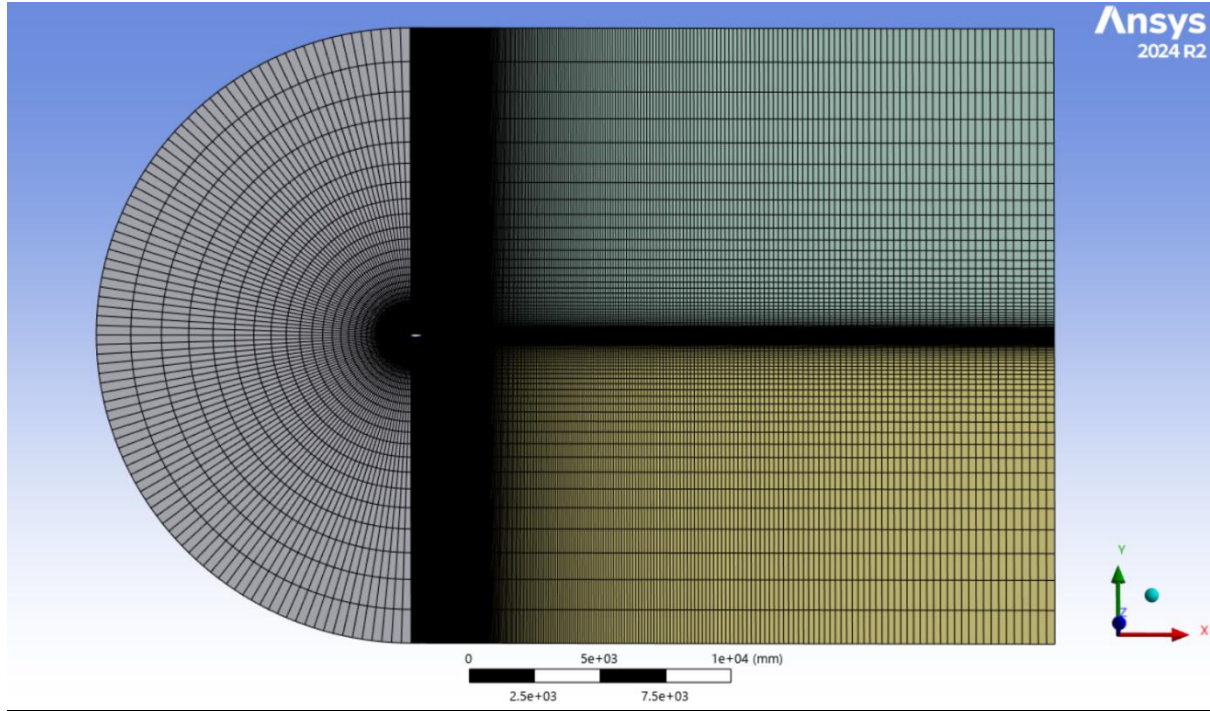


Figure 9: Computational mesh

A C-type structured mesh was used as shown in Figure 9 for the analysis as it can be used to refine the near wall mesh while keeping a relatively coarse mesh away from surface. This approach is computationally efficient as it minimizes the number of cells at far field where flow property gradients are negligible. The diameter of the C domain is $40C$ and the wake region (Downstream) length is taken as $40C$ where the C is the chord length of the aerofoil as seen in Figure 2: Computational domain dimensions.

3.4 Near wall Y^+ value

In the original study, initially two meshes have been generated with wall $y^+ = 1$ and $y^+ = 30$ since four turbulence models were used. But considering both accuracy and the computational time, $k-\epsilon$ model with standard wall functions was chosen as the turbulence model. The wall $y^+ = 30$ was maintained for all the simulations. This not only reduces the number of cells but also reduces the computational time while maintaining a good accuracy level by capturing the boundary layer effects at the near wall surface. The use of inflation layers is not possible as this structured mesh uses a face mesh. To get the effect of inflation layers, bias factor was used as

seen in Figure 10, 11 and 12. The first layer height was adjusted to 1mm using the bias factor of 500 with 200 divisions normal to the aerofoil surface.

First layer height calculation: [1]

$$\text{Chord length} = 0.6m$$

$$Re = 1 \times 10^6$$

Skin friction coefficient can be calculated using an empirical formular:

$$C_f = [2 \log_{10}(Re) - 0.65]^{-2.3} = [2 \log_{10}(1 \times 10^6) - 0.65]^{-2.3} = 3.7454 \times 10^{-3}$$

$$\text{Wall shear stress } \tau_w = C_f \times \frac{1}{2} \rho u^2 = 3.7454 \times 10^{-3} \times \frac{1}{2} \times 1.225 \times 24.34^2 = 1.35911 Pa$$

$$\text{Friction velocity } u^* = \sqrt{\frac{\tau_w}{\rho}} = \sqrt{\frac{1.35911}{1.225}} = 1.05331 ms^{-1}$$

$$y_p = \frac{y^+ \mu}{u^*} = \frac{30 \times 1.7894 \times 10^{-5}}{1.05331} = 5.0965 \times 10^{-4} \text{ where } y^+ = 30$$

$$\text{First layer height} = 2y_p = 1.0193 \text{ mm}$$

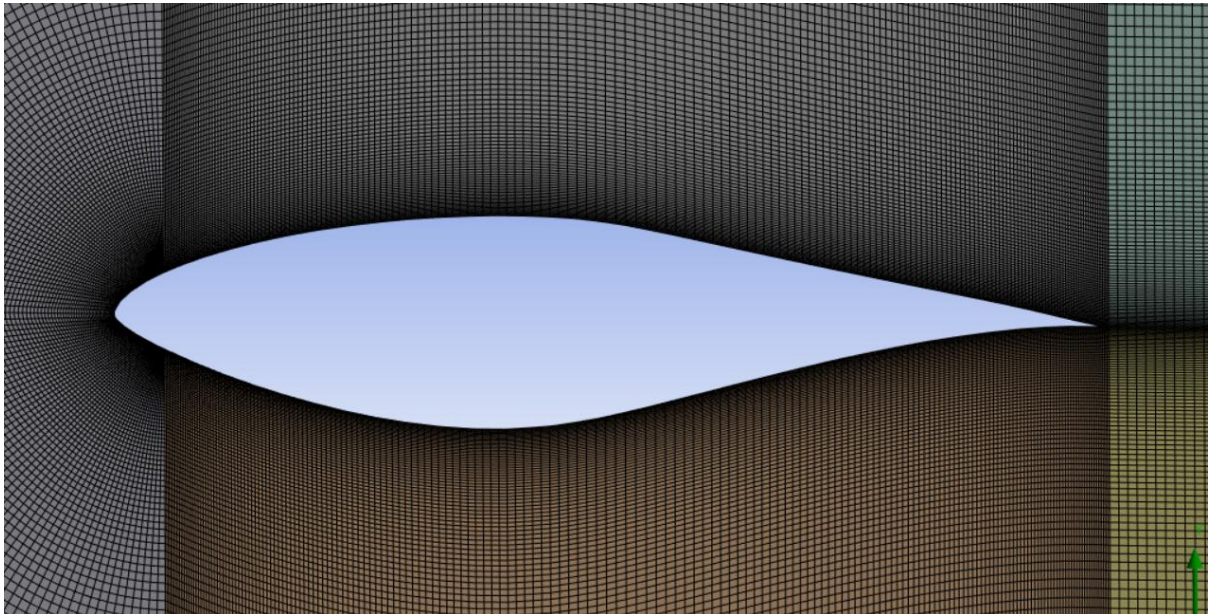


Figure 10: Near wall surface mesh

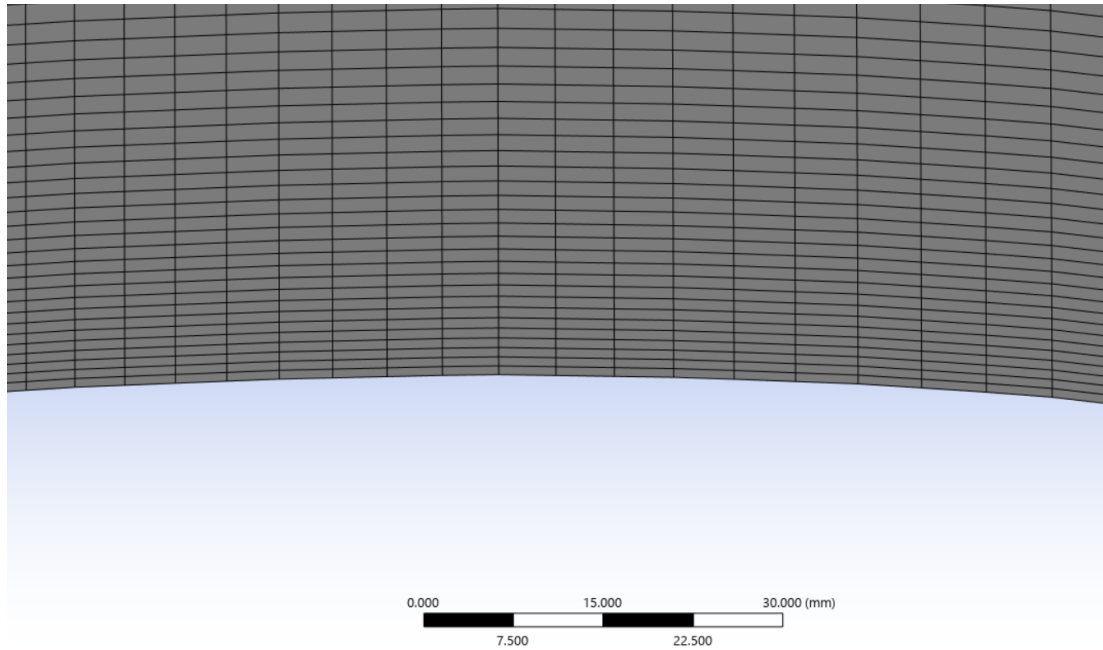


Figure 11: Near wall surface mesh (Zoomed)

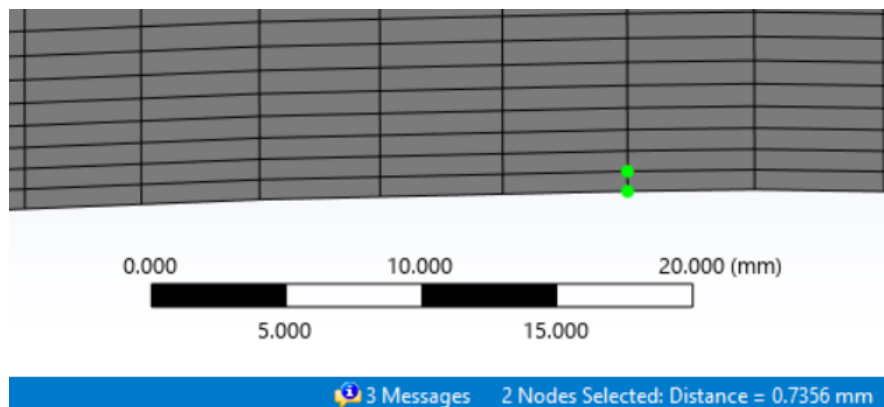


Figure 12: First layer height at near wall surface

Measured first layer height at the near wall surface is 0.7356mm which is well below than required 1.0193mm. so the $y^+ = 30$ is achieved.

3.5 Basic mesh statistics

Statistics	
☐ Nodes	235422
☐ Elements	234400
Show Detailed Statistics	No

Figure 13: Mesh statistics

3.6 Mesh metrics

3.6.1 Orthogonality

Details of "Mesh"	
Display	
Display Style	Use Geometry Setting
Defaults	
Physics Preference	CFD
Solver Preference	Fluent
☐ Element Size	Default (2187.1 mm)
Export Format	Standard
Export Preview Surface Mesh	No
Sizing	
Quality	
Check Mesh Quality	Mesh Quality Worksheet
☐ Target Skewness	Default (0.9)
Smoothing	Medium
Mesh Metric	Orthogonal Quality
☐ Min	0.69215
☐ Max	1.
☐ Average	0.99245
☐ Standard Deviation	1.665e-002
Inflation	
Batch Connections	

Figure 14: Orthogonal quality

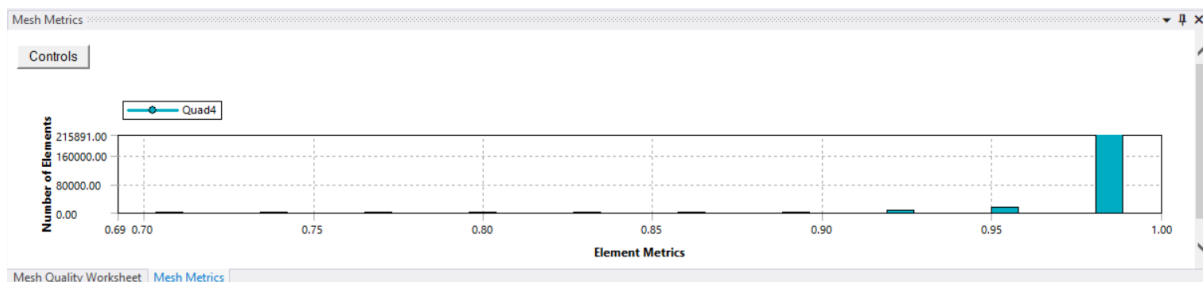


Figure 15: Orthogonal quality distribution of the mesh

3.6.2 Skewness

Details of "Mesh" ▾ ⚙ □ ×

[-] Display	
Display Style	Use Geometry Setting
[-] Defaults	
Physics Preference	CFD
Solver Preference	Fluent
☐ Element Size	Default (2187.1 mm)
Export Format	Standard
Export Preview Surface Mesh	No
[+] Sizing	
[-] Quality	
Check Mesh Quality	Mesh Quality Worksheet
☐ Target Skewness	Default (0.9)
Smoothing	Medium
Mesh Metric	Skewness ▾
☐ Min	1.3057e-010
☐ Max	0.51794
☐ Average	3.7431e-002
☐ Standard Deviation	6.3148e-002
[+] Inflation	
[+] Batch Connections	

Figure 16: Skewness

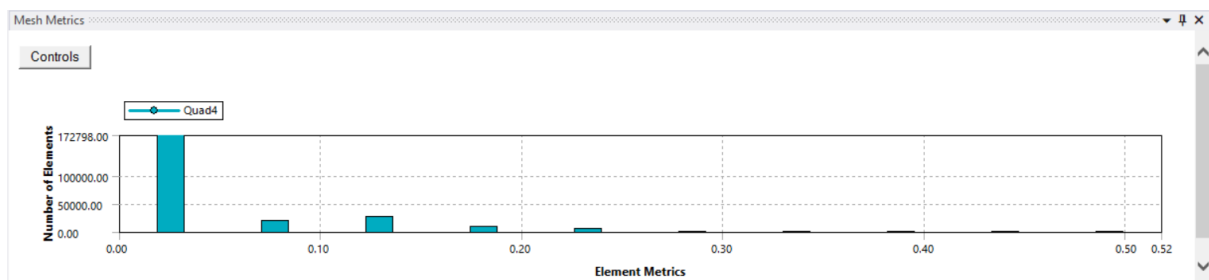


Figure 17: Skewness distribution of the mesh

3.6.3 Element quality

Details of "Mesh" ▾ 🔍 □ ×

[-] Display	
Display Style	Use Geometry Setting
[-] Defaults	
Physics Preference	CFD
Solver Preference	Fluent
☐ Element Size	Default (2187.1 mm)
Export Format	Standard
Export Preview Surface Mesh	No
[+] Sizing	
[-] Quality	
Check Mesh Quality	Mesh Quality Worksheet
☐ Target Skewness	Default (0.9)
Smoothing	Medium
Mesh Metric	Element Quality ▾
☐ Min	5.4181e-003
☐ Max	1.
☐ Average	0.47215
☐ Standard Deviation	0.35765
[+] Inflation	
[+] Batch Connections	

Figure 18: Element quality

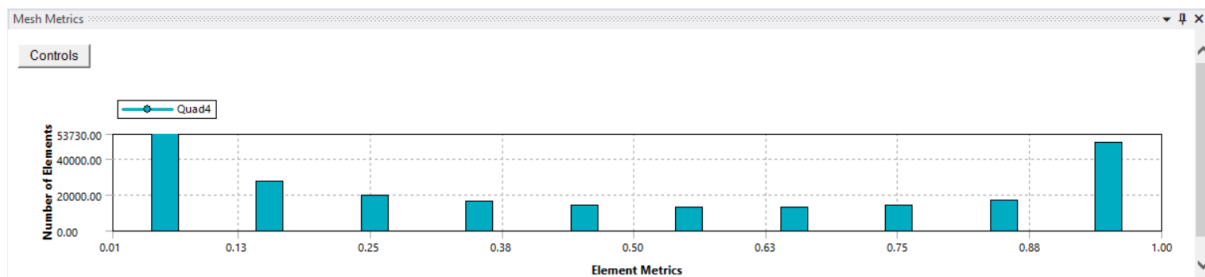


Figure 19: Element quality distribution of the mesh

3.6.4 Aspect ratio

Details of "Mesh" ▾ ↕ □ ×

[-] Display	
Display Style	Use Geometry Setting
[-] Defaults	
Physics Preference	CFD
Solver Preference	Fluent
☐ Element Size	Default (2187.1 mm)
Export Format	Standard
Export Preview Surface Mesh	No
[+] Sizing	
[-] Quality	
Check Mesh Quality	Mesh Quality Worksheet
☐ Target Skewness	Default (0.9)
Smoothing	Medium
Mesh Metric	Aspect Ratio ▾
☐ Min	1.
☐ Max	369.13
☐ Average	17.373
☐ Standard Deviation	31.901
[+] Inflation	
[+] Batch Connections	

Figure 20: Aspect ratio

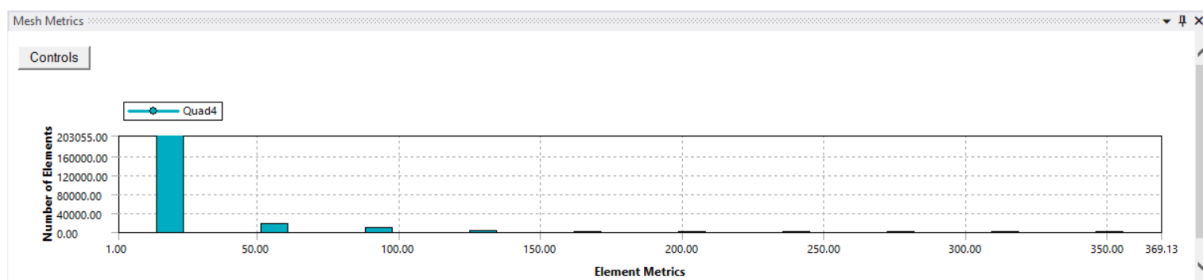


Figure 21: Aspect ratio distribution of the mesh

3.6.5 Recommended values for mesh metrics

Skewness mesh metrics spectrum

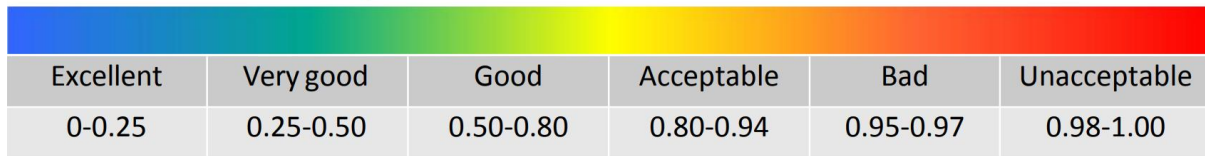


Figure 22: Skewness recommended values

Orthogonal Quality mesh metrics spectrum

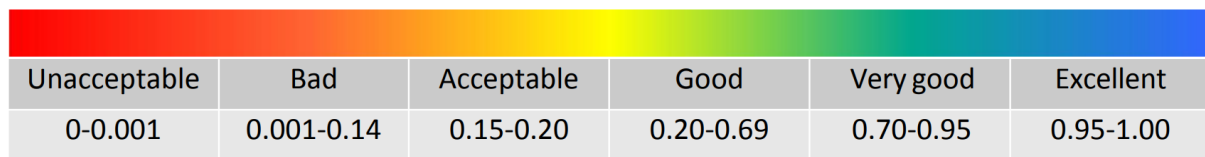


Figure 23: Orthogonal recommended values

Skewness of the mesh is ranging from 1.3057×10^{-10} to 0.51794. The maximum value of them is in the good region of the recommended values as seen in Figure 22.

Orthogonal quality of the mesh is ranging from 0.69215 to 1. The minimum value of them is in the very good region of the recommended values as seen in Figure 23.

4 Simulation Settings

4.1 Convective flux calculation scheme

The authors have not specified what scheme they use to perform this analysis. In my study, Second-Order Upwind Scheme has been used for the discretization of convective flux terms in governing equations.

First-Order Upwind Scheme: Although this scheme is numerically stable, it introduces excessive numerical diffusion (or artificial viscosity), which tends to smear sharp gradients and smooth out important flow features such as the separation point or pressure recovery regions. This results in underprediction of the lift coefficient and overprediction of drag, leading to less accurate aerodynamic performance results.

The Second-Order Upwind Scheme: The second order upwind scheme improves the solution accuracy by using a quadratic interpolation of the flow variables between neighbouring cells. It reconstructs the variable values at the control volume faces based on the local flow gradients (computed from the upstream direction). This approach significantly reduces numerical diffusion and captures the pressure and velocity gradients more precisely, which is essential for accurately predicting boundary layer behaviour, flow separation, and stall onset in aerodynamic simulations [2].

QUICK (Quadratic Upstream Interpolation for Convective Kinematics) scheme: It is of higher order accuracy. But for application to this work, it is not possible because it will generate numerical oscillations and instability in flows with strong pressure or velocity gradients, such as near an aerofoil. It also requires a smooth cell alignment along the direction of flow in a structured mesh, which cannot be easily obtained in a C-type grid near curved geometries such as the leading edge of the aerofoil.

Although the second-order scheme is more computationally expensive and less stable in the case of poor mesh quality or severe skewness, it is well worth employing here since the flow over the S809 aerofoil has strong pressure gradients and curvature effects that require higher-order accuracy. Moreover, since the experiment is revolving around validation of aerodynamic coefficients (C_l and C_d) and comparison against experimental outcomes, the increased accuracy of the second-order scheme is crucial in being able to obtain valuable results.

4.2 Diffusive flux calculation scheme

Again, the authors have not specified the diffusive flux calculation scheme. Least Squares Cell-Based was selected in my research to solve for the spatial gradients of the flow variables required for calculating the diffusive fluxes of the governing equations. This was selected because it offers second order accuracy and greater stability [3].

This method estimates the gradients by least-square minimization of variable value differences across neighbouring cell centres. This allows it to handle distorted cell shapes and skewed grids more effectively than simpler gradient reconstruction techniques. As a result, it smooths the diffusion calculations and produces physically more realistic results near the boundary layer and wake regions where mesh orthogonality is lost.

Some other commonly used schemes for the computation of gradients are,

Green-Gauss Cell-Based Method: This method computes gradients from the face-averaged integral form of Gauss's theorem. While computationally faster, it is best on very orthogonal and structured meshes. On skewed or curved grids such as the leading and trailing edges of the aerofoil, it produces false gradients and artificial diffusion effects.

Green-Gauss Node-Based Method: It improves precision by interpolating nodal values before applying Gauss's theorem. However, it increases computational cost and is most effectively used for tetrahedral or polyhedral unstructured meshes. For the present quadrilateral C-type grid, it lacks significant advantages over the Least Squares Cell-Based method.

On the other hand, the Least Squares Cell-Based method provides a mesh-independent, higher-order accurate gradient reconstruction that remains valid in regions of poor orthogonality. It minimizes errors of gradient estimation without inducing false diffusion, leading to better prediction of viscous stresses and heat fluxes in aerodynamic flow calculations.

The settings used for convective and diffusive flux scheme is shown in Figure 24.

Solution Methods ?

Pressure-Velocity Coupling

Scheme
Coupled ▼

Flux Type
Rhie-Chow: momentum based ▼ ☒ Auto Select

Spatial Discretization

Gradient
Least Squares Cell Based ▼

Pressure
Second Order ▼

Momentum
Second Order Upwind ▼

Turbulent Kinetic Energy
First Order Upwind ▼

Turbulent Dissipation Rate
First Order Upwind ▼

Pseudo Time Method
Off ▼

Transient Formulation
▼

☐ Non-Iterative Time Advancement

☐ Frozen Flux Formulation

☐ Warped-Face Gradient Correction

☐ High Order Term Relaxation

Default

Figure 24: Convective and Diffusive flux calculation schemes

4.3 Turbulence modelling

In this research paper, four turbulence models were initially used to accurately capture the flow behaviour: k - ϵ , k - ω Standard, k - ω SST, and Spalart Allmaras.

k - ϵ is a two-equation turbulence model that solves for the turbulent kinetic energy (k) and its dissipation rate (ϵ). This model is great for predicting far field conditions of a flow with a greater solution stability. So this is widely used for external flow simulations. Because this uses a comparatively large y^+ value, it is computationally more efficient. But this model has limitations in predicting flow separations in complex turbulence such as stalled conditions of an aerofoil.

k - ω Standard is also a two-equation model, but it solves for turbulent kinetic energy (k) and specific dissipation rate (ω). This model performs well at the near wall regions due to more accurate representation of boundary layers. This is great for predicting adverse pressure gradients and separation at low Reynolds numbers where wall effects are dominant. This model is used with a $y^+ \approx 1-5$. This means more smaller mesh elements near the surfaces to represent the boundary layer. This makes the total mesh element count high which results in higher computational demand for solving.

k - ω SST is a turbulence model that combines both k - ϵ and k - ω Standard. It uses k - ϵ for the far field and k - ω Standard for the near wall regions. So, this combines the accurate near wall treatment of k - ω Standard and free stream flow prediction of k - ϵ . This model is excellent for predicting flow separations and adverse pressure gradient flows in aerospace applications. This also uses a boundary layer of $y^+ \approx 1-5$, demanding for higher computational cost.

The **k - ϵ turbulence model** has been selected for this study because it is computationally efficient and well suited for simulating external, steady-state turbulent flows, particularly over airfoils. It is recommended to use the wall $y^+ \geq 30$ for simulating this model. It means that it uses relatively taller first layer cells near the wall surfaces. Comparatively, the other models such as k - ω Standard, k - ω SST, and Spalart Allmaras need a wall $y^+ \approx 1-5$ which makes the first layer heights much smaller. This means required mesh resolution near the wall is high resulting in higher mesh element counts. So, the k - ϵ model uses comparatively coarse mesh resulting in lower mesh counts which is more computationally efficient. This research paper focuses on wind turbine airfoils and the k - ϵ model is selected after comparing it with other models, such as Spalart-Allmaras, k - ω , and k - ω SST, based on its accuracy in predicting lift and drag

coefficients without requiring excessive computational resources [4]. Used settings for this turbulence model is shown in Figure 25.

The $k-\epsilon$ model was thus suitable for this study as it offered an efficient and effective means to capture the primary turbulence effects accurately while remaining within manageable computational resource limits.

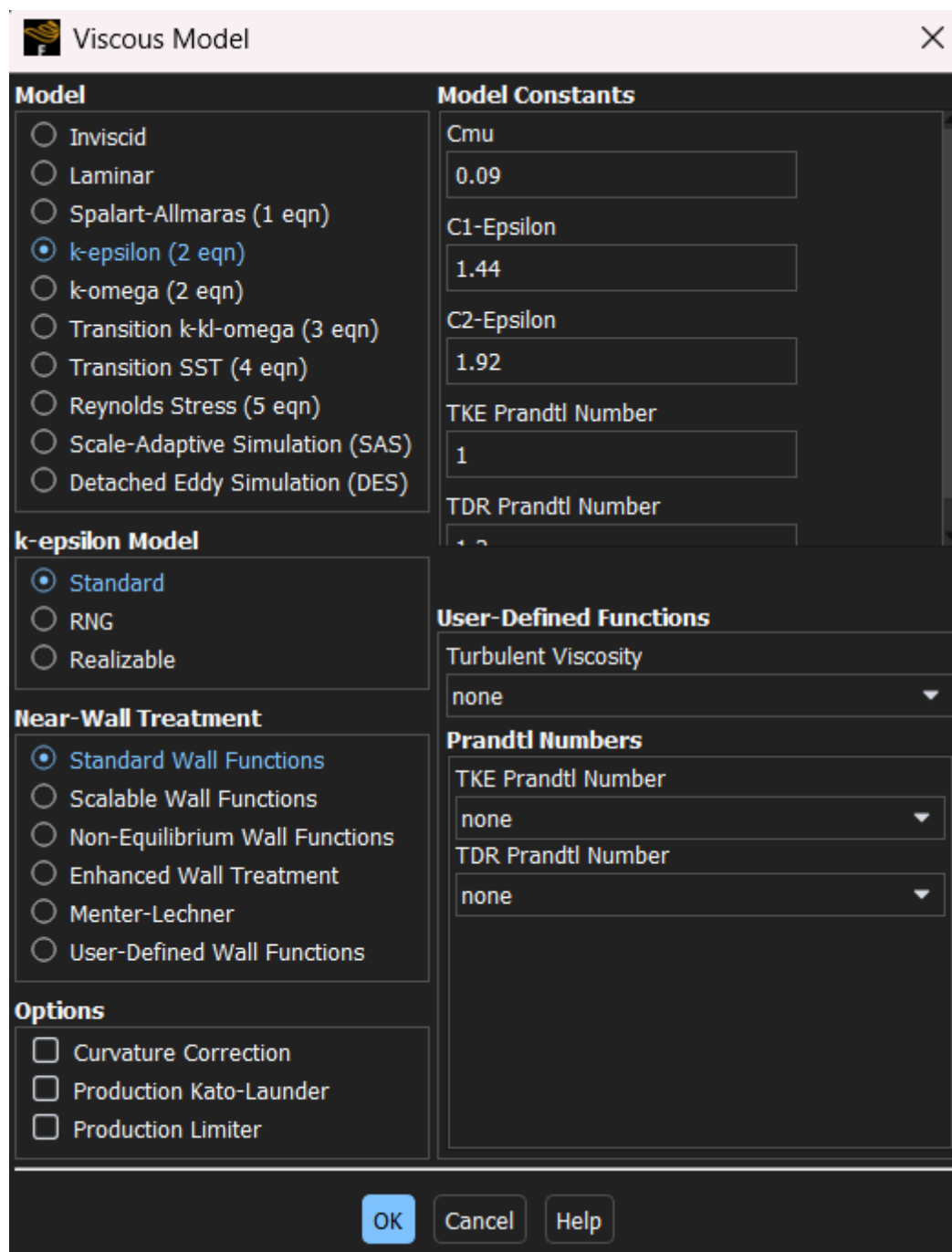


Figure 25: Turbulence model settings

4.4 Pressure velocity coupling

In the incompressible flow simulations, the pressure and velocity fields are strongly coupled because the pressure serves as a constraint to enforce mass conservation, while velocity governs momentum transport. So, suitable pressure-velocity coupling scheme is essential to ensure numerical stability and convergence in the solution of the NS equations. Several algorithms are being used in CFD for this. As examples, SIMPLE, SIMPLEC, PISO and Coupled methods can be taken.

SIMPLE (Semi-Implicit Method for Pressure Linked Equations): This is the most widely used and stable for steady-state problems but converges relatively slowly because it solves pressure and velocity in a segregated manner.

SIMPLEC (Semi-Implicit Method for Pressure Linked Equations Consistent): This improves the convergence rate of SIMPLE by modifying the correction step, reducing under-relaxation effects.

PISO (Pressure-Implicit with Splitting of Operators): This is designed for transient or unsteady simulations, performing multiple correction steps per iteration to enhance accuracy but at the cost of additional computational time per step.

Coupled: The **Coupled pressure-velocity coupling method** was employed in this study to solve the steady-state, incompressible flow around the S809 aerofoil. Compared to the other methods, the Coupled method solves pressure and momentum equations simultaneously in a fully implicit manner, significantly improving convergence rates for steady external flows. Although it requires slightly more memory per iteration, the overall computational time to reach convergence is often lower due to fewer total iterations. The Coupled scheme is particularly advantageous for flows involving strong pressure-velocity coupling, such as external aerodynamic problems with boundary layer development and separation regions. Therefore, following the original study, the Coupled algorithm was chosen as it provides a stable, accurate, and computationally efficient framework for reproducing the aerodynamic characteristics of the baseline S809 aerofoil under steady, incompressible conditions.

5 Results and Validation

5.1 Grid independency analysis

Grid independence test is conducted to ensure that the results obtained from the CFD simulations are not significantly affected by the computational mesh resolution. In this study, a series of grid refinements were done using parametric analysis. For each design point, the lift coefficient and drag coefficient were obtained. The resolution of the computational mesh is governed mainly by three aspects. The number of divisions in,

- Normal direction to the aerofoil
- Downstream of the aerofoil
- Aerofoil shape

By keeping them as parameters, design points were created by varying the number of divisions by increment of 10 as shown in Table 2. The angle of attack was kept at 16 degrees.

Table 2: Grid independence test results

Design Point	Number of Divisions			Mesh Elements	Drag Coefficient	Lift Coefficient
	Aerofoil	Downstream	Normal			
DP 17	190	240	90	84780	0.098417472	1.0911021
DP 16	200	250	100	98000	0.097082151	1.106069
DP 15	210	260	110	111980	0.096235239	1.1166139
DP 14	220	270	120	126720	0.095861631	1.122527
DP 13	230	280	130	142220	0.095639873	1.1267712
DP 12	240	290	140	158480	0.095683871	1.1280391
DP 4	270	320	170	169320	0.095633199	1.1301249
DP 3	280	330	180	200880	0.095415865	1.1389272
DP 2	290	340	190	217360	0.095314212	1.1430488
DP 1	300	350	200	234400	0.095269989	1.1464498
DP 7	310	360	210	251580	0.09526695	1.1491259
DP 8	320	370	220	269720	0.095270242	1.1516706
DP 9	330	380	230	288420	0.095341294	1.1533666
DP 10	340	390	240	307200	0.095348118	1.1556055
DP 11	350	400	250	326500	0.095456363	1.1567295
DP 18	360	410	260	346840	0.09553749	1.1580602
DP 19	370	420	270	367200	0.09562191	1.1592303
DP 20	380	430	280	388080	0.095719331	1.1603135
DP 21	390	440	290	431520	0.095757136	1.1625081
DP 22	400	450	300	454800	0.095896479	1.1629552

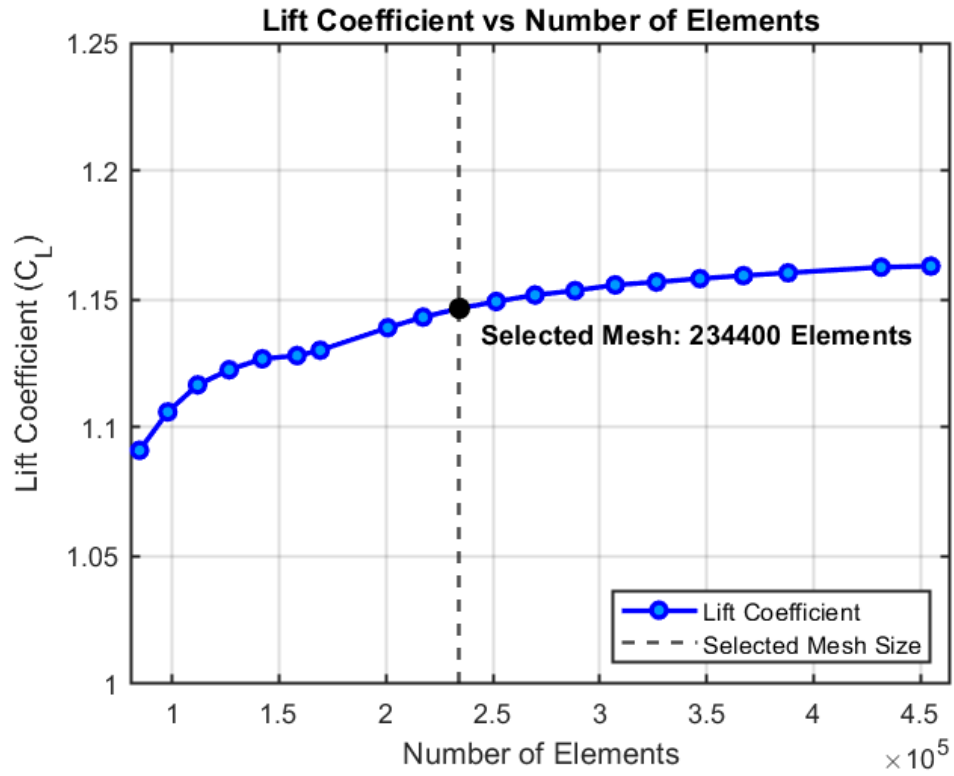


Figure 26: Grid Independence for Lift Coefficient

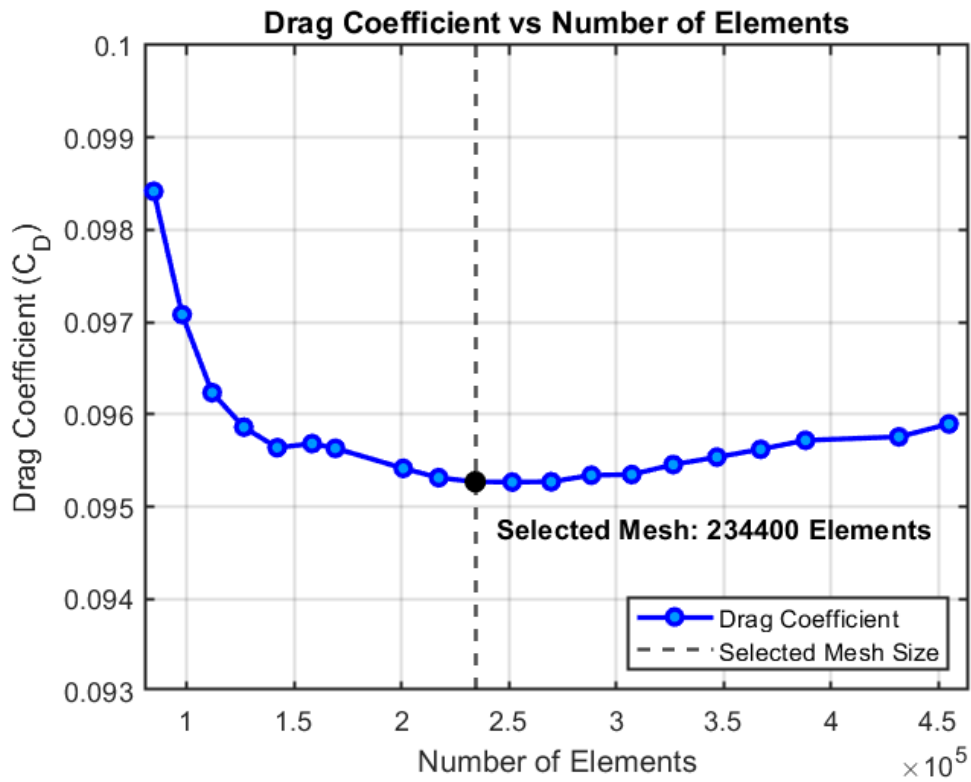


Figure 27: Grid Independence for Drag Coefficient

By observing the results of the parametric design points, both lift coefficient and drag coefficient began to converge after around 200000 mesh count as seen in Figure 26 and 27. The mesh with 234400 number of elements was selected for the study as a converged solution. The selected mesh point is coloured in black in the lift and drag coefficient curves.

5.2 Solution convergency study

To ensure the reliability and stability of the simulation results, the convergence criteria were refined to achieve higher solution accuracy. The residual convergence limits for continuity, x-velocity, y-velocity, turbulent kinetic energy (k), and turbulent dissipation rate (ϵ) were set to 1×10^{-4} rather than keeping it in 1×10^{-3} as seen in Figure 28. This ensures better numerical stability and allows the flow field variables to reach a consistent steady-state condition.

In addition, report definitions were created to monitor the convergence behaviour of the lift coefficient and drag coefficient, with an absolute convergence criterion of 1×10^{-4} as seen in Figure 29. These aerodynamic coefficients serve as critical indicators of overall solution convergence, ensuring that both flow and performance parameters stabilize. The solution calculation was set to stop only when all the above convergence conditions are met.

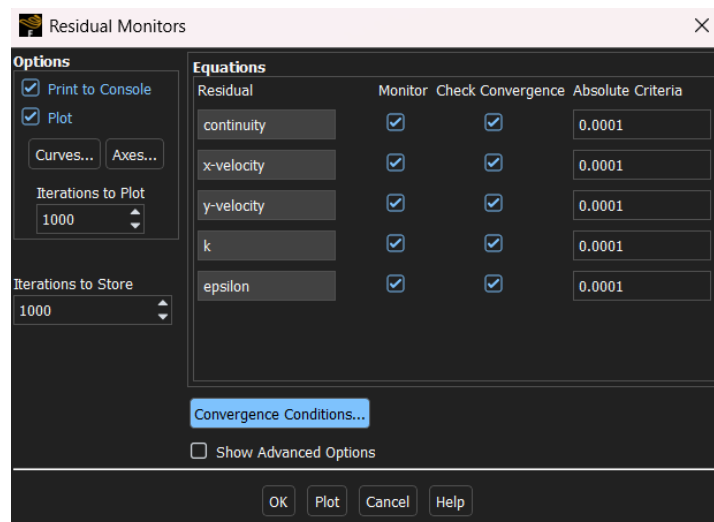


Figure 28: Residuals convergence criteria

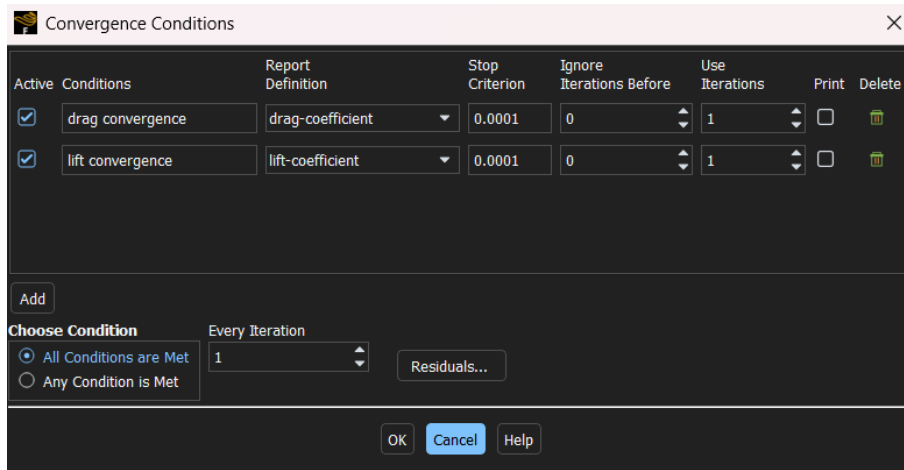


Figure 29: Lift and drag convergence criteria

For each angle of attack, four report definitions were generated: lift coefficient, drag coefficient, lift-to-drag ratio, and net mass inflow rate. This study was conducted for angle of attacks ranging from 0° to 20° , with increments of 2° . All simulations were performed according to the convergence criteria described above to ensure solution accuracy and consistency.

Presented below are the plots of lift coefficient, drag coefficient, lift-to-drag ratio, net mass inflow rate and residuals for selected angle of attacks: 0° , 10° , and 20° representing key points across the tested range as examples. For the other angles, lift coefficient and drag coefficient convergence plots are presented.

20° Angle of Attack

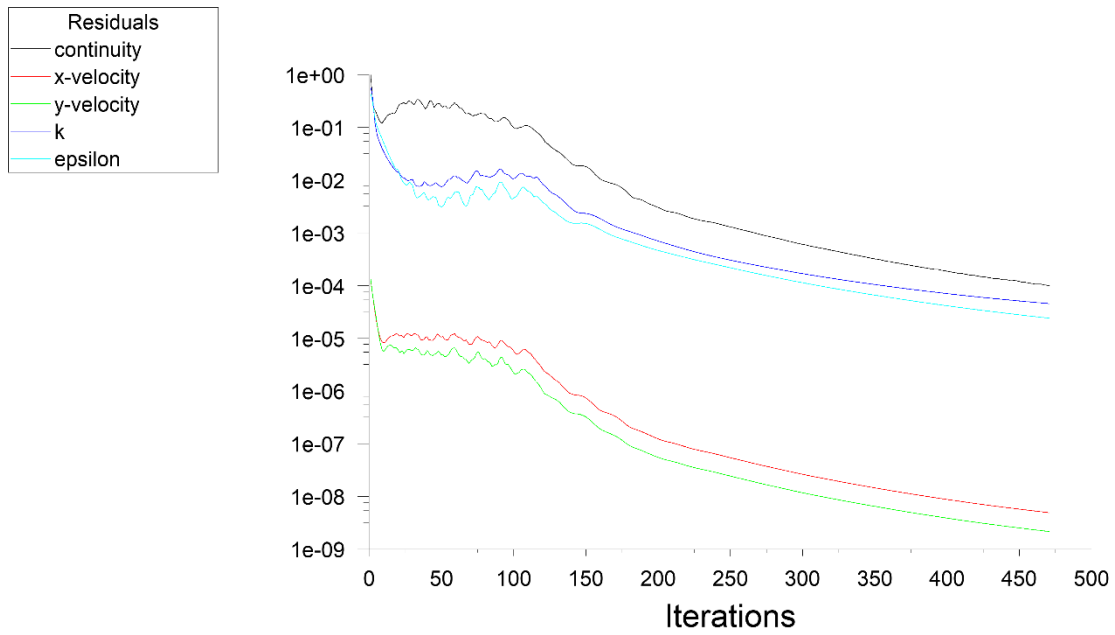


Figure 30: Residuals monitor (20 AOA)

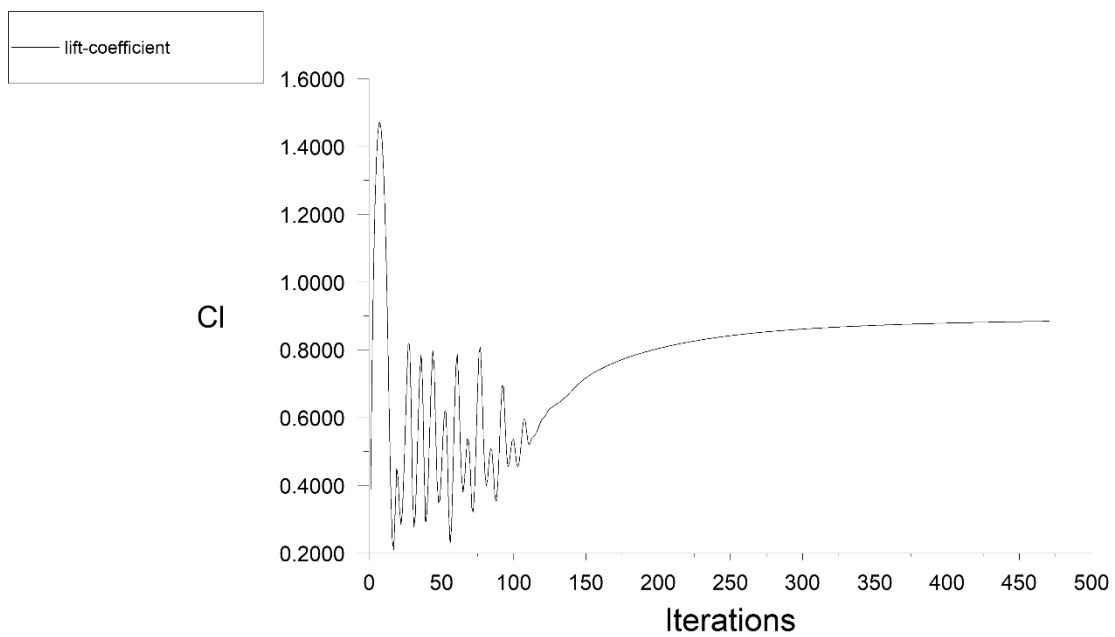


Figure 31: Lift coefficient vs Iterations (20 AOA)

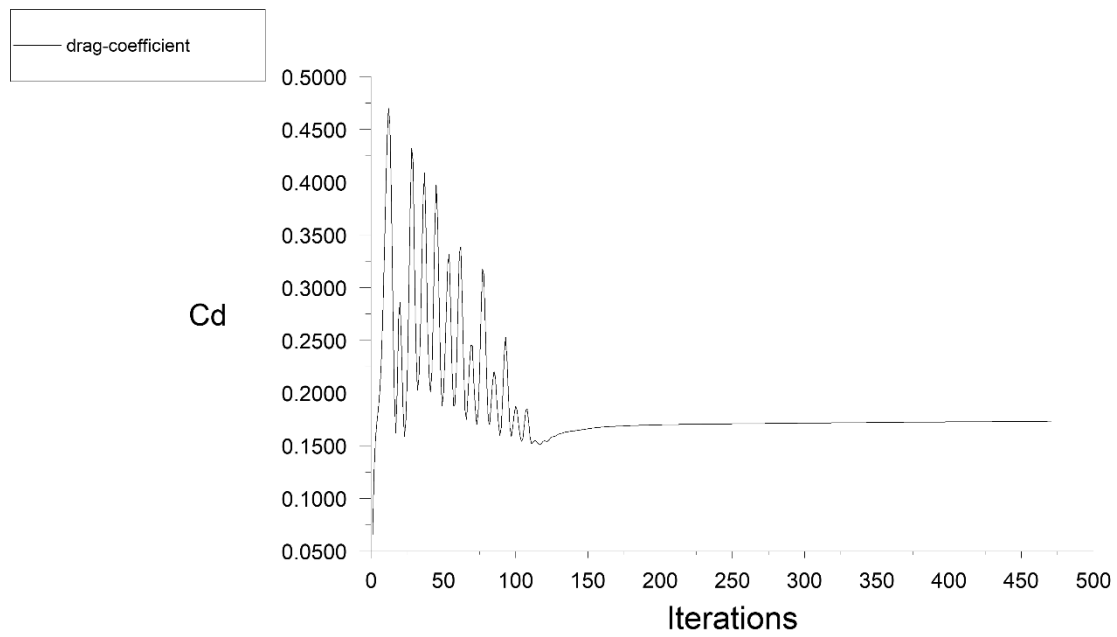


Figure 32: Drag coefficient vs Iterations (20 AOA)

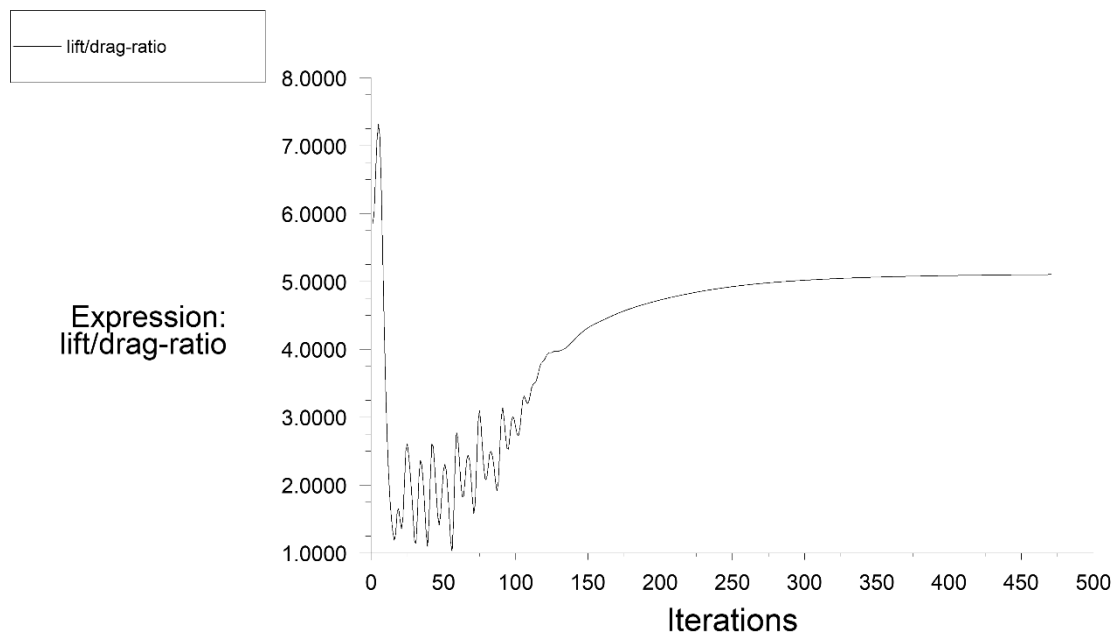


Figure 33: Lift/Drag vs Iterations (20 AOA)

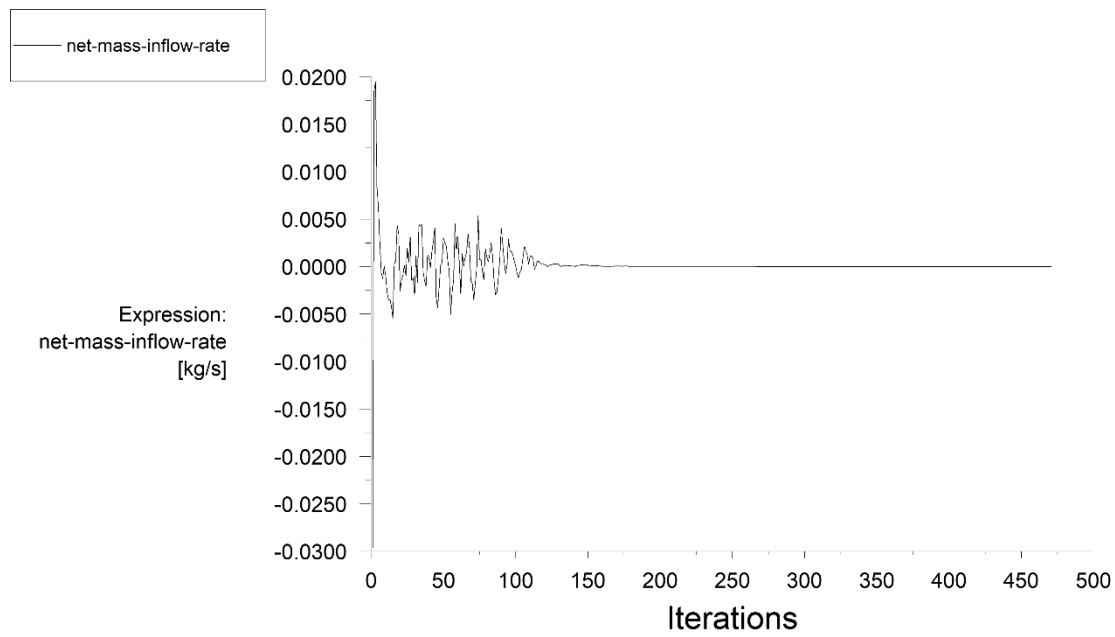


Figure 34: Net mass flow rate vs Iterations (20 AOA)

18° Angle of Attack

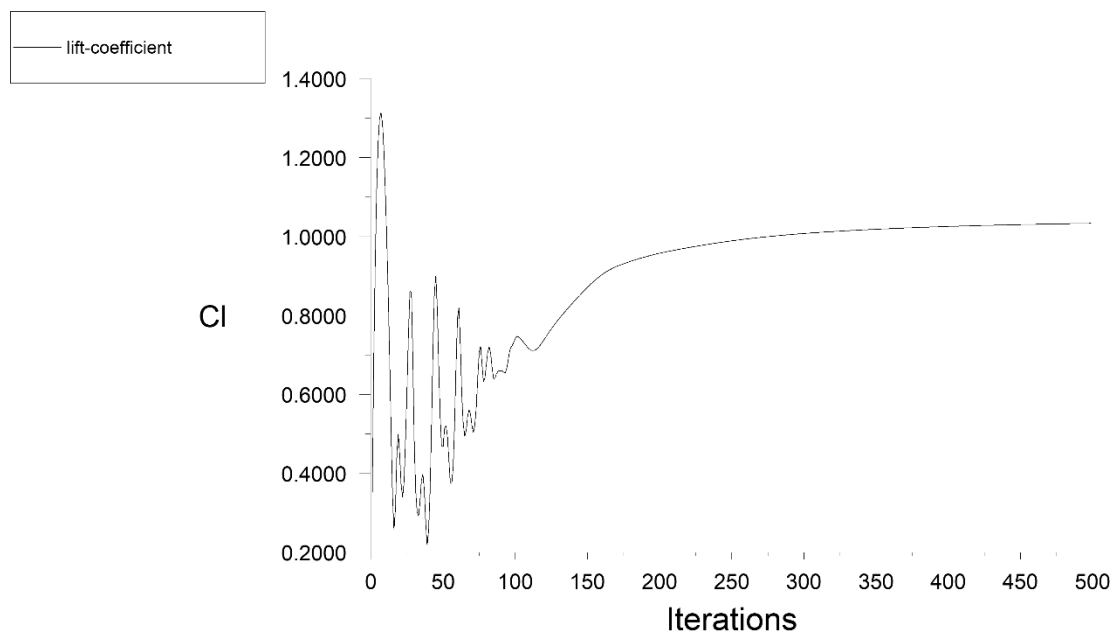


Figure 35: Lift coefficient vs Iterations (18 AOA)

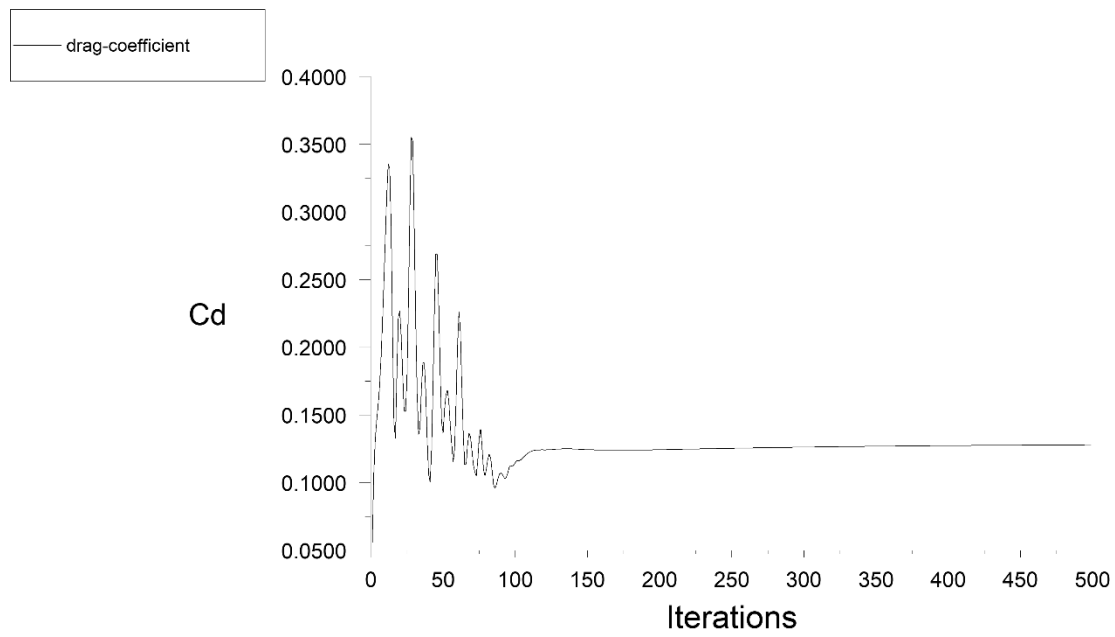


Figure 36: Drag coefficient vs Iterations (18 AOA)

16° Angle of Attack

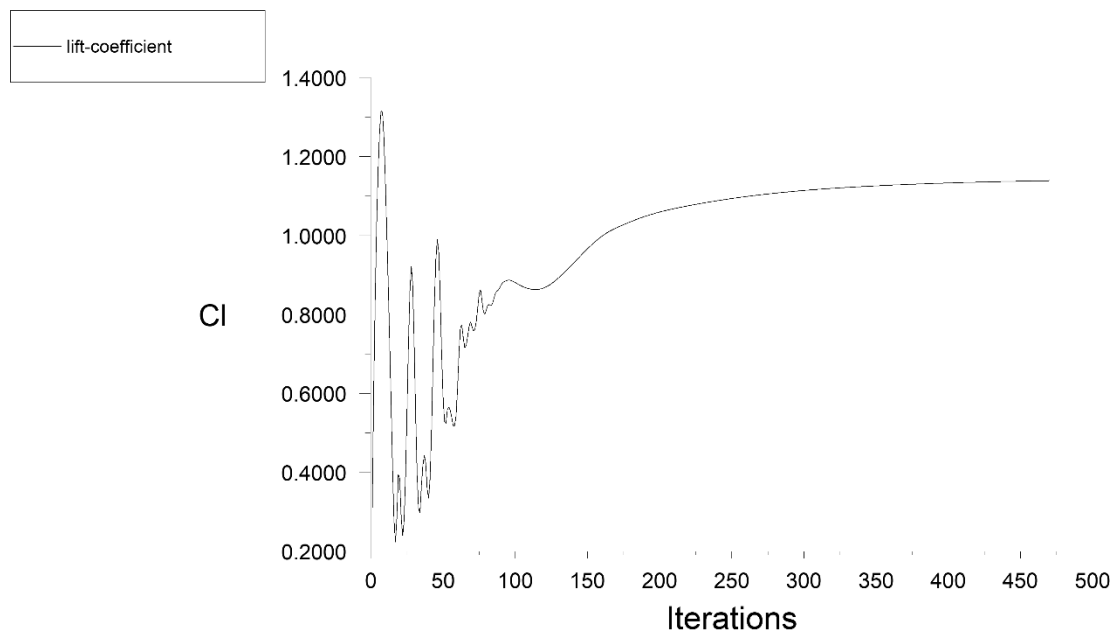


Figure 37: Lift coefficient vs Iterations (16 AOA)

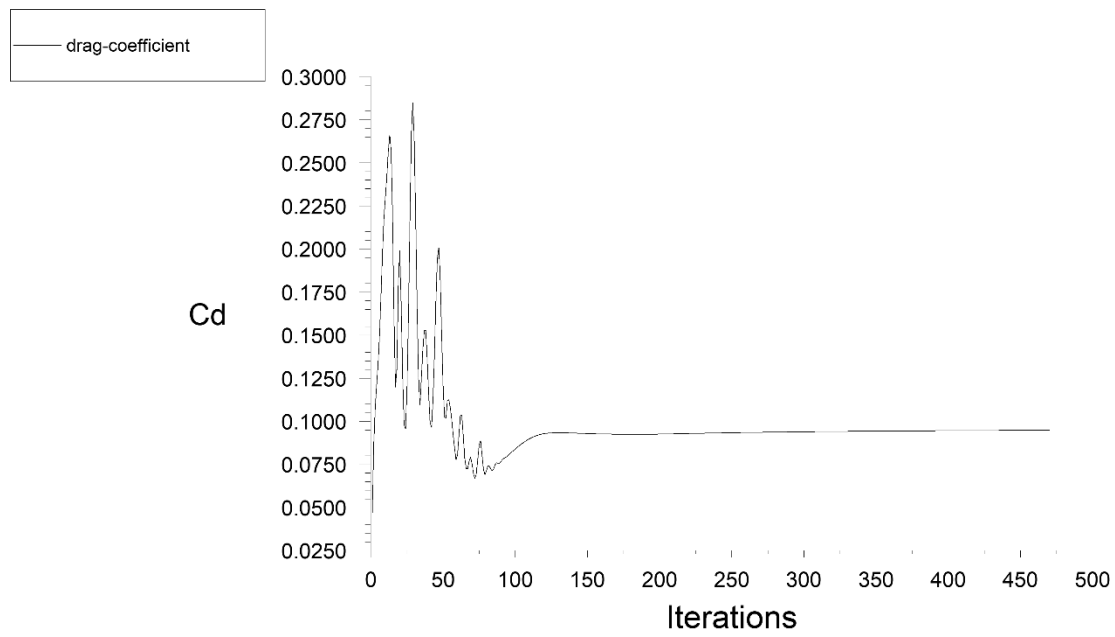


Figure 38: Drag coefficient vs Iterations (16 AOA)

14° Angle of Attack

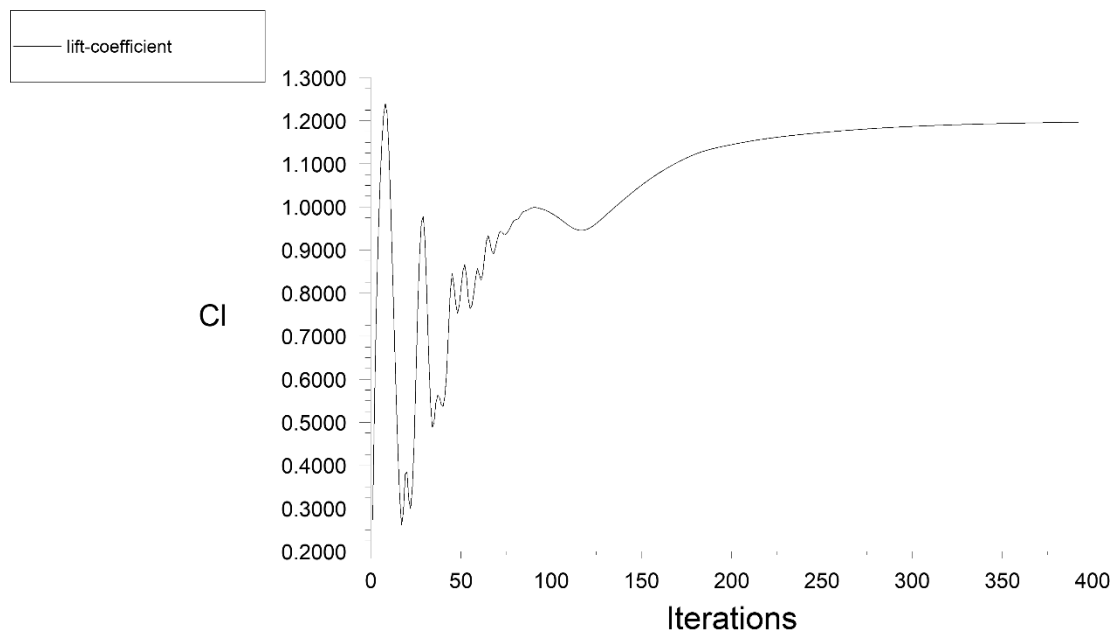


Figure 39: Lift coefficient vs Iterations (14 AOA)

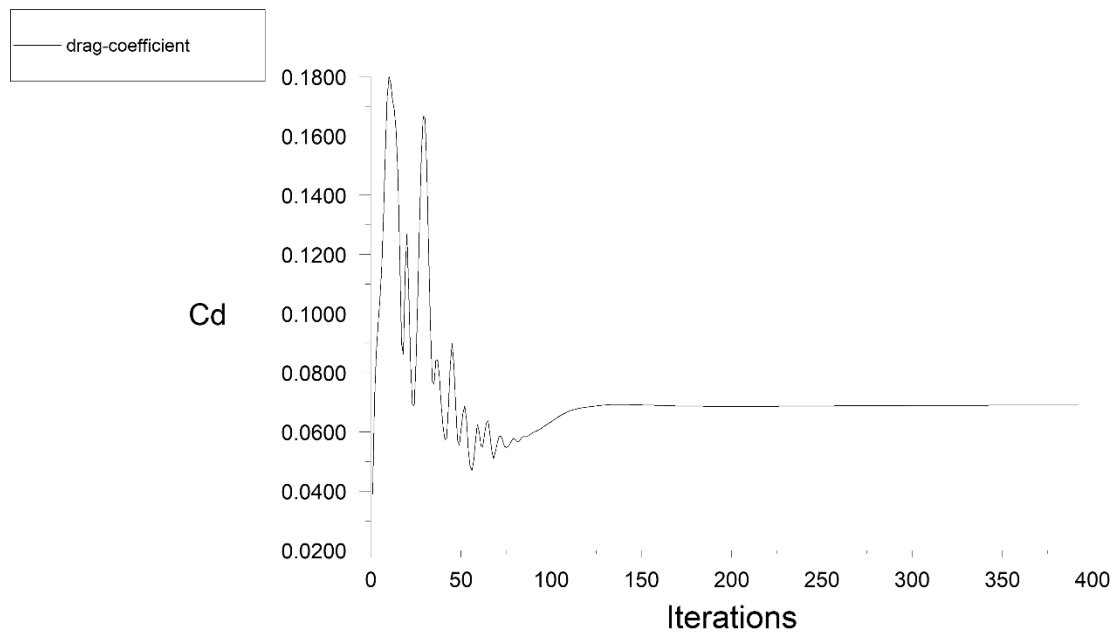


Figure 40: Drag coefficient vs Iterations (14 AOA)

12° Angle of Attack

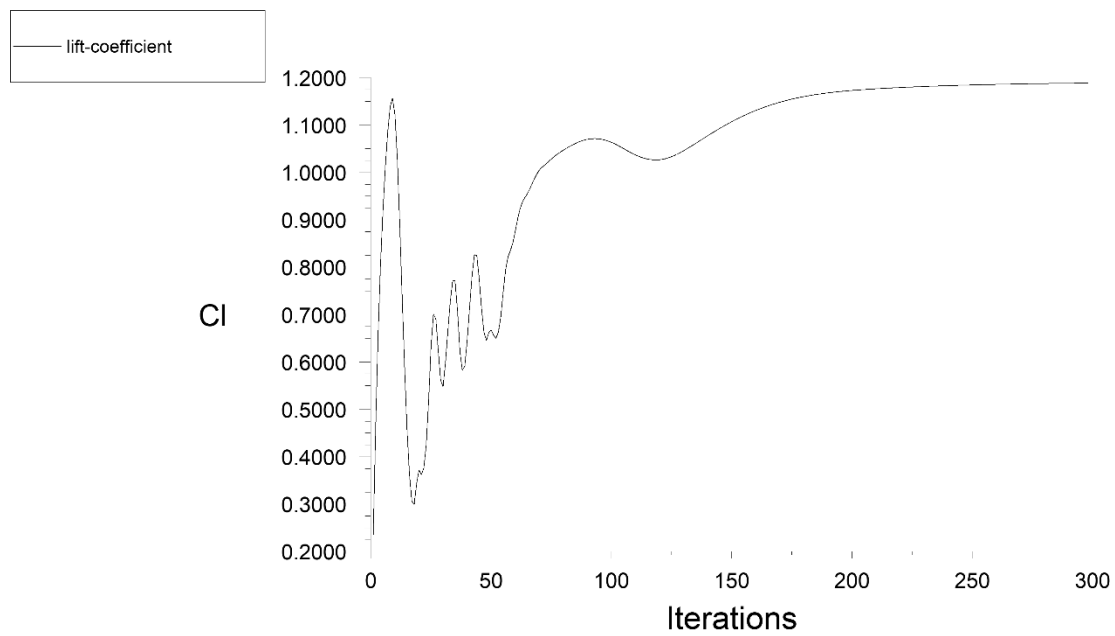


Figure 41: Lift coefficient vs Iterations (12 AOA)

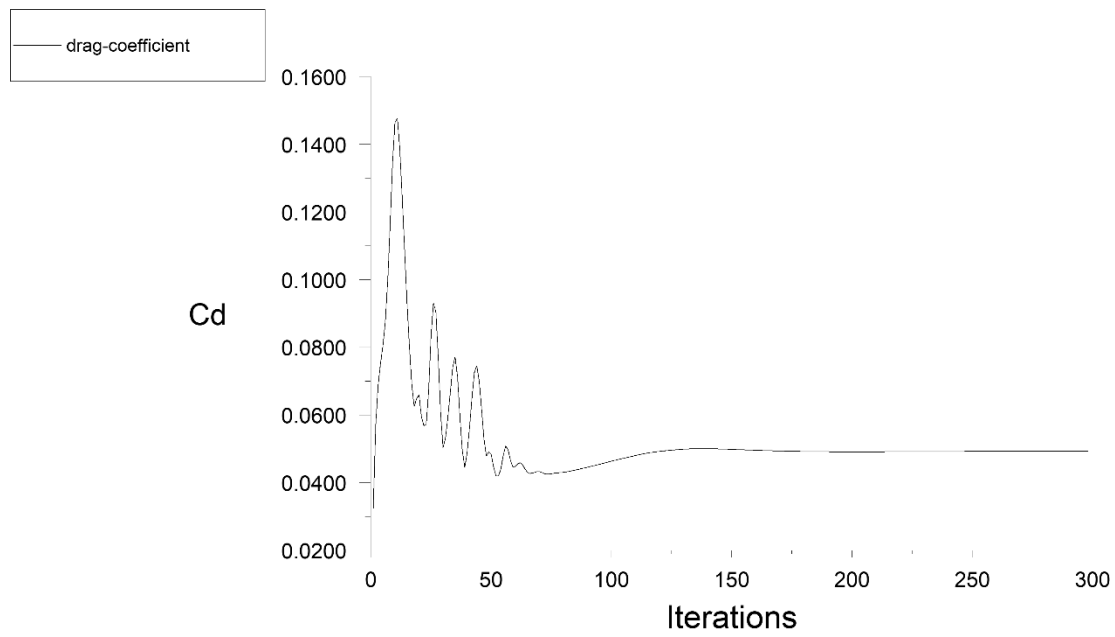


Figure 42: Drag coefficient vs Iterations (12 AOA)

10° Angle of Attack

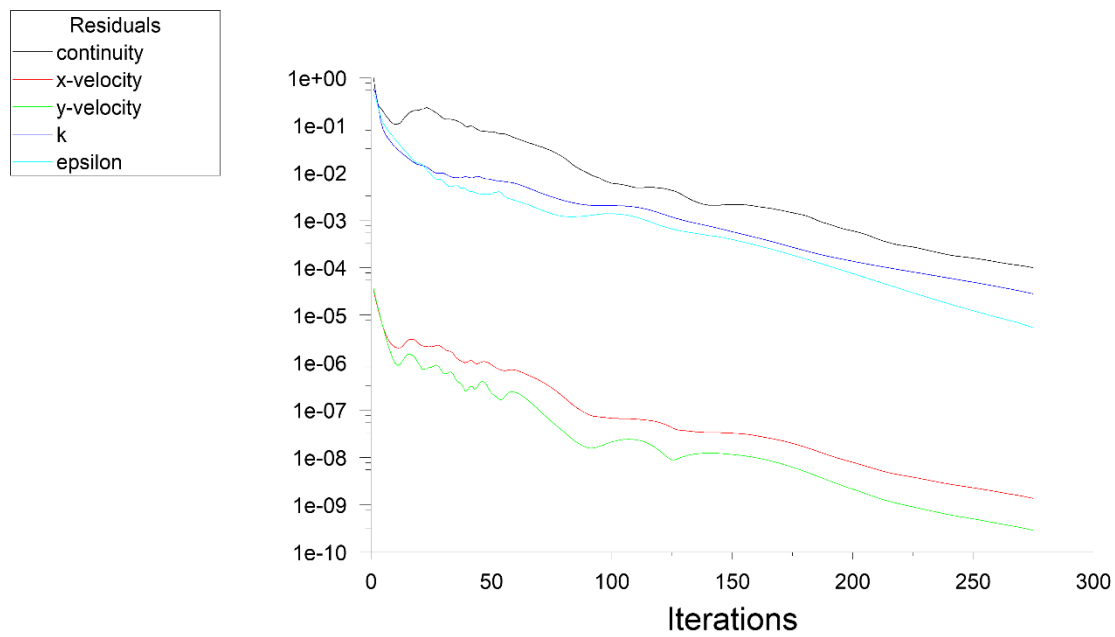


Figure 43: Residuals monitor (10 AOA)

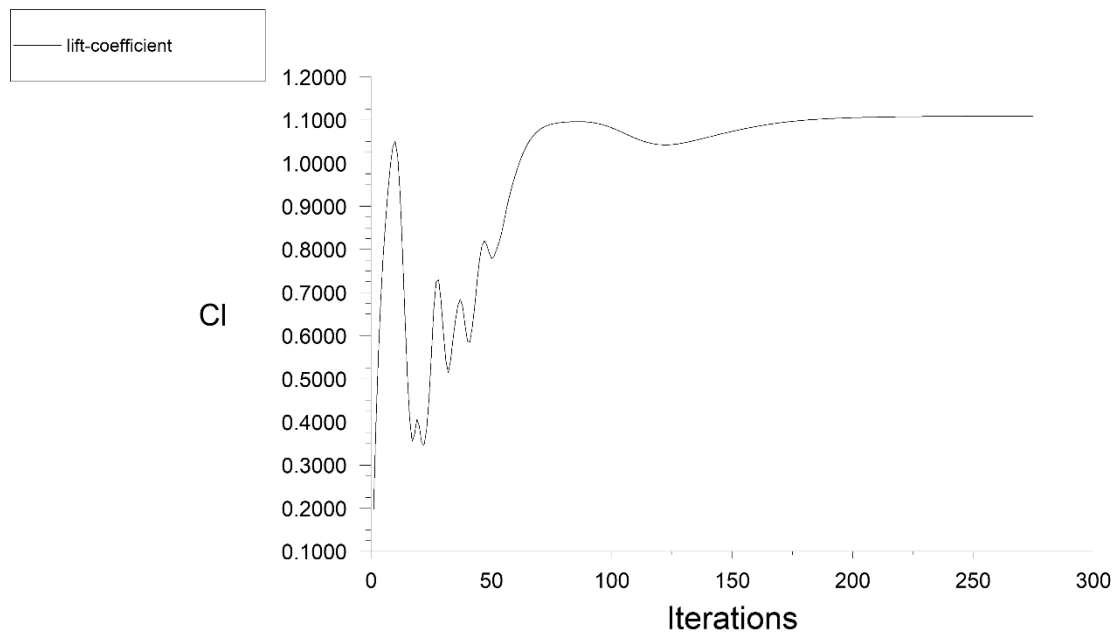


Figure 44: Lift coefficient vs Iterations (10 AOA)

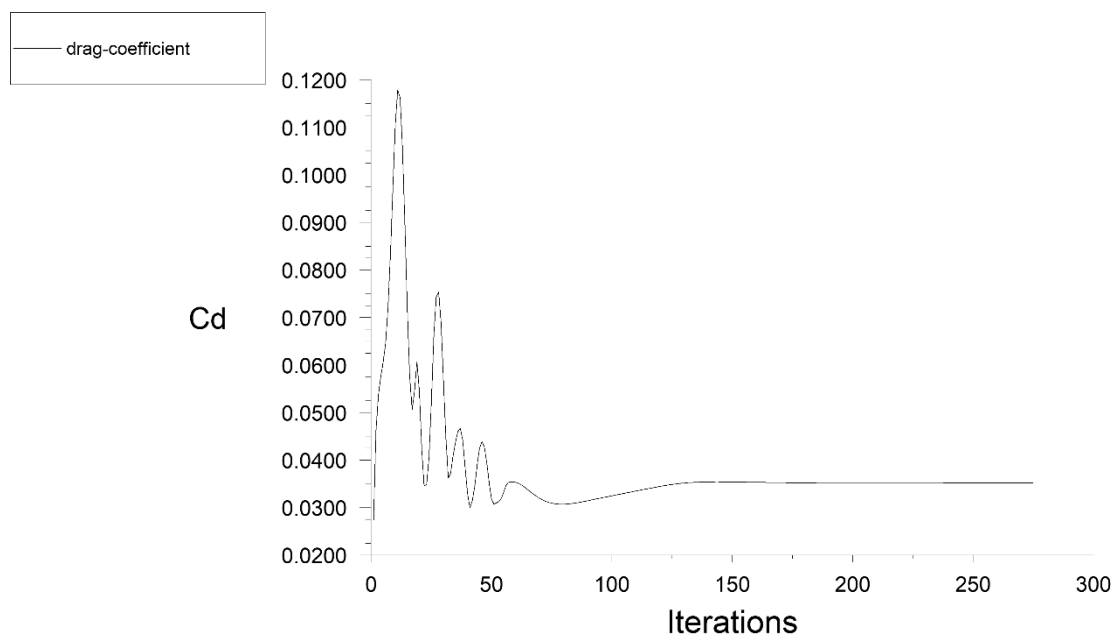


Figure 45: Drag coefficient vs Iterations (10 AOA)

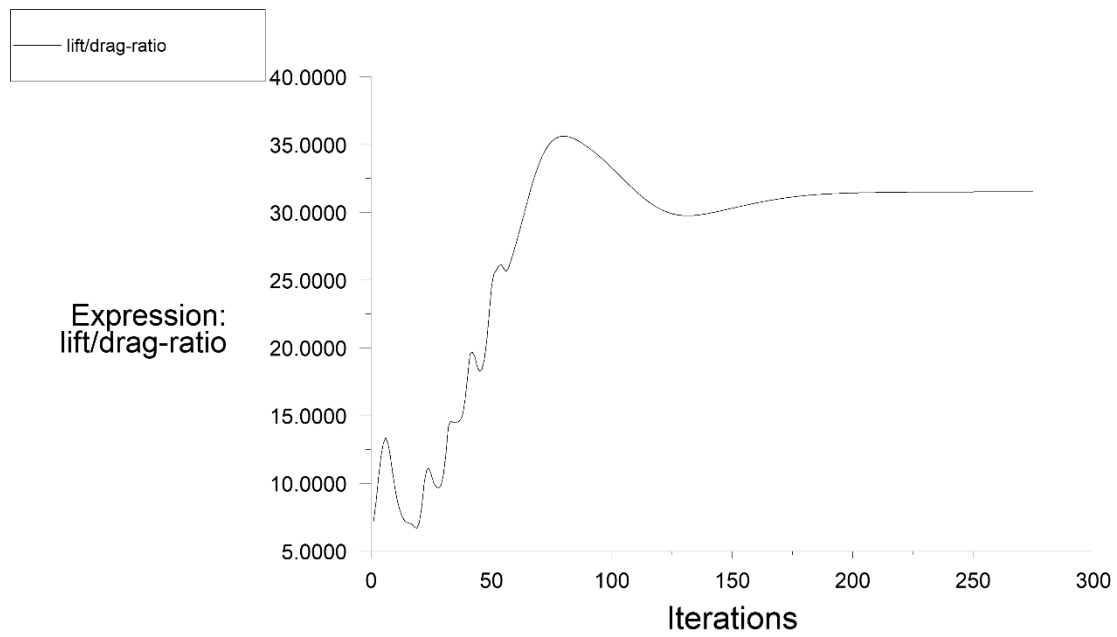


Figure 46: Lift/Drag vs Iterations (10 AOA)

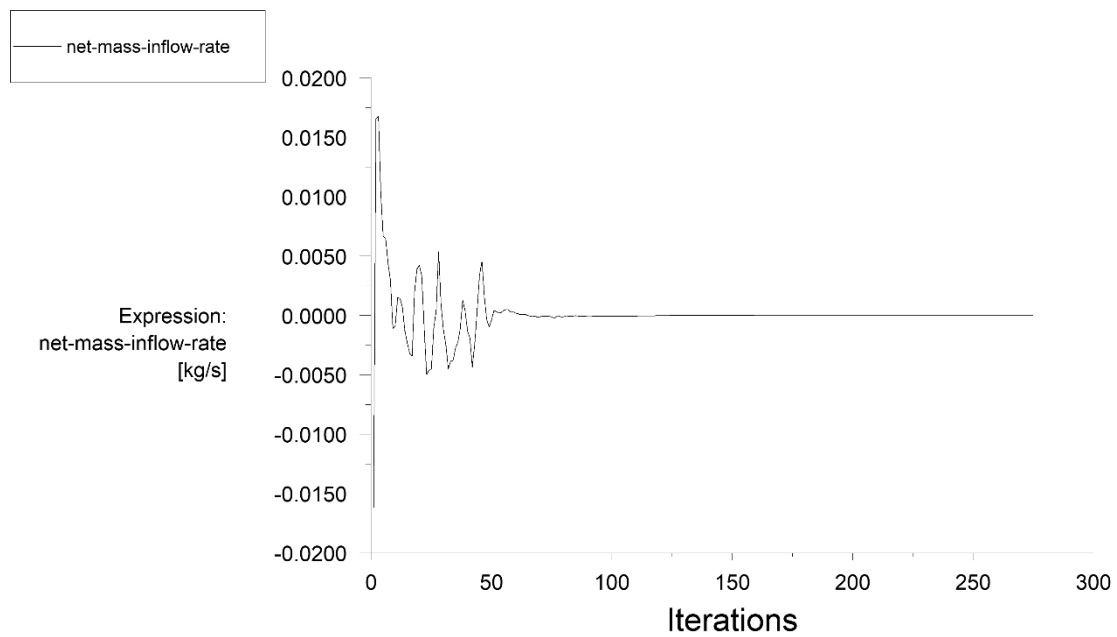


Figure 47: Net mass flow rate vs Iterations (10 AOA)

8° Angle of Attack

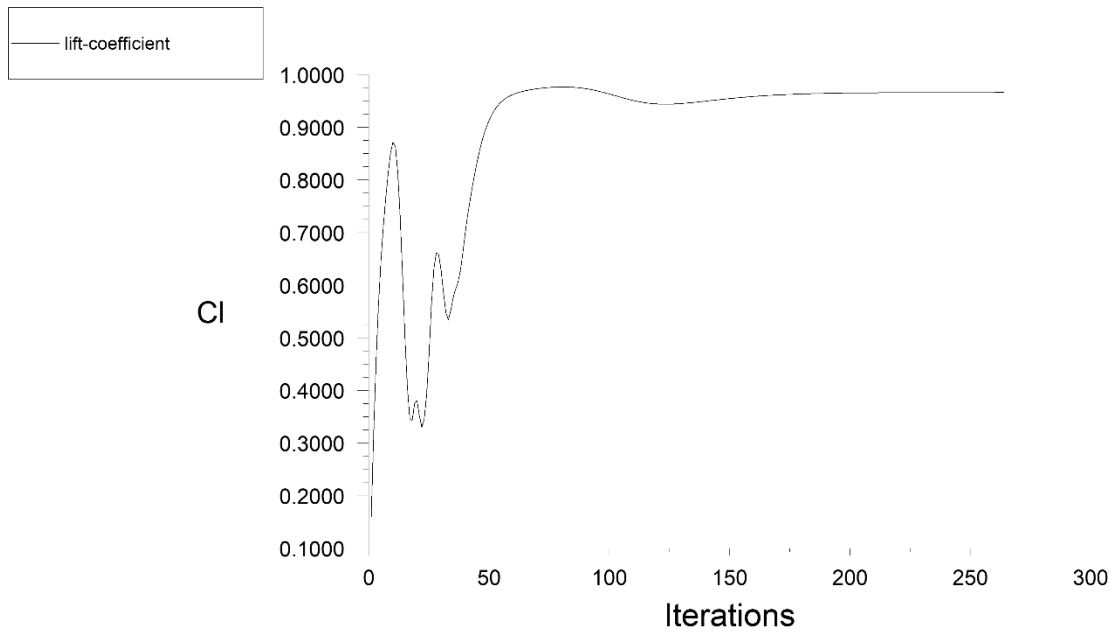


Figure 48: Lift coefficient vs Iterations (8 AOA)

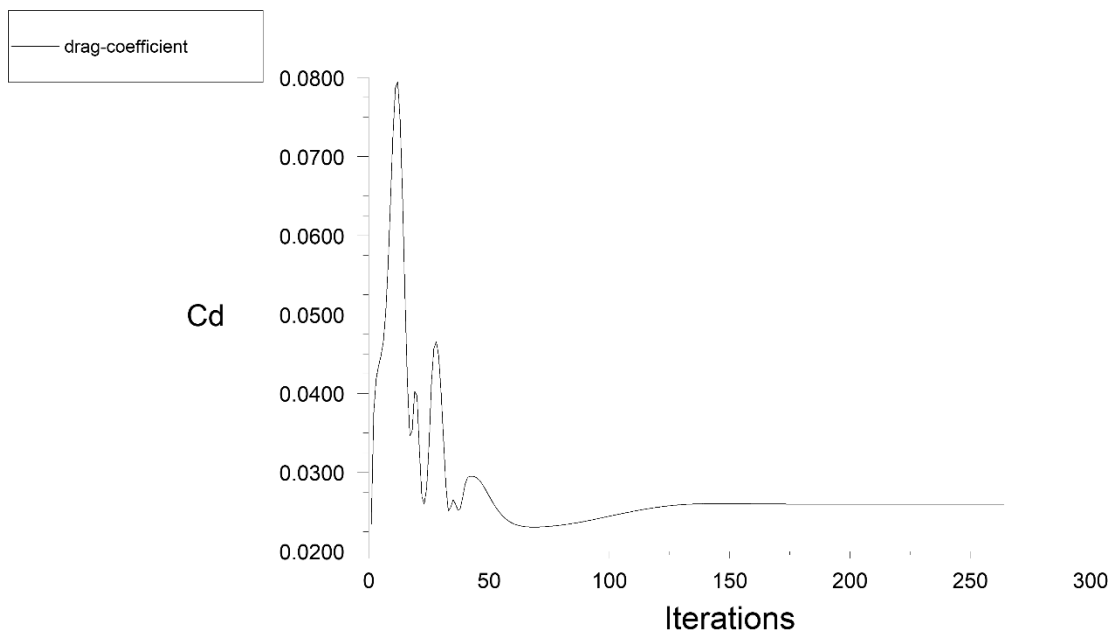


Figure 49: Drag coefficient vs Iterations (8 AOA)

6° Angle of Attack

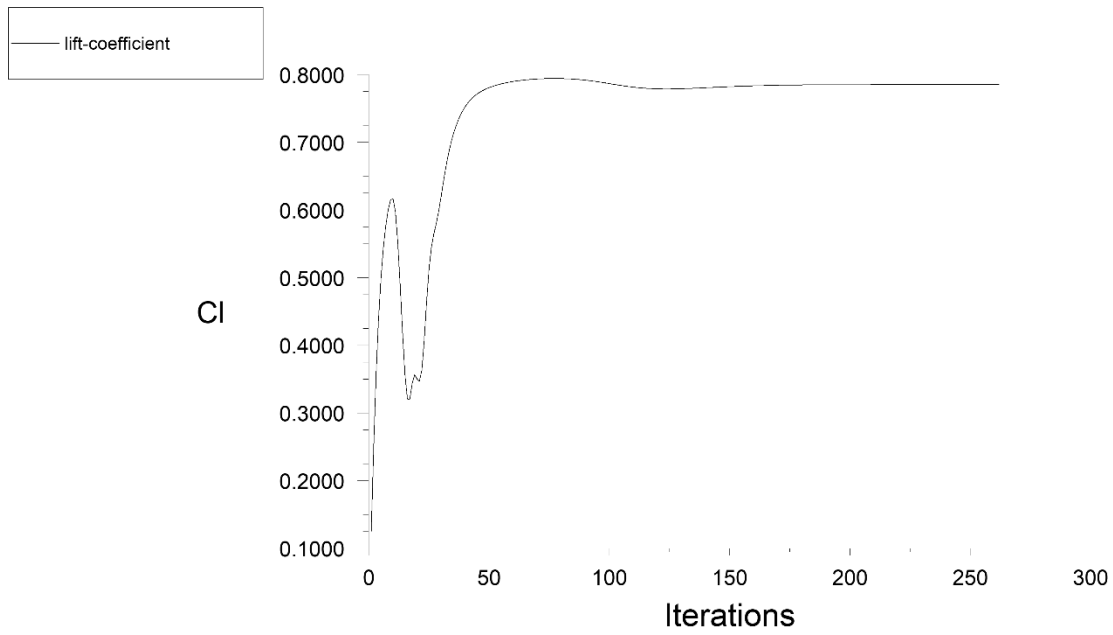


Figure 50: Lift coefficient vs Iterations (6 AOA)

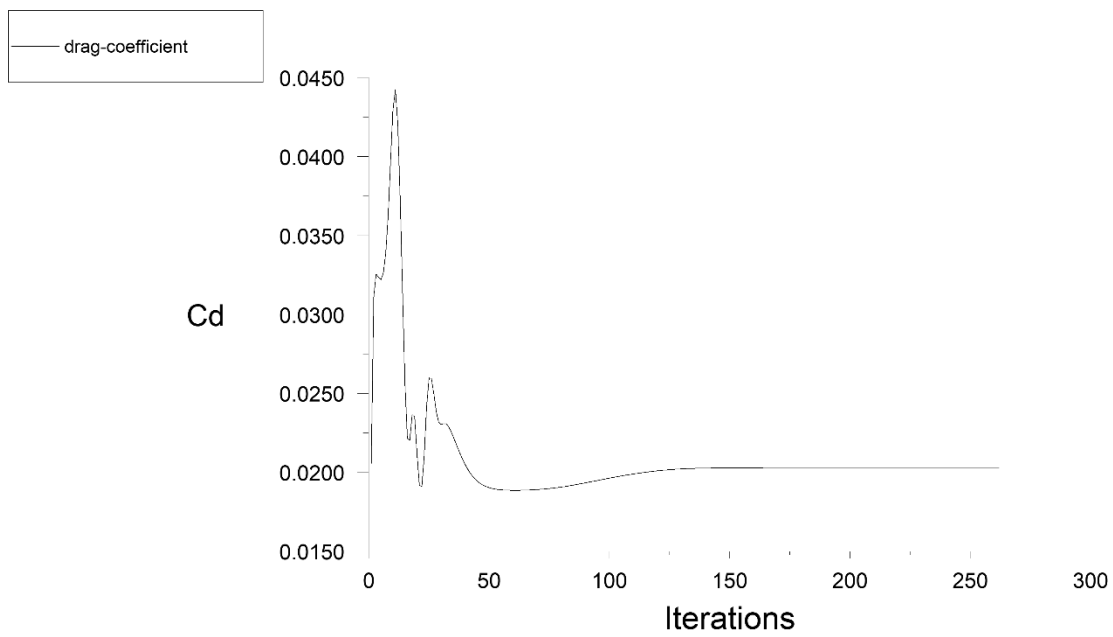


Figure 51: Drag coefficient vs Iterations (6 AOA)

4° Angle of Attack

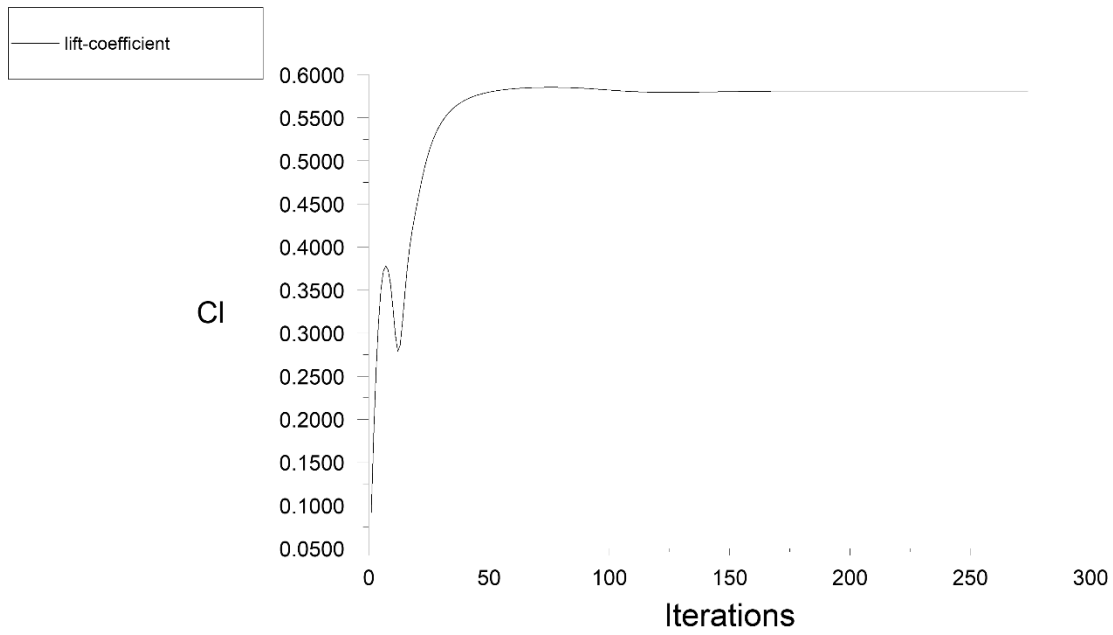


Figure 52: Lift coefficient vs Iterations (4 AOA)

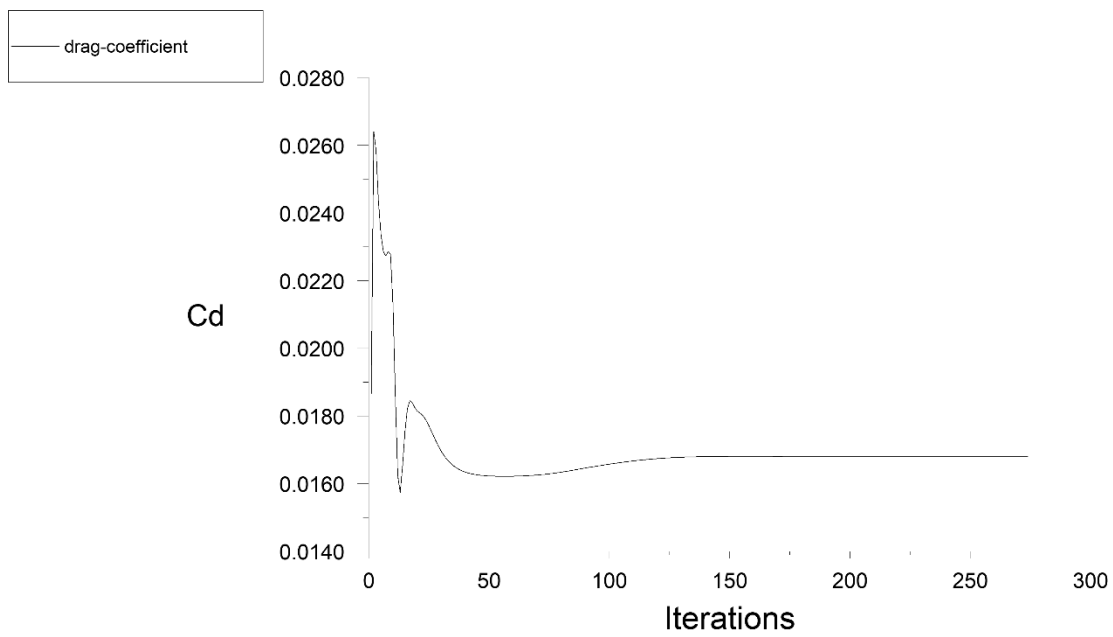


Figure 53: Drag coefficient vs Iterations (4 AOA)

2° Angle of Attack

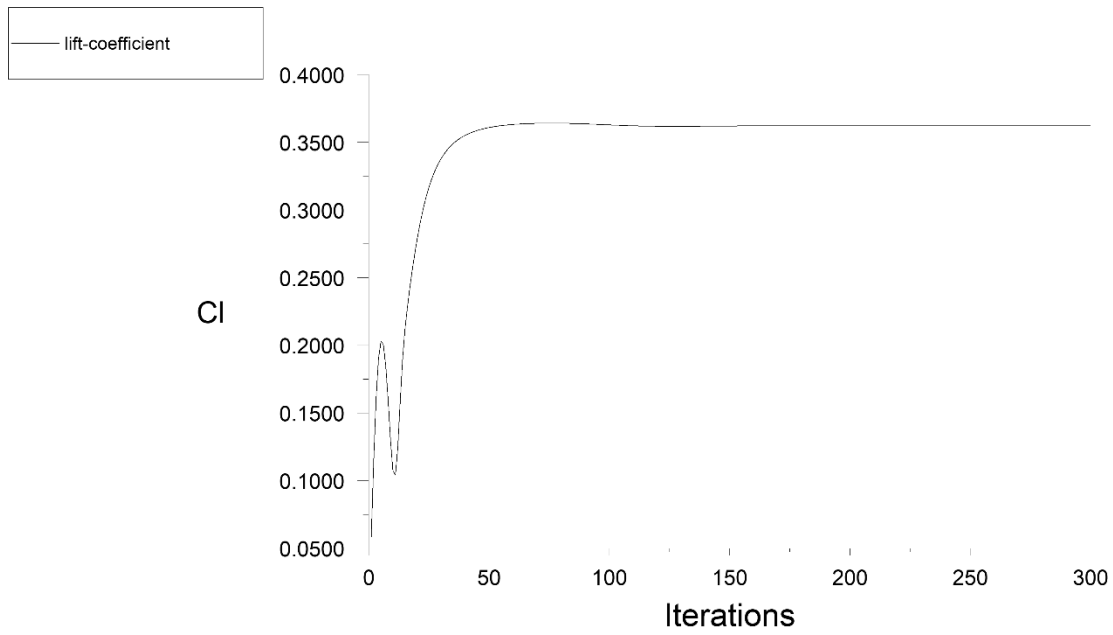


Figure 54: Lift coefficient vs Iterations (2 AOA)

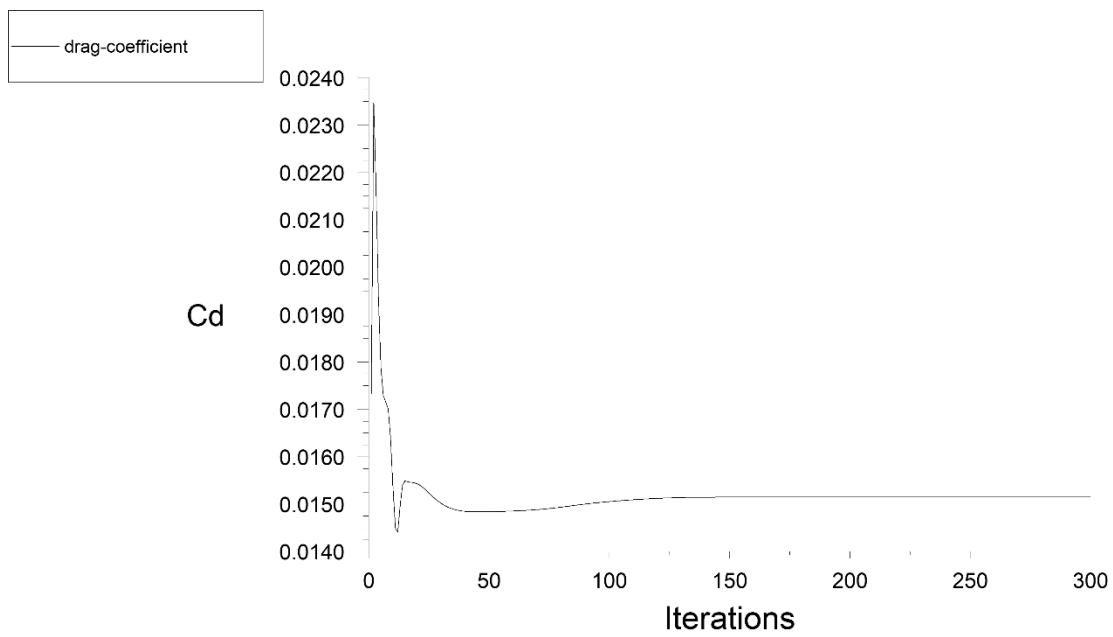


Figure 55: Drag coefficient vs Iterations (2 AOA)

0° Angle of Attack

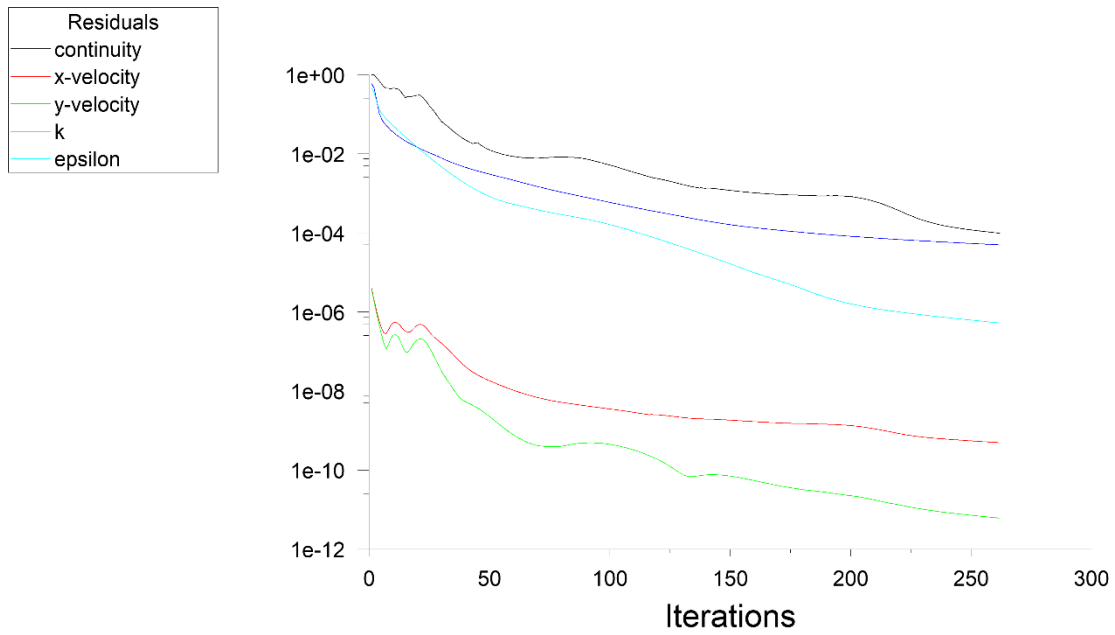


Figure 56: Residuals monitor (0 AOA)

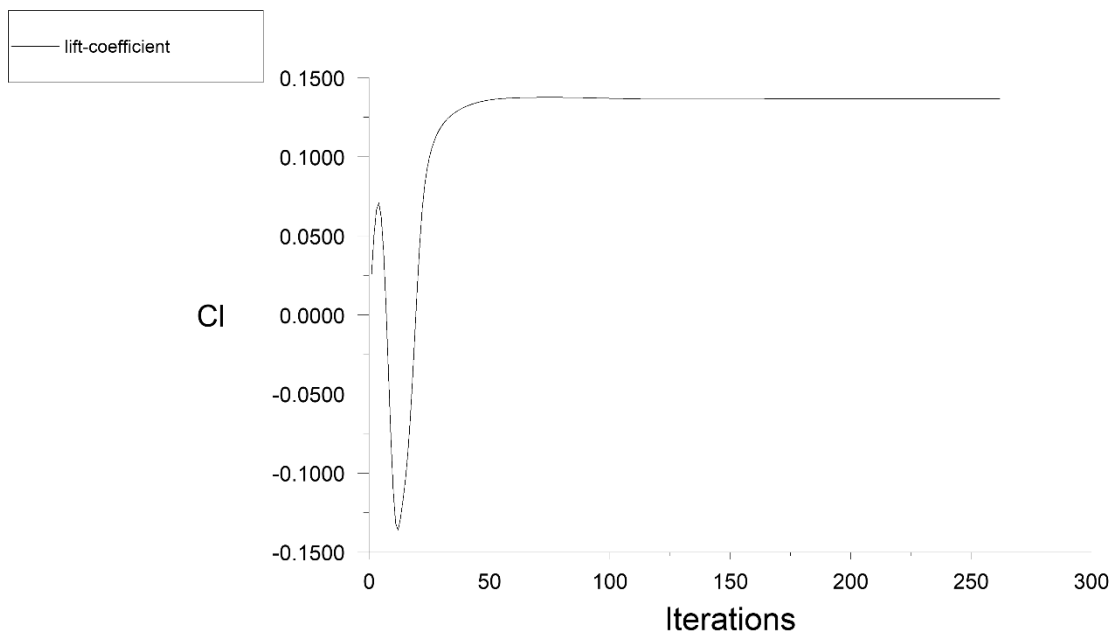


Figure 57: Lift coefficient vs Iterations (0 AOA)

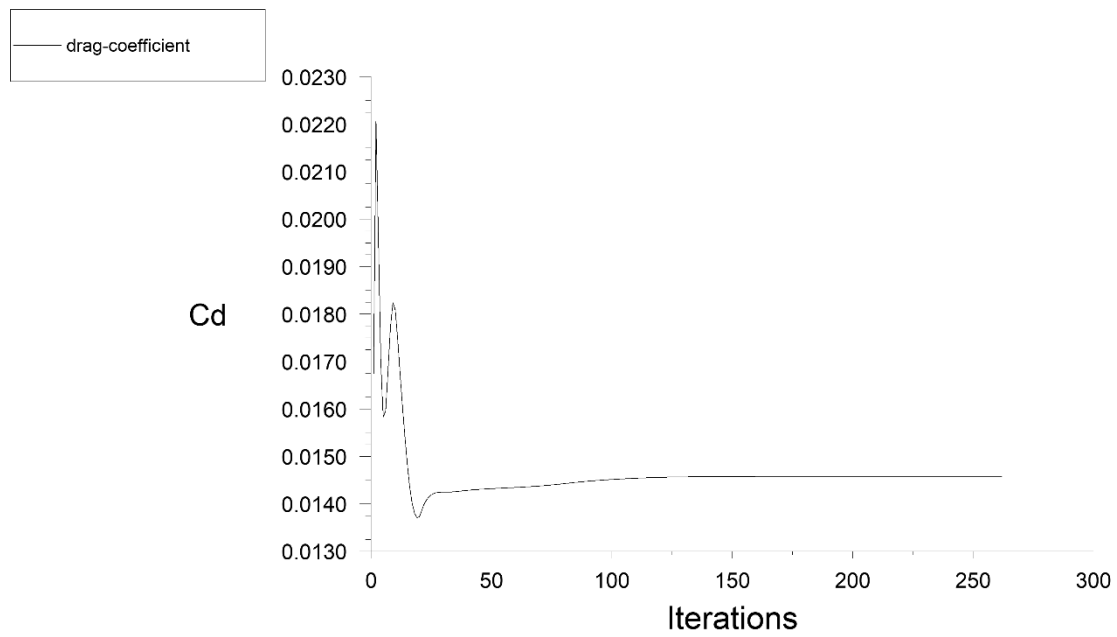


Figure 58: Drag coefficient vs Iterations (0 AOA)

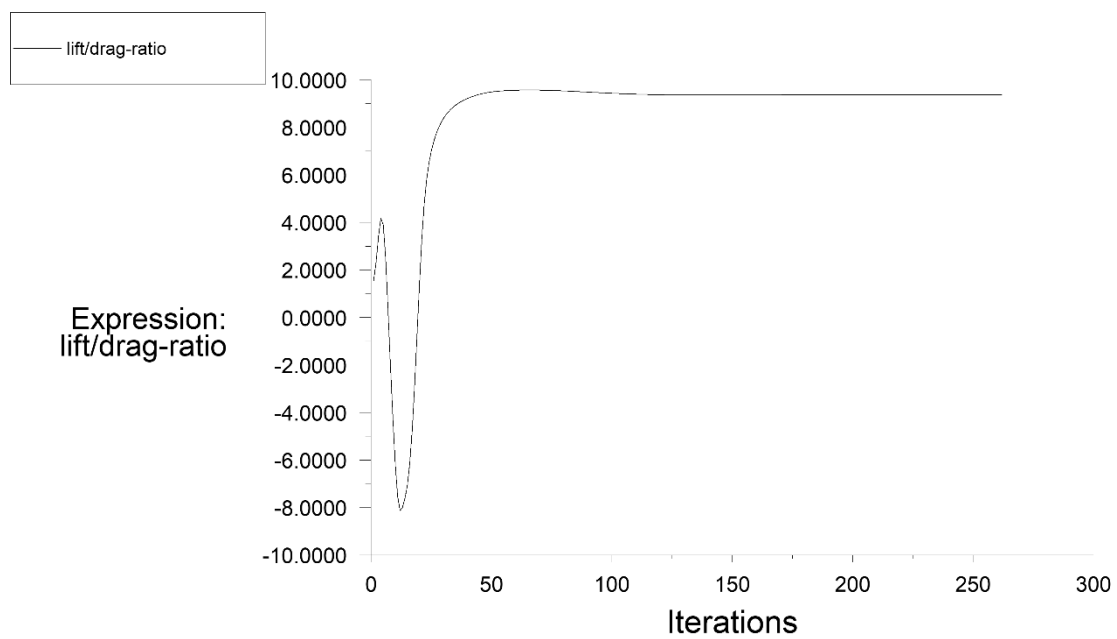


Figure 59: Lift/Drag vs Iterations (0 AOA)

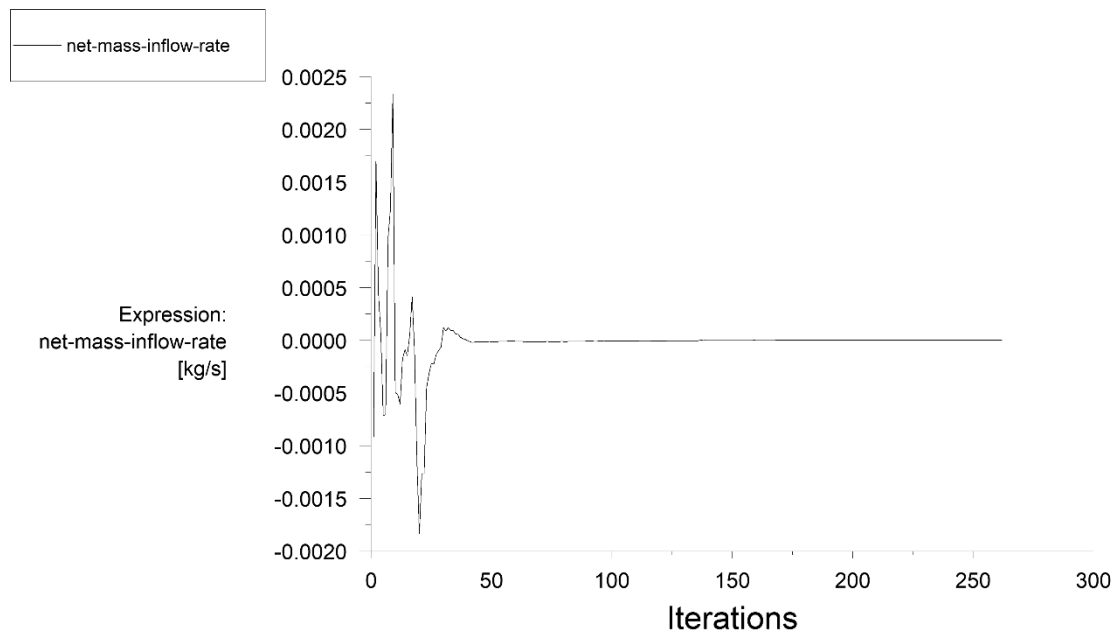


Figure 60: Net mass flow rate vs Iterations (0 AOA)

5.3 Results

The primary objective of this original research is to compare baseline NREL S809 aerofoil with slotted aerofoil that incorporate passive boundary layer control. However, in this CFD study, the focus is limited to simulating the baseline aerofoil for angles of attack ranging from 0° to 20° . Lift coefficient (In Figure 61), drag coefficient (In Figure 62) and lift/drag ratio variation (In Figure 63) with Angle of attack can be plotted as below.

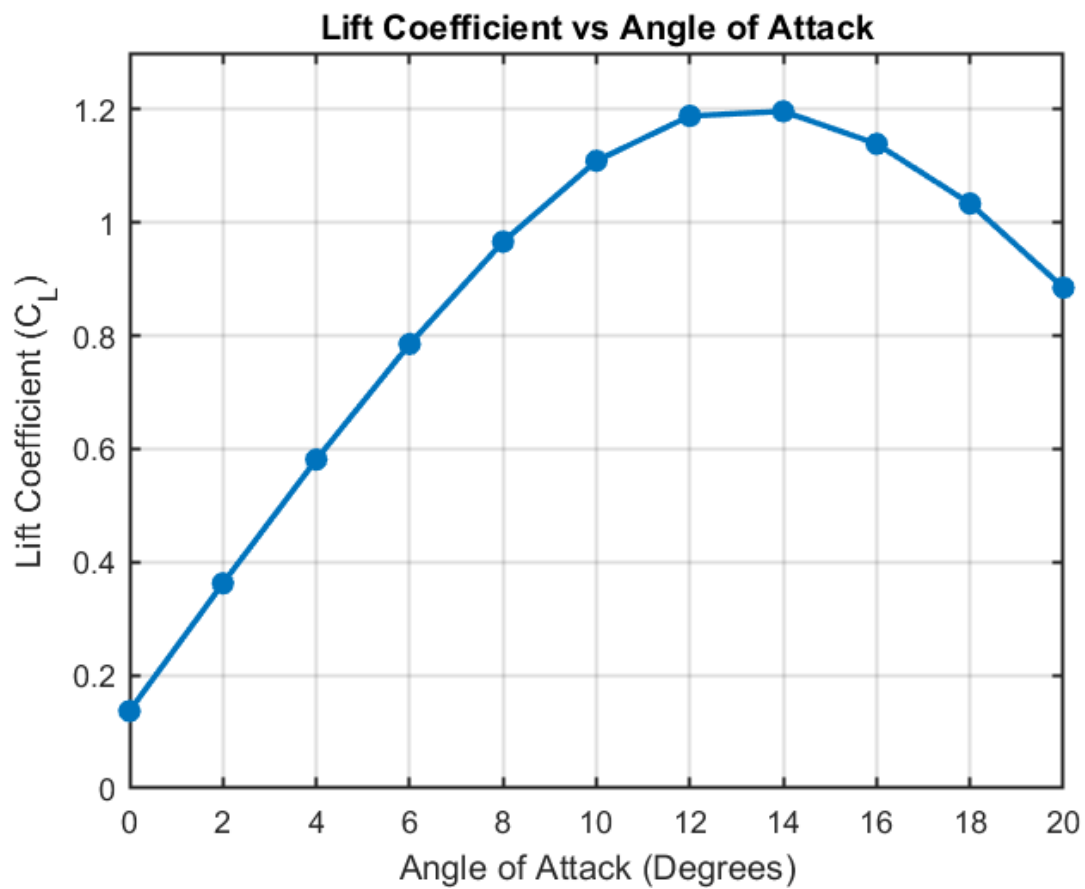


Figure 61: Lift Coefficient vs AOA (Simulated)

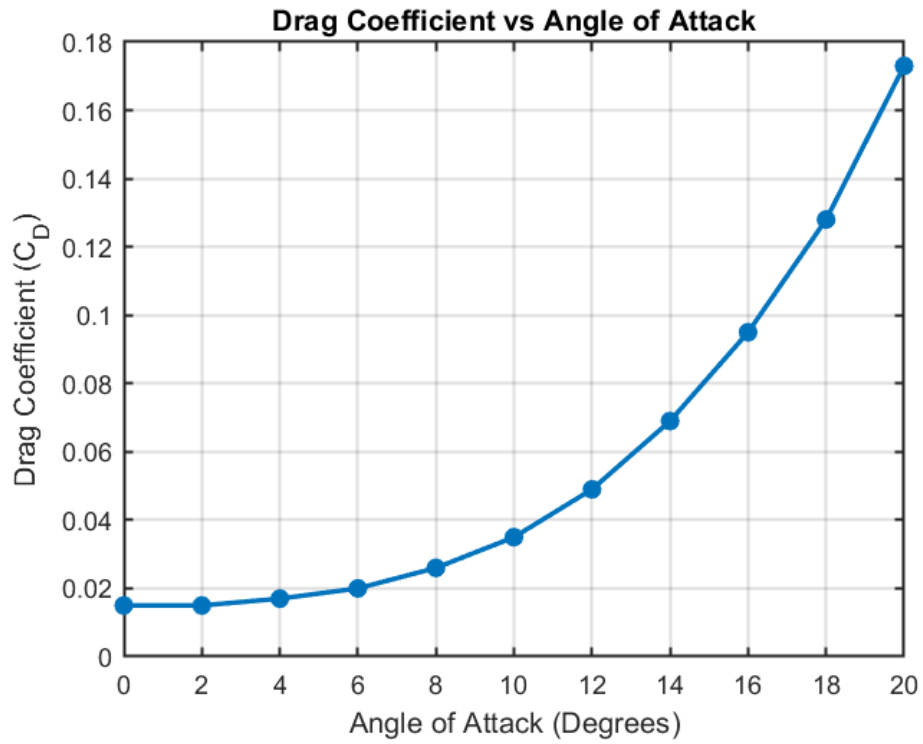


Figure 62: Drag Coefficient vs AOA (Simulated)

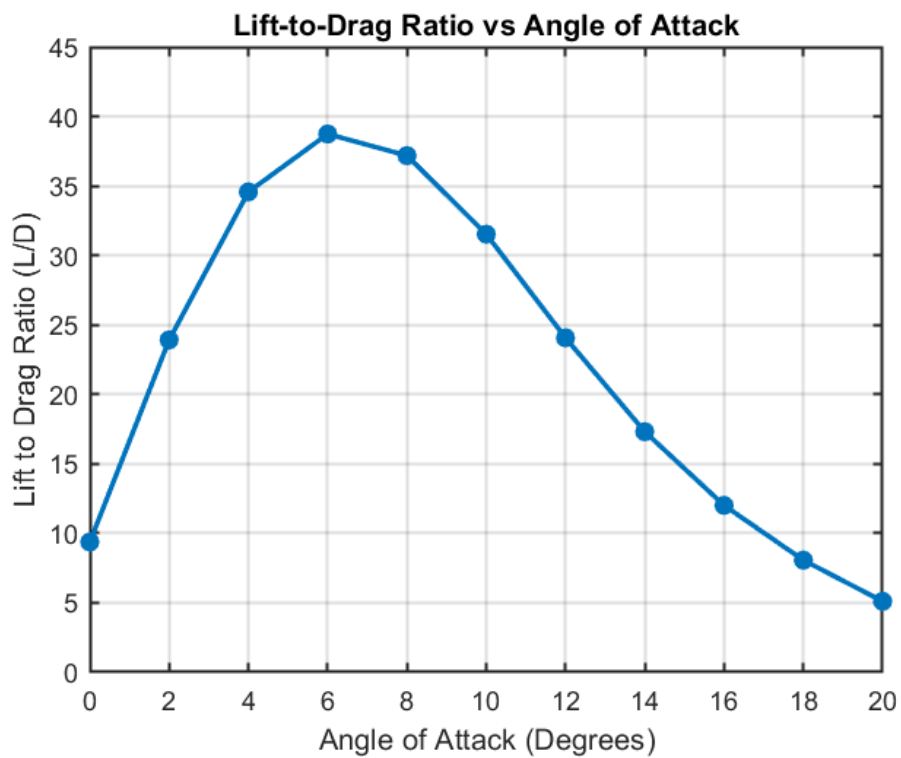


Figure 63: Lift/Drag Ratio vs AOA (Simulated)

The corresponding velocity and static pressure contours for these angles of attack have been extracted as below.

0° Angle of Attack

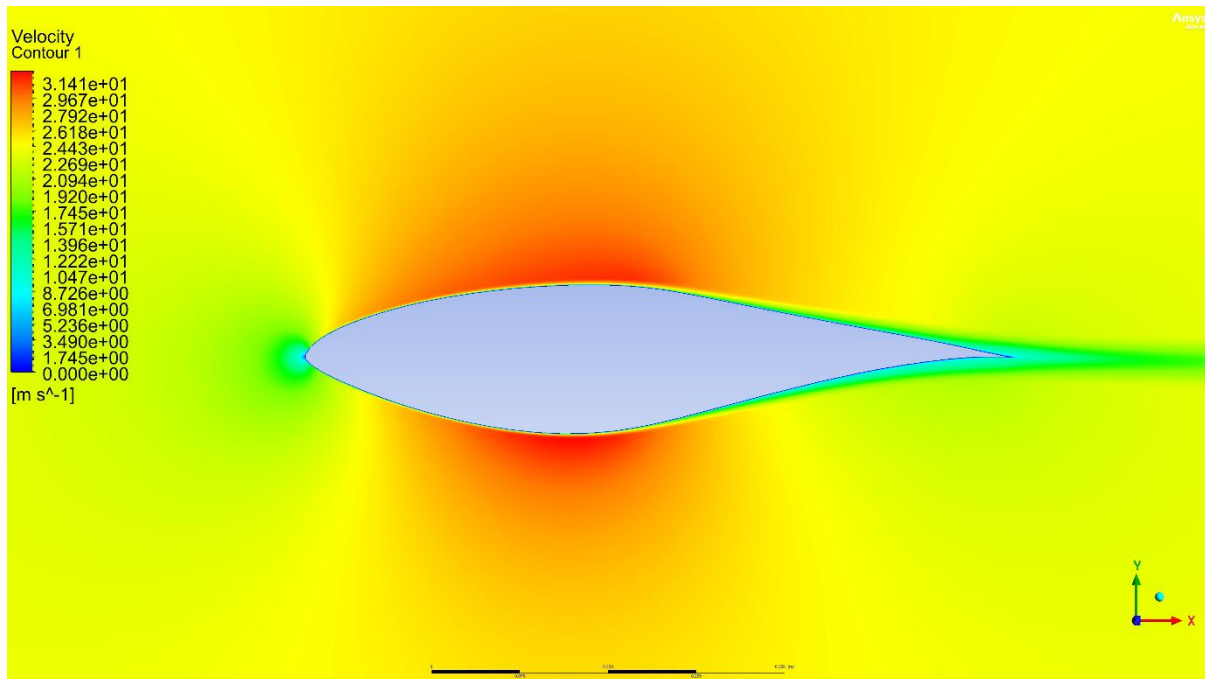


Figure 64: Velocity contour (0 AOA)

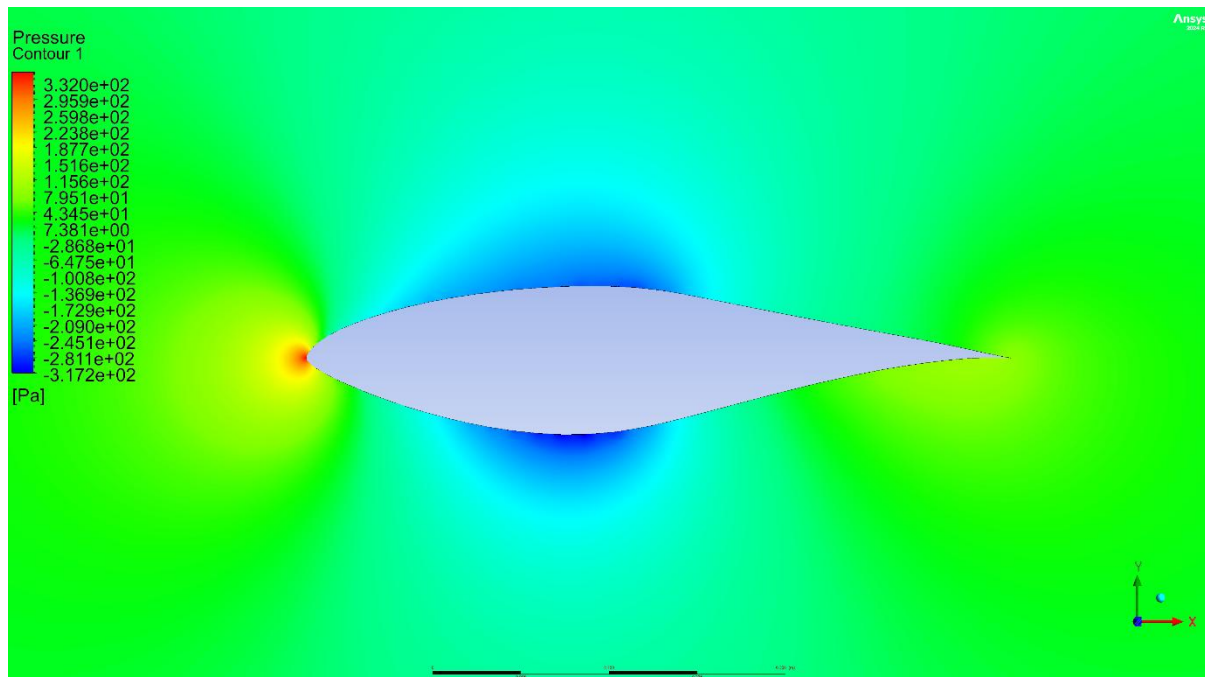


Figure 65: Pressure contour (0 AOA)

2° Angle of Attack

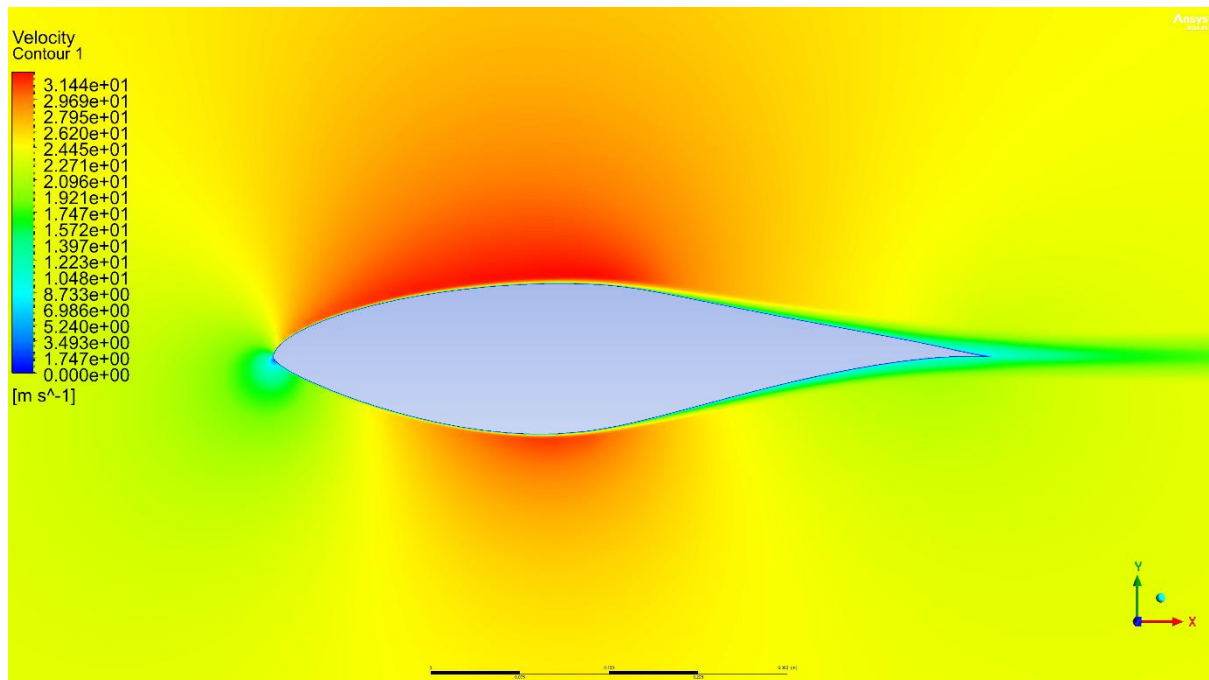


Figure 66: Velocity contour (2 AOA)

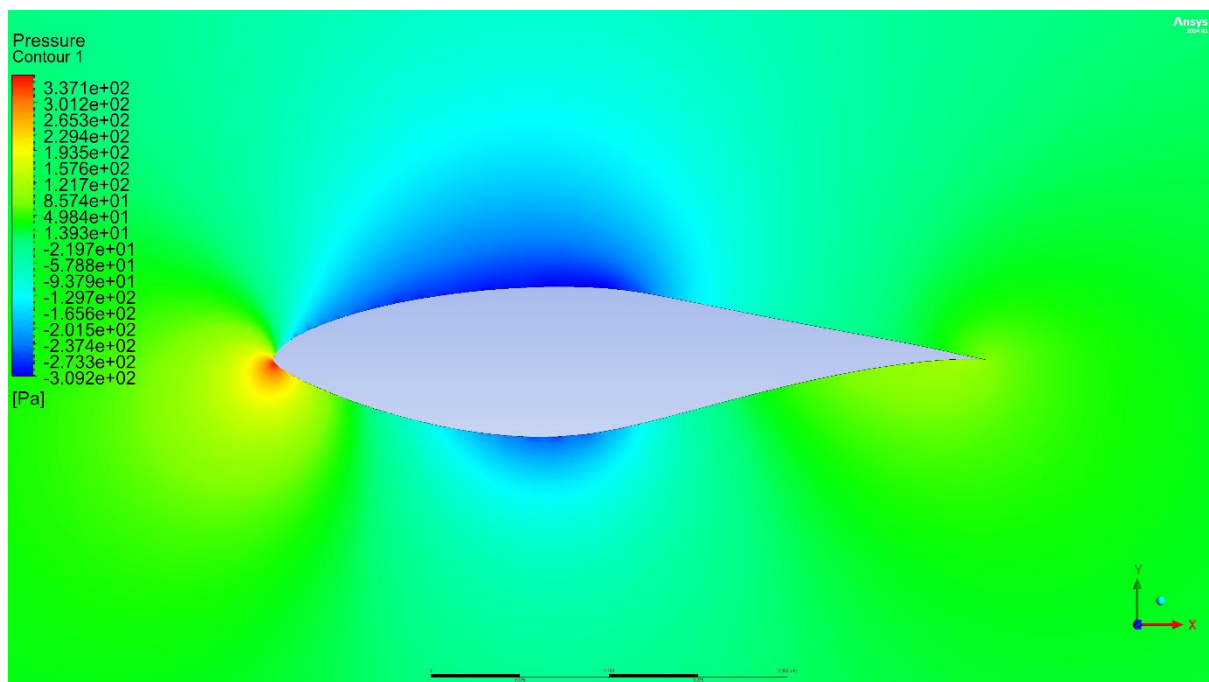


Figure 67: Pressure contour (2 AOA)

4° Angle of Attack

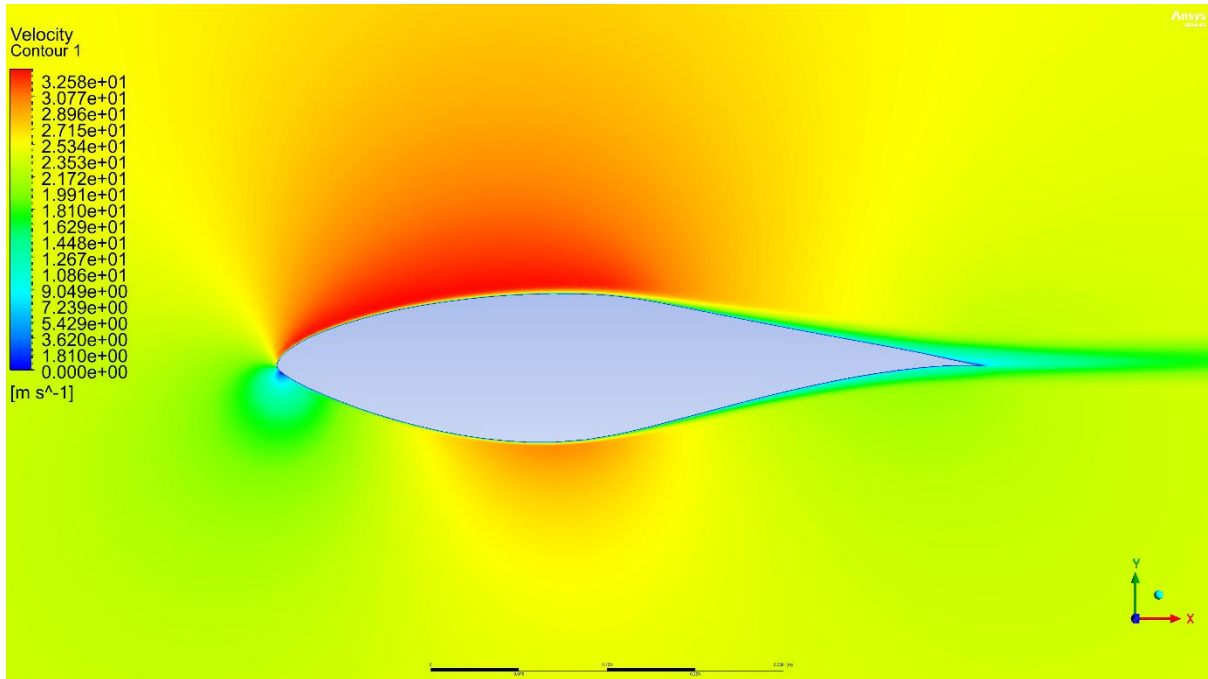


Figure 68: Velocity contour (4 AOA)

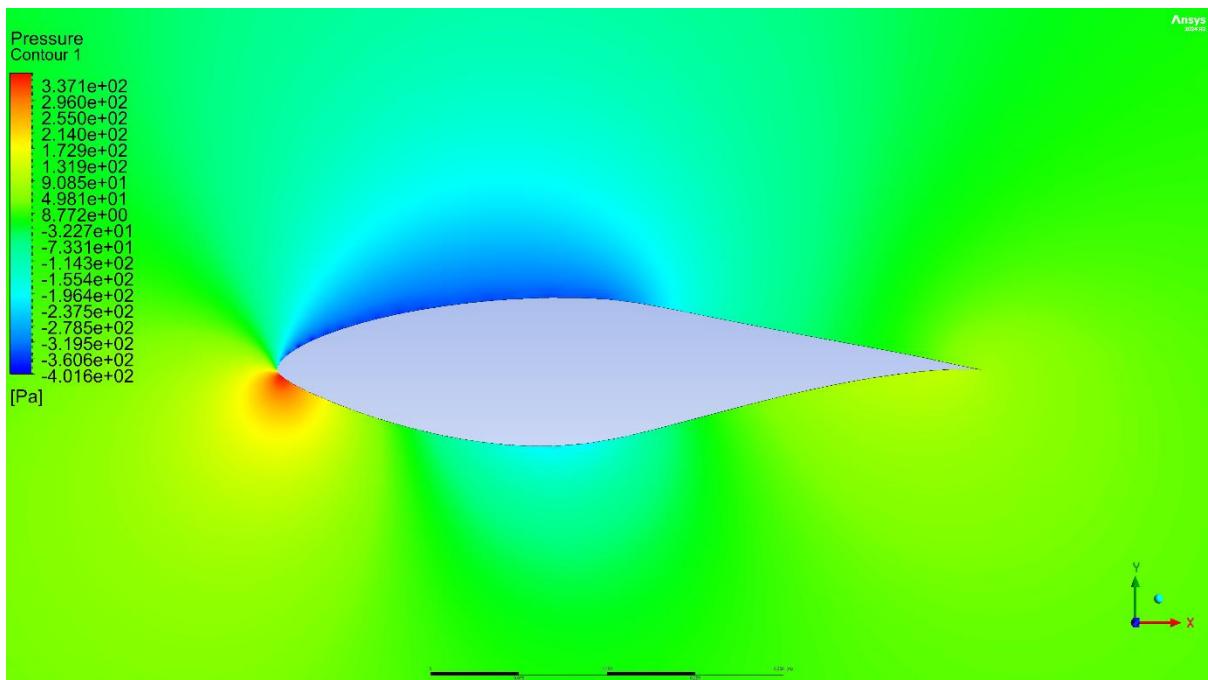


Figure 69: Pressure contour (4 AOA)

6° Angle of Attack

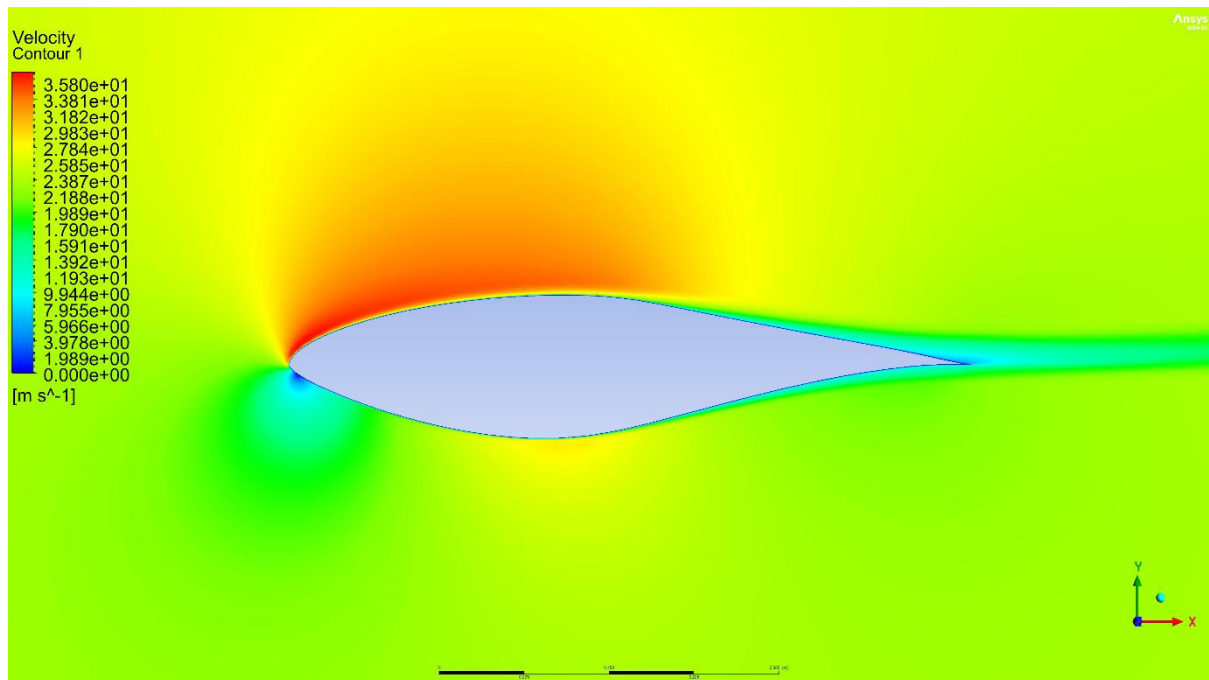


Figure 70: Velocity contour (6 AOA)

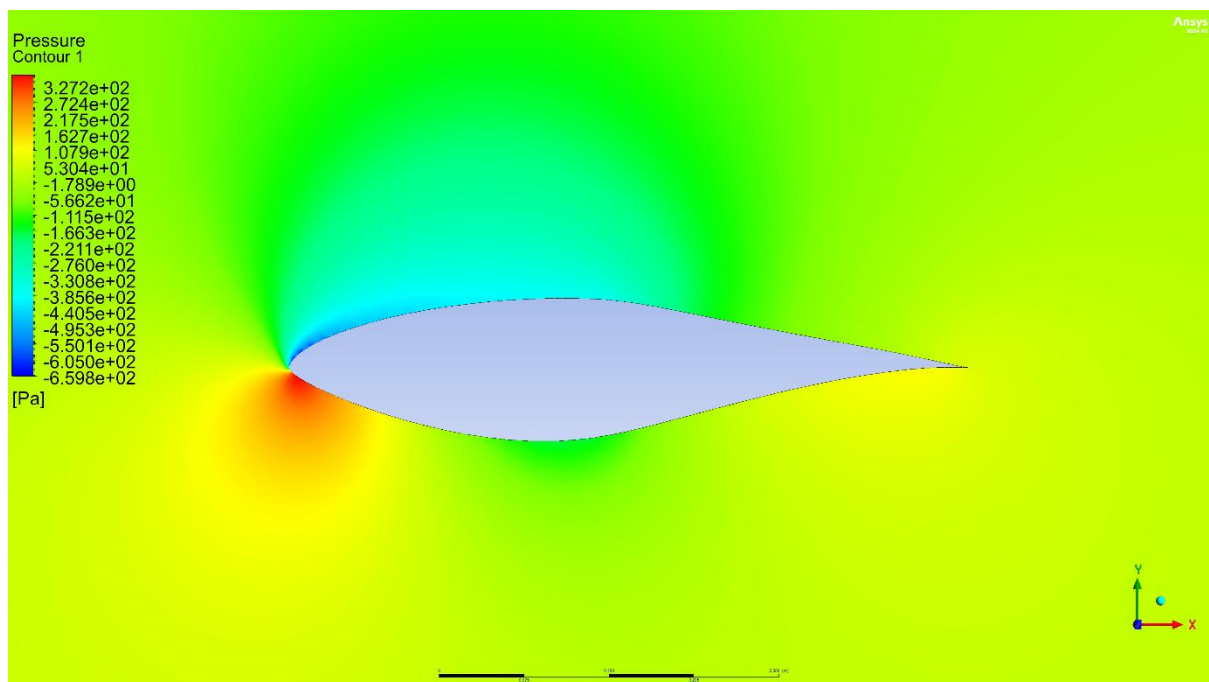


Figure 71: Pressure contour (6 AOA)

8° Angle of Attack

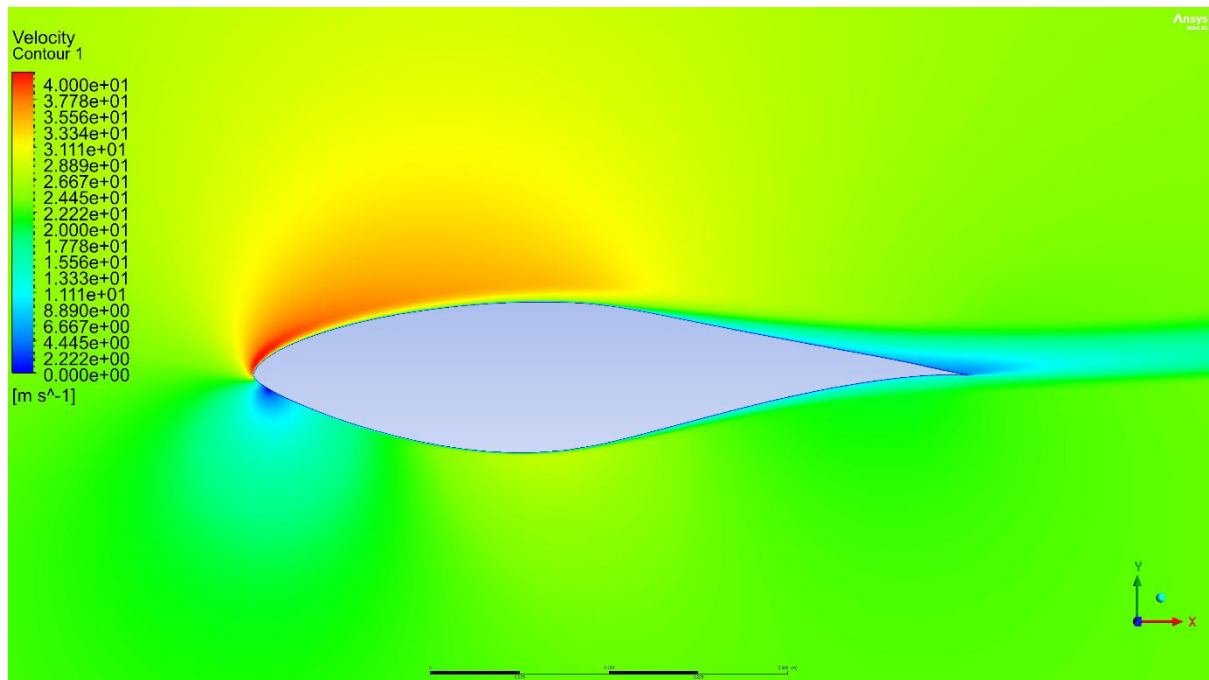


Figure 72: Velocity contour (8 AOA)

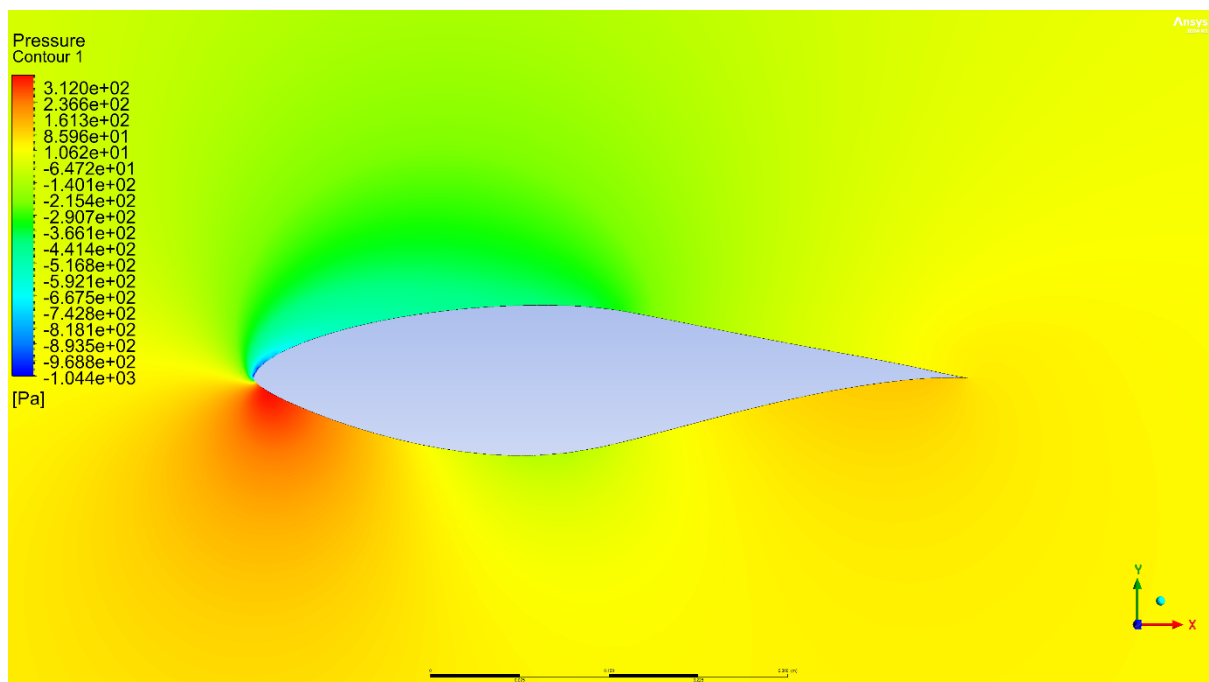


Figure 73: Pressure contour (8 AOA)

10° Angle of Attack

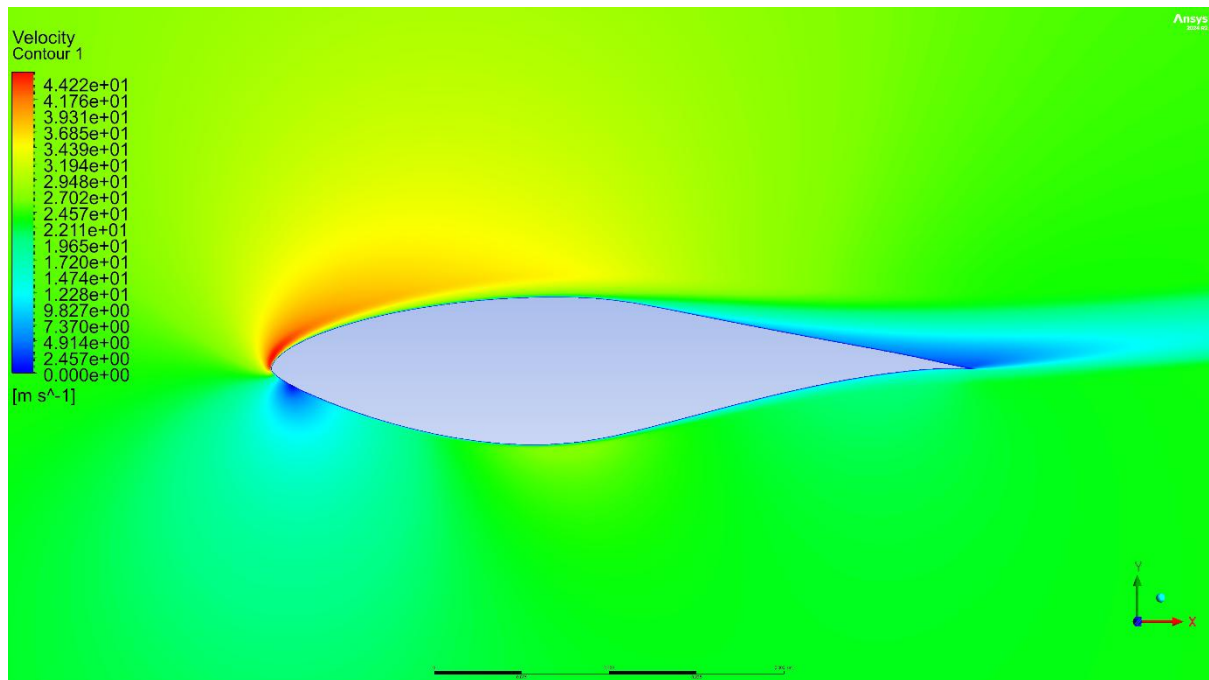


Figure 74: Velocity contour (10 AOA)

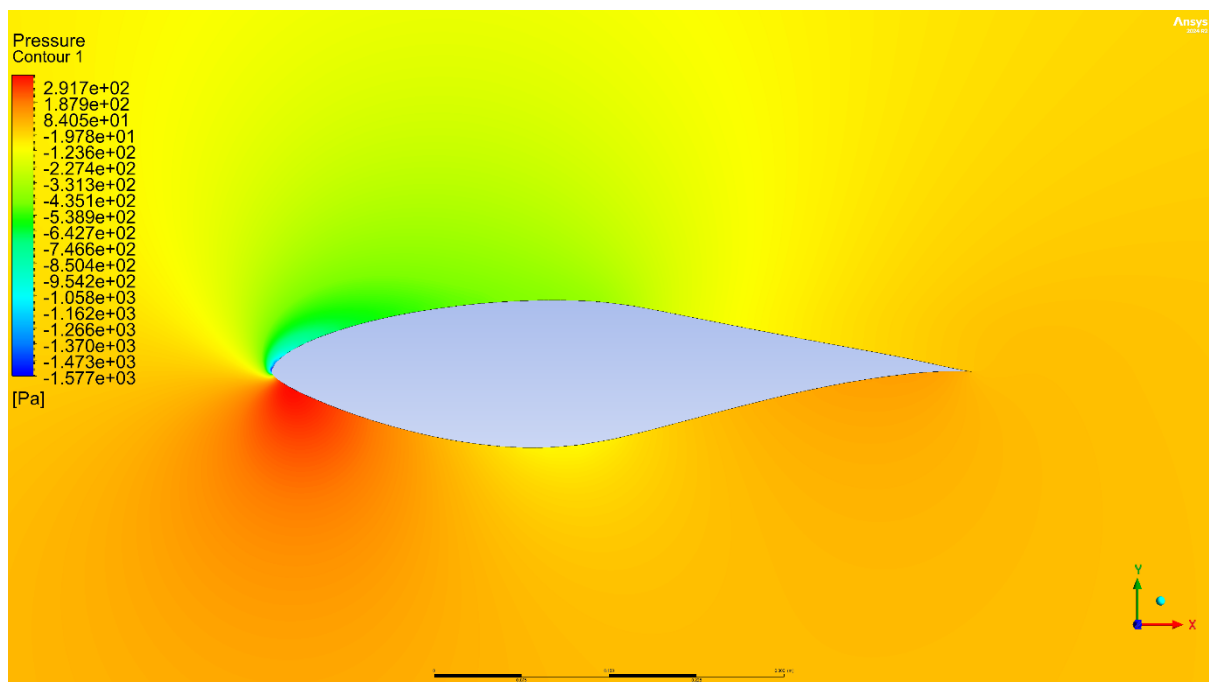


Figure 75: Pressure contour (10 AOA)

12° Angle of Attack

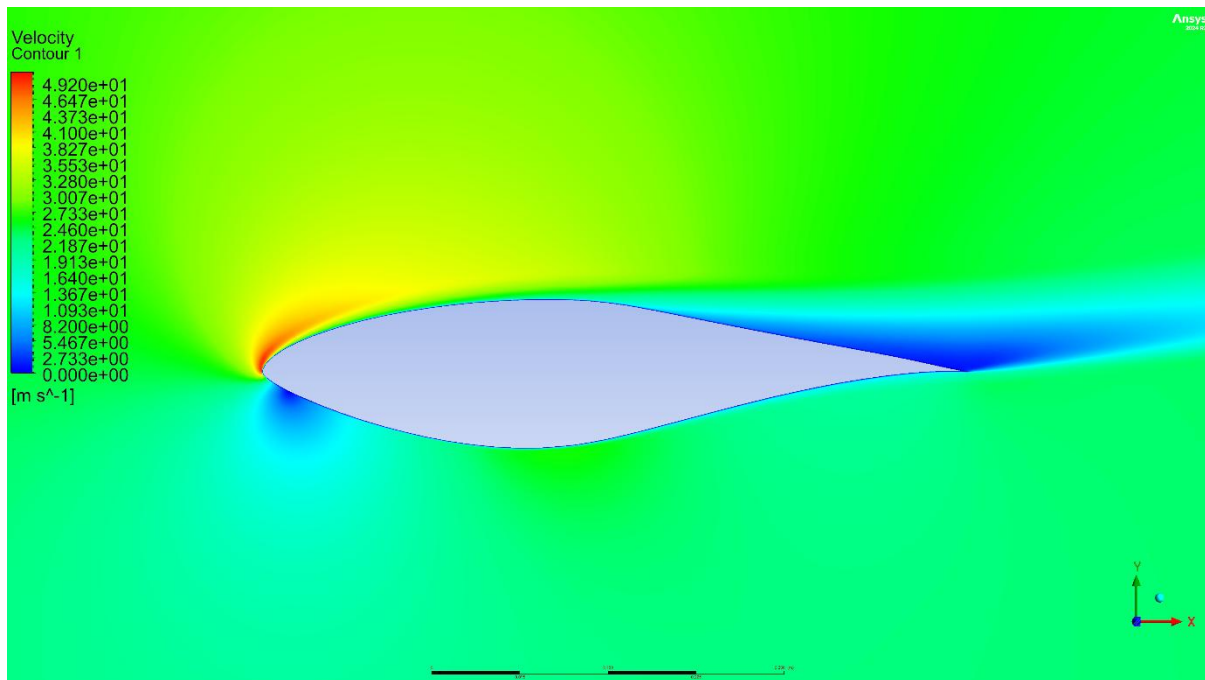


Figure 76: Velocity contour (12 AOA)

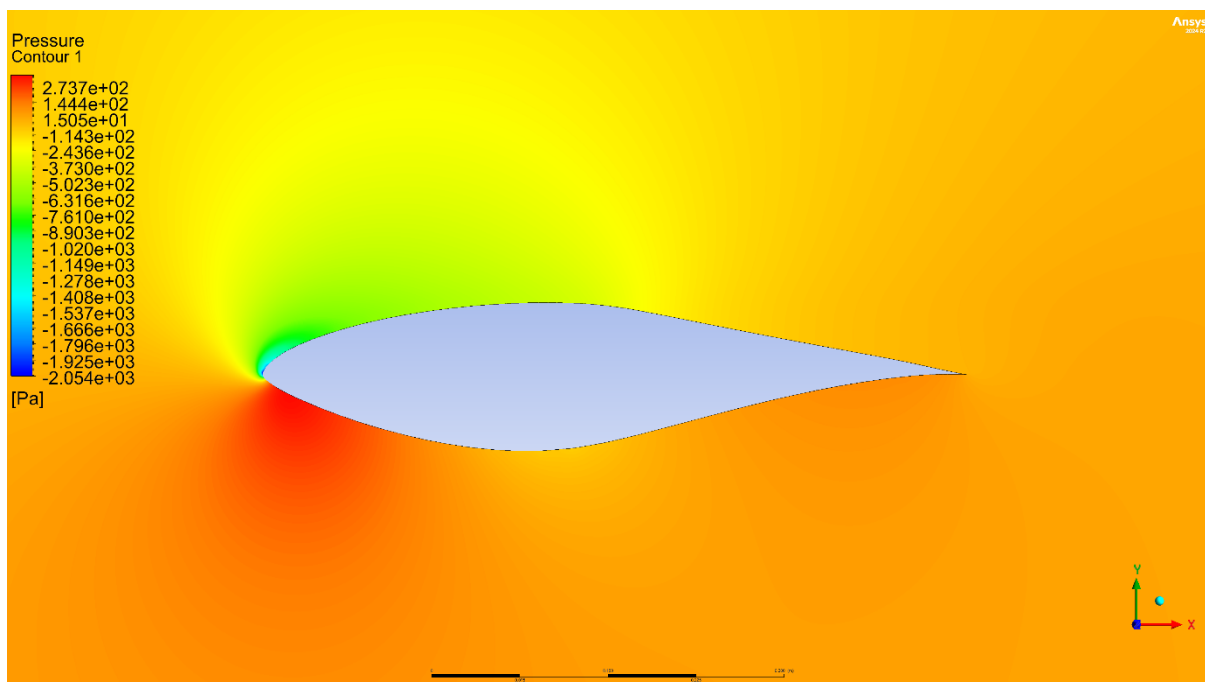


Figure 77: Pressure contour (12 AOA)

14° Angle of Attack

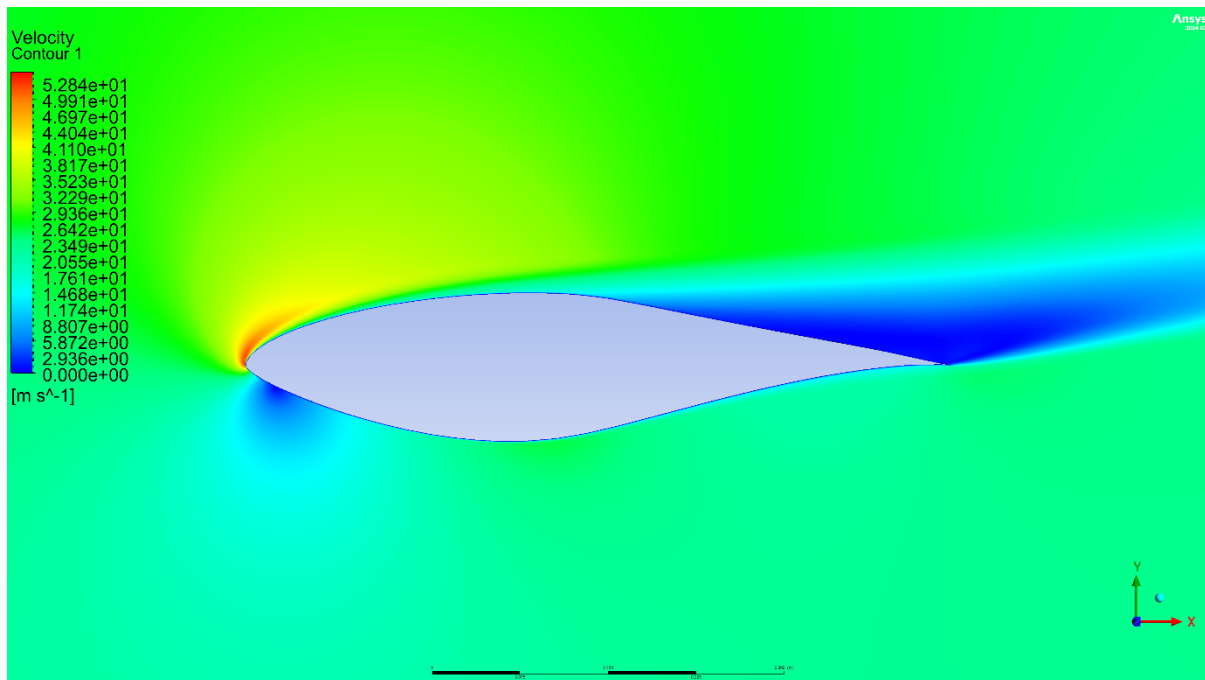


Figure 78: Velocity contour (14 AOA)

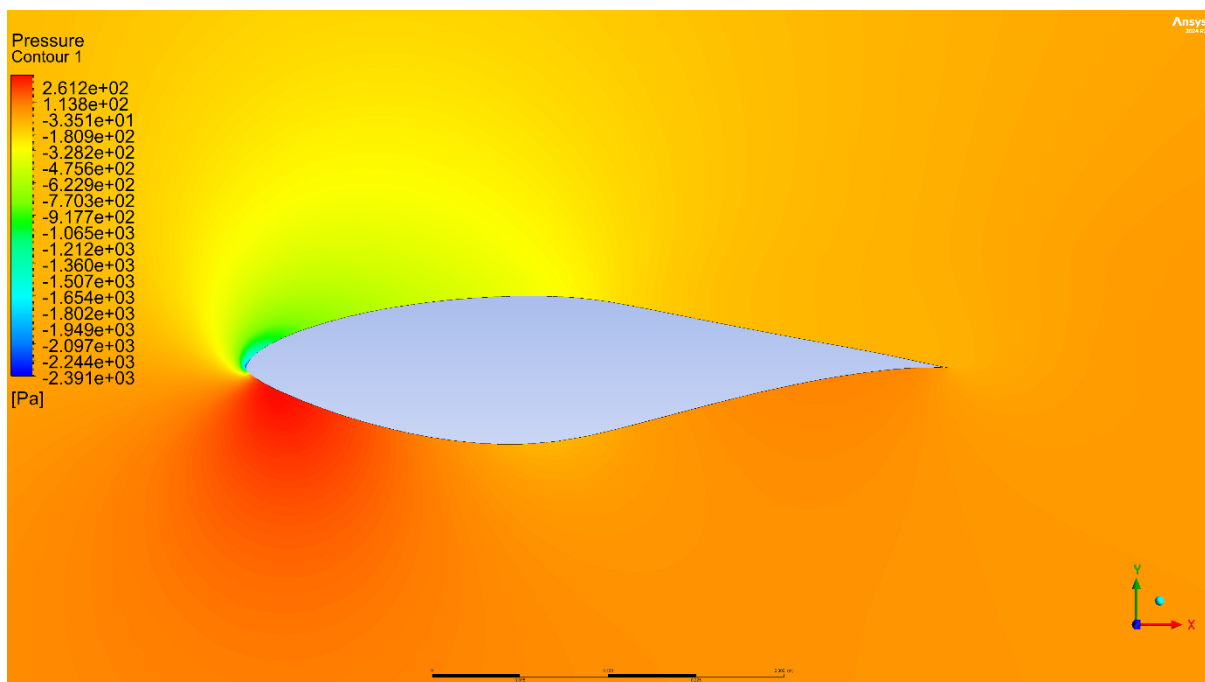


Figure 79: Pressure contour (14 AOA)

16° Angle of Attack

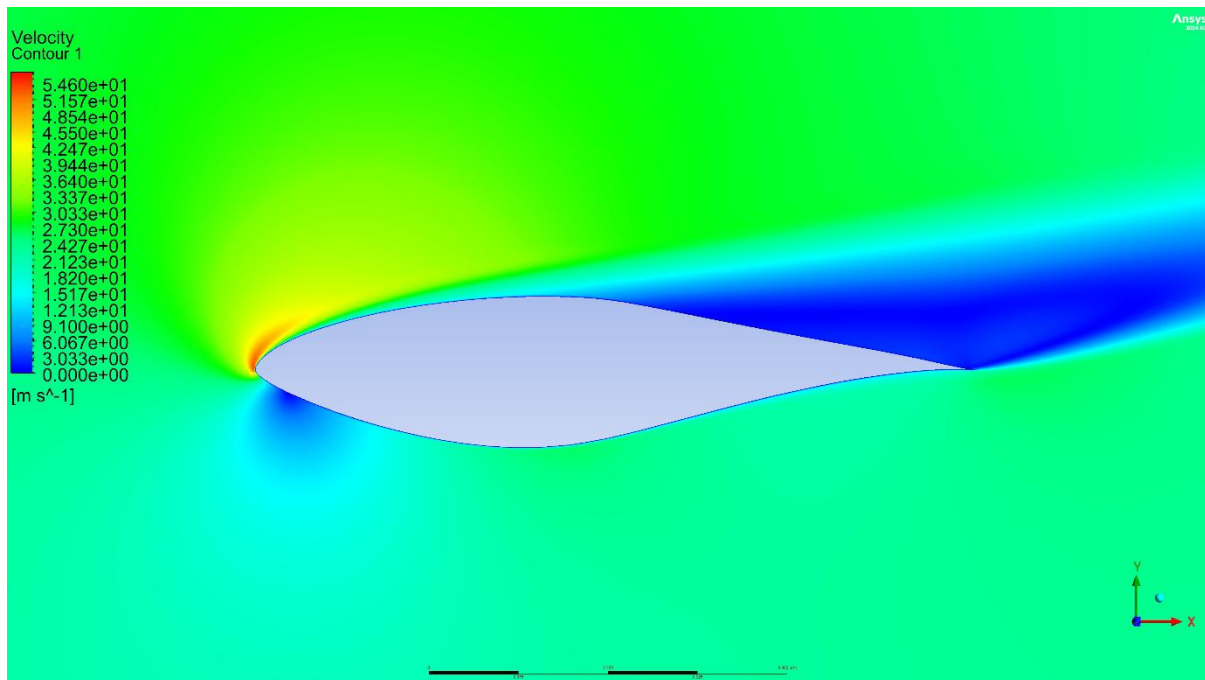


Figure 80: Velocity contour (16 AOA)

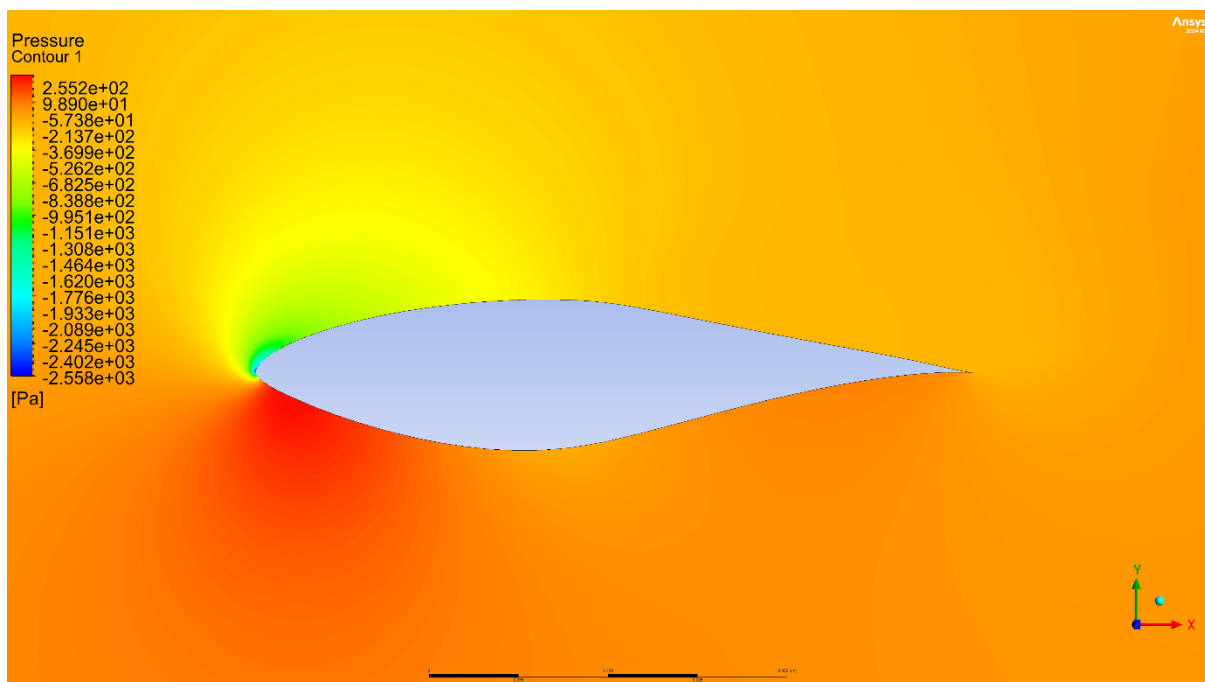


Figure 81: Pressure contour (16 AOA)

18° Angle of Attack

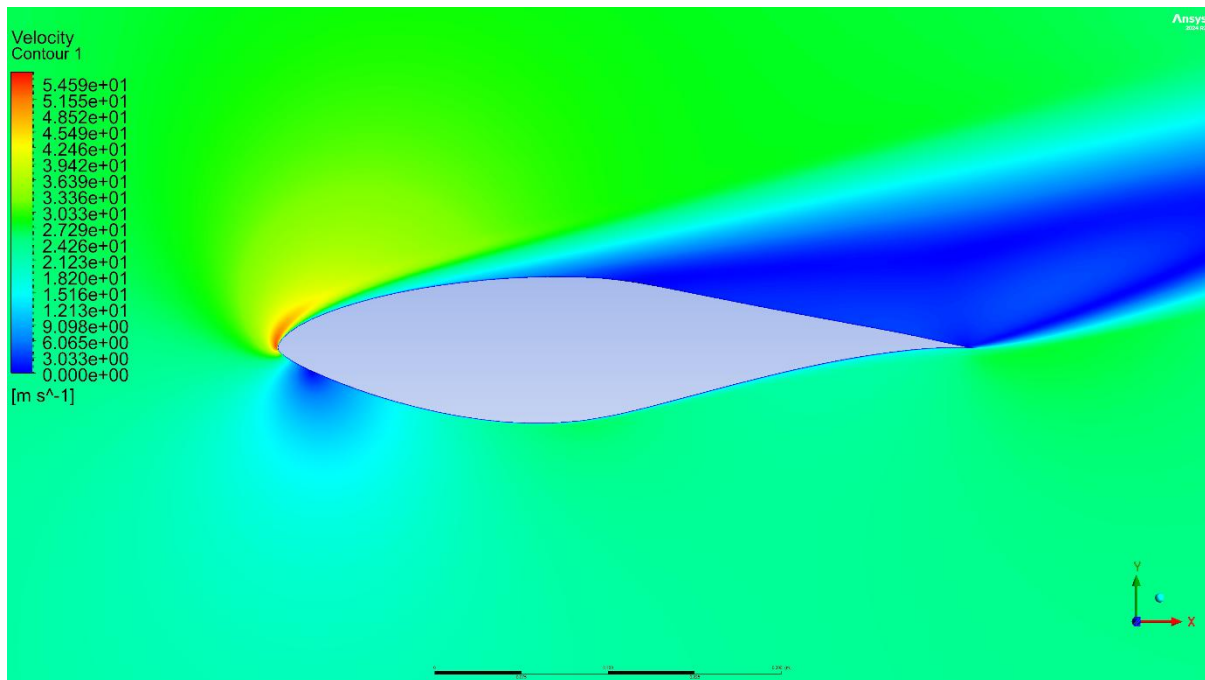


Figure 82: Velocity contour (18 AOA)

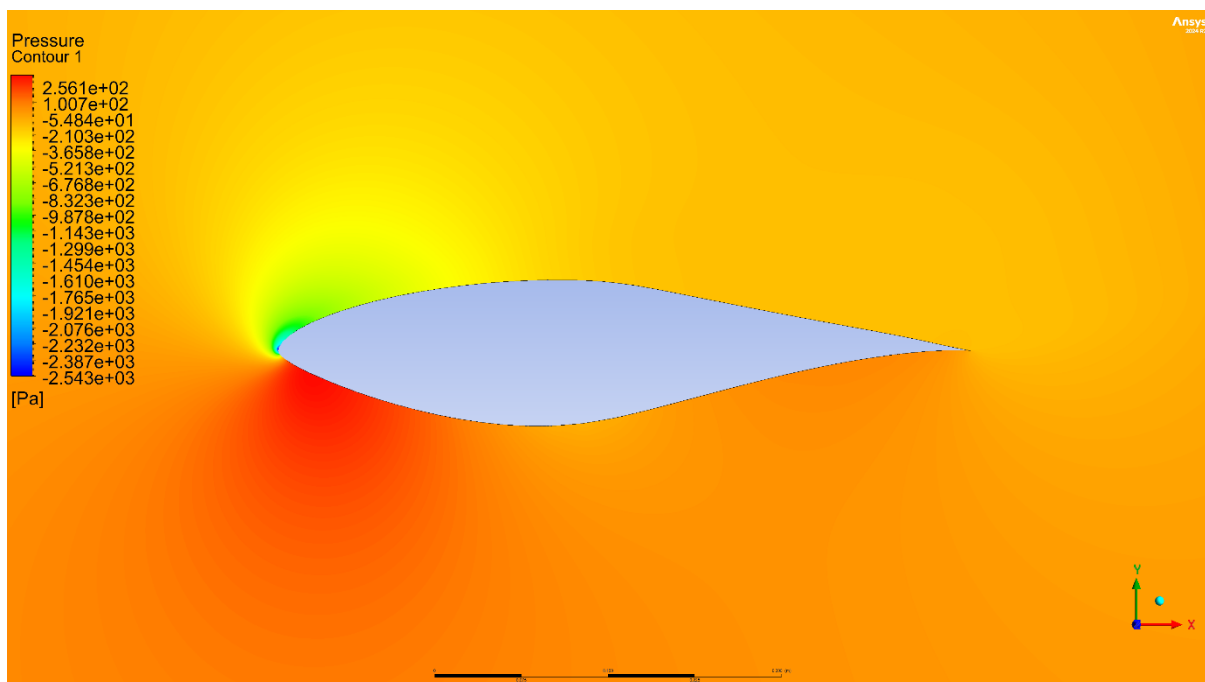


Figure 83: Pressure contour (18 AOA)

20° Angle of Attack

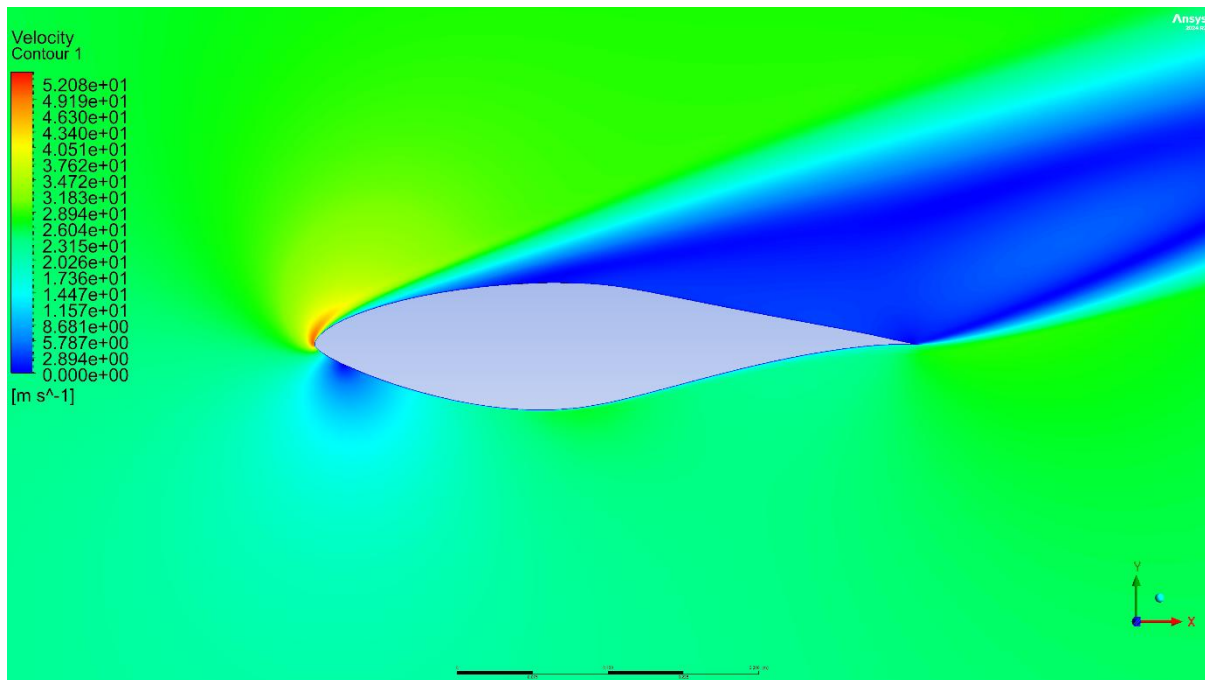


Figure 84: Velocity contour (20 AOA)

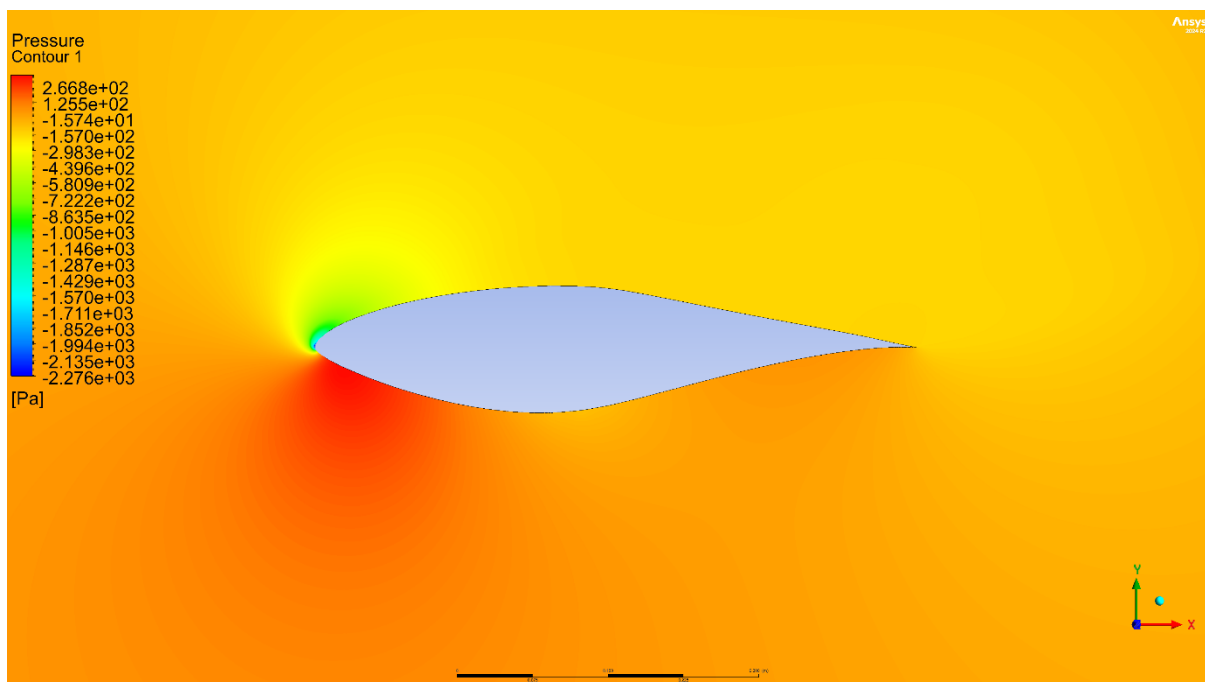


Figure 85: Pressure contour (20 AOA)

5.4 Result validation

The following Figure 86, 87 and 88 shows the deviation between simulated data and research paper data.

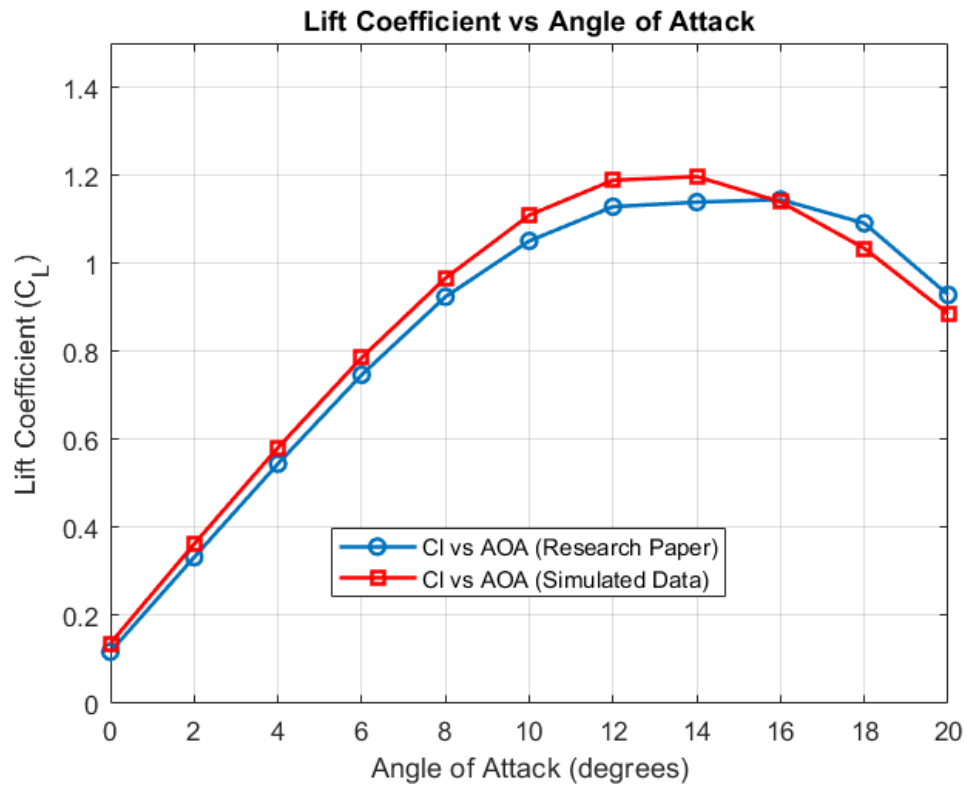


Figure 86: Lift Coefficient vs AOA Comparison

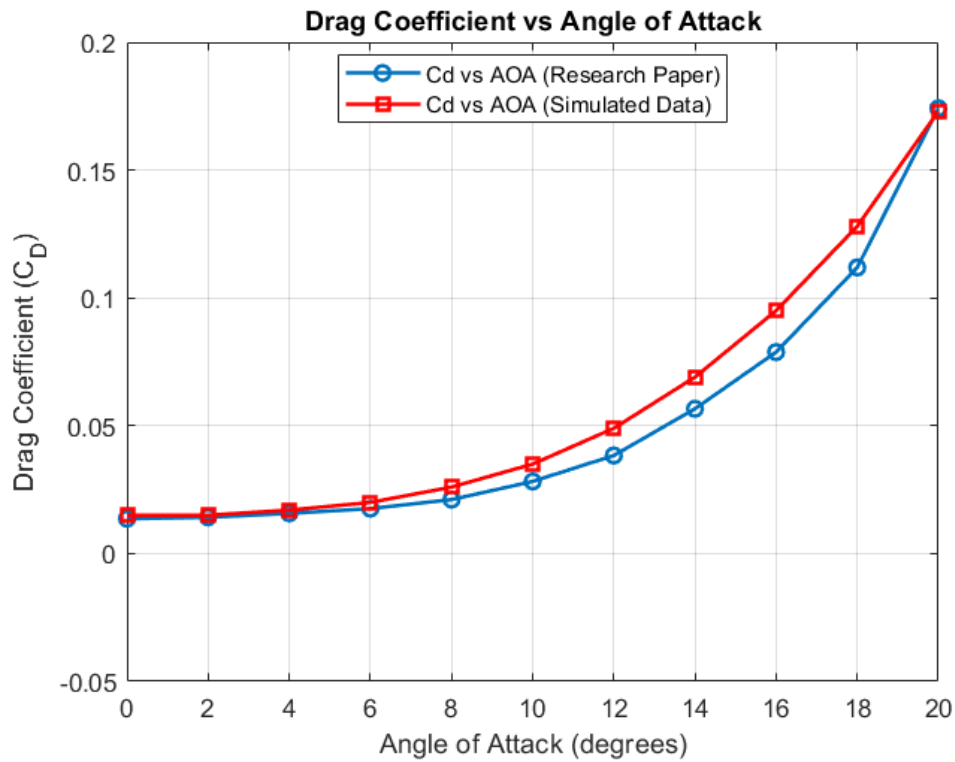


Figure 87: Drag Coefficient vs AOA Comparison

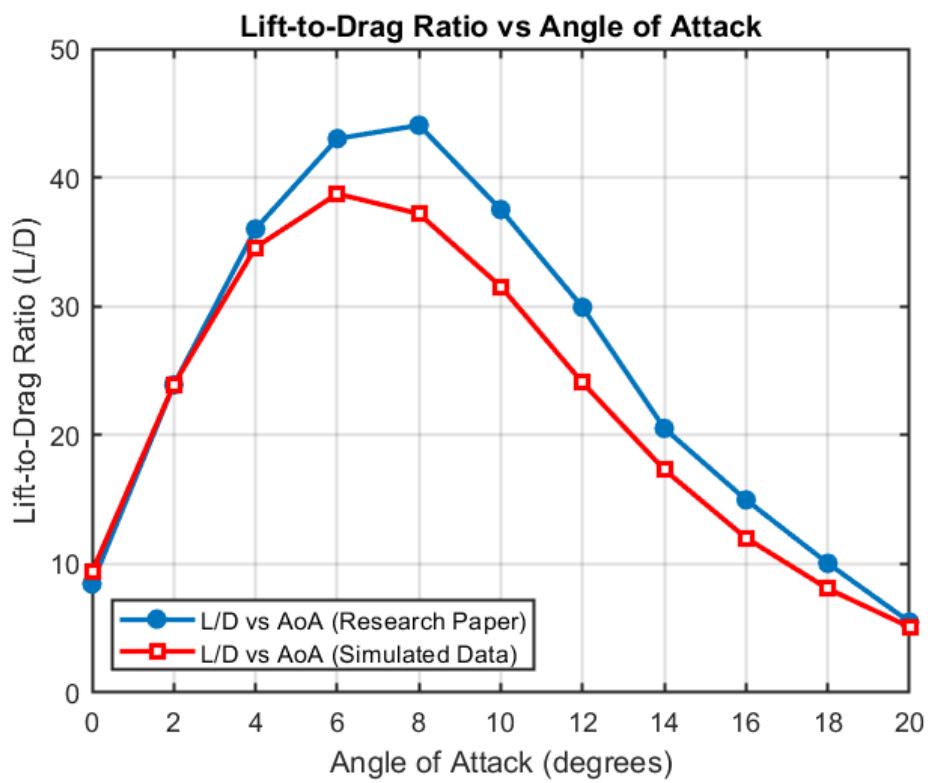


Figure 88: Lift to Drag Ratio vs AOA Comparison

6 Discussion

6.1 Accuracy of the results

Table 3: Error Percentage of lift coefficient values

AOA	Reference Values	Simulated Values	Error Percentage (%)
0	0.118	0.137	16.299
2	0.332	0.362	9.029
4	0.545	0.581	6.732
6	0.747	0.786	5.215
8	0.924	0.966	4.602
10	1.051	1.109	5.575
12	1.128	1.189	5.340
14	1.139	1.197	5.089
16	1.145	1.139	0.460
18	1.091	1.034	5.236
20	0.928	0.885	4.650

Table 4: Error Percentage of drag coefficient values

AOA	Reference Values	Simulated Values	Error Percentage (%)
0	0.014	0.015	6.593
2	0.014	0.015	6.518
4	0.016	0.017	6.582
6	0.018	0.020	13.331
8	0.021	0.026	18.791
10	0.028	0.035	19.863
12	0.038	0.049	22.348
14	0.057	0.069	18.093
16	0.079	0.095	16.937
18	0.112	0.128	12.682
20	0.174	0.173	0.444

Table 5: Error Percentage of lift/drag ratio values

AOA	Reference Values	Simulated Values	Error Percentage (%)
0	8.440	9.373	11.052
2	23.890	23.915	0.106
4	36.000	34.570	3.973
6	43.010	38.735	9.940
8	44.050	37.182	15.591
10	37.500	31.528	15.925
12	29.910	24.067	19.535
14	20.520	17.312	15.635
16	14.960	12.010	19.717
18	10.040	8.067	19.648
20	5.490	5.105	7.021

Tables 3, 4, and 5 present the percentage differences between the current study and the reference values (From the research paper) for the lift coefficient, drag coefficient, and lift-to-drag ratio, respectively. It shows that the drag coefficient values have deviated from the reference values compared to lift coefficient values which results in a higher error percentage around 20%. This deviation has been occurred in angle of attacks around 8 to 16 degrees.

In conclusion, although a few data points show noticeable deviations, the overall accuracy of the present numerical results is satisfactory, with error percentages remaining below 23% when compared to the reference data. These variations are acceptable within the context of CFD, as the accuracy of numerical simulations is strongly influenced by several factors including mesh quality, boundary layer treatment and solver settings.

6.2 Possible errors in the simulation and what can be done to reduce the errors

The minor deviations observed can primarily be due to differences in the mesh resolution used in the two studies. The reference study used a mesh with approximately 60,000 cells, while the present simulation used a finer mesh of about 234,400 to achieve a converged solution. This discrepancy may be occurred due to variations in mesh quality and refinement strategy. Although the original study used fewer elements, its mesh was likely of higher quality, with better element distribution near the aerofoil surface and within the boundary layer. As a result, it was able to produce a mesh independent solution with a lower cell count, whereas the current mesh, despite being denser, may contain elements of lower orthogonality or higher skewness, affecting convergence behaviour and overall accuracy. To avoid these issues, the near wall mesh can be refined and adjust the mesh settings such that it doesn't produce highly skewed cells. Having highly skewed cells causes the mesh quality to drop.

Moreover, the current study adopted a near-wall mesh with a y^+ value slightly below 30, while the reference paper used a y^+ value of around 30. This variation in near wall treatment can influence how the boundary layer is resolved around the aerofoil, contributing to the small deviations in the predicted aerodynamic coefficients. To avoid this, the near wall mesh can be modelled exactly as the reference model with 1mm of first layer height rather than setting the first layer height lower than 1mm.

6.3 Difficulties faced

As observed from the error percentage tables, at higher angles of attack the flow becomes more turbulent, as it approaches the stall region. Due to this complex and highly unsteady flow behaviour, the solver required a greater number of iterations to achieve convergence compared to lower angles of attack. Near stall, the flow is inherently transient and unstable, making it challenging for a steady state solver to accurately capture the flow characteristics. As a result of that, the iterative process becomes longer and more demanding, resulting in an increased number of iterations before convergence is reached.

As mentioned earlier, inflation layers cannot be activated simultaneously with face meshing when generating a structured mesh. To address this limitation, a bias factor was applied to

replicate the effect of inflation layers by gradually refining the mesh toward the airfoil surface. Achieving the correct first-layer height corresponding to the desired y^+ value required an iterative approach, where multiple bias factors were tested until the appropriate near-wall resolution was obtained.

Maintaining low skewed, high orthogonal mesh elements was challenging due to the usage of high bias factors. Achieving an optimum balance between the mesh element quality and the required wall y^+ condition was challenging.

6.4 What can be learned from the simulation results?

- This study focused on mesh quality rather than quantity. Although my CFD study used a higher resolution mesh (Around 200,000 cells) than the 60,000-cell mesh used in the initial publication, it was harder to achieve convergence. This demonstrated that element orthogonality, skewness, and near-wall refinement are more influencing factors for the accuracy of the results than simply cell numbers.
- Maintaining the y^+ value appropriate for the turbulence model near the aerofoil surface was essential to properly simulate viscous effects, especially when in stall conditions. The research again emphasized the need for finding the right balance between mesh density and wall function requirements, while using turbulence models like $k-\epsilon$ and standard wall functions.
- The reproduction confirmed that the original study's justification for employing the basic $k-\epsilon$ model as it presents a mathematically steady, computationally efficient approach to the computation of steady, incompressible external flow. It also showed the necessity to select a turbulence model based on the regime of the flow and the expected separation behaviour.
- Carrying out a mesh independence study demonstrated how grid convergence ensures result reliability. The test revealed that smooth aerodynamic coefficients (C_l and C_d) across various mesh densities indicate a physically consistent solution, as opposed to a numerically driven solution.
- This reproduction of results brought out the sensitivity of the CFD solutions to the solver parameters such as pressure-velocity coupling (Coupled scheme), spatial discretization schemes, and convergence. The same conditions used in the original paper enabled improved observation of how such parameters affect simulation stability and accuracy.

- Recreating another researcher's CFD setup prompted further appreciation of verification and reproducibility in computational simulations. It demonstrated how slight variations in meshing approach, solver settings, or turbulence treatment can influence aerodynamic predictions.

7 Suggestions for Future Improvement

One of the key improvements that can be implemented in future work is the selection of a more suitable turbulence model. The current study used the standard $k-\epsilon$ model, which performs well for predicting flow variables in the far-field region. However, for aerodynamic applications such as aerofoils, the $k-\omega$ SST (Shear Stress Transport) model is generally more appropriate. This hybrid model combines the advantages of both $k-\epsilon$ and $k-\omega$ formulations. It applies the $k-\omega$ approach near the wall to accurately capture boundary layer effects and switches to $k-\epsilon$ in the free stream to ensure numerical stability. Consequently, it provides a more accurate representation of flow separation, adverse pressure gradients, and near-wall turbulence. To use the $k-\omega$ SST model effectively, the y^+ value should be maintained within the range of 1-5, which demands finer near-wall meshing and higher computational resources but results in significantly improved accuracy [6].

Another improvement relates to the boundary condition setup used in the C-type computational domain. In the current study, three separate inlet boundaries were defined, and the angle of attack was simulated by adjusting the x and y components of the inlet velocity vector. However, this approach can introduce non-physical flow behaviour. For an example, at the upper inlet, the flow may exit the domain instead of entering, even though it is defined as an inlet boundary. This leads to unrealistic flow representation near the inlet region. A more physically accurate approach would be to rotate the aerofoil geometry itself to achieve the desired angle of attack, while keeping a uniform inlet velocity. However, this can be done by parameterizing the angle of attack so that for each angle, a new computational mesh is generated. This method, particularly important in a C-type domain, prevents wake interference near the boundaries and ensures that the flow physics around the aerofoil are captured more realistically, improving the overall reliability of the simulation results.

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Link to the ANSYS files: <https://dms.uom.lk/s/Hm6R3Be2XtyJcyM>